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Feature reduction using swarm optimization and random forest classifiers for early diabetes risk prediction

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Diabetes is a chronic metabolic disorder caused by excessive blood sugar levels, which leads to severe damage to other organs. Type 2 diabetes is, more often than others, a long-term metabolic disorder in which the body resists insulin or does not produce enough of it. Early diabetes detection with fewer features reduces patient burden, and machine learning makes the process more time-efficient. This study proposes three machine learning approaches that achieve both high performance and effective feature reduction using the Early Stage Diabetes Risk Prediction dataset. Multiple research works have been published. However, they have struggled to achieve efficient feature reduction while maintaining high accuracy and have not provided detailed explanations of the models' nature or misclassifications. This research resolves the issues by showing outstanding performance, including a remarkable feature reduction. Three different swarm-based metaheuristic algorithms: Fox Optimizer, Honey Badger Algorithm, and Tuna Swarm Optimization, have been used, wrapped with a Random Forest Classifier. SHAP, as an explainable AI, is used to present the model's nature and feature importance, including individual predictions. FOX_RF, HBA_RF, and TSO_RF have gained an Accuracy of 99.36%, 99.36%, and 100% without cross-validation. However, TSO_RF has achieved the highest mean 10-fold crossvalidation accuracy of 98.14%, F-score of 98.47%, and 98.54% of Precision using only 14 features out of 16. And, FOX_RF has achieved the highest mean Precision of 98.43%. HBA_RF has shown the highest number of feature reduction by selecting 10 features out of 16, maintaining a moderate performance. SHAP has confirmed that Polyuria, Polydipsia, and Gender are the most impacted features for diabetes prediction. SHAP-based individual prediction analysis has revealed that even small changes in these features can influence the model's decisions. This research analyzes the Early Stage Diabetes Risk Prediction dataset, which includes 520 individuals with 16 predictors and one target class, where TSO_RF has outperformed other models by achieving 100% and 98.14% of Accuracy, respectively, without cross-validation and using cross-validation.

Diabetes is a chronic metabolic disorder caused by elevated blood sugar (glucose) levels, which leads to severe damage to other organs such as the heart, eyes, kidneys, and so on¹. It arises from insufficient insulin production by the pancreas or the body's inability to utilize insulin effectively². The prevalence of diabetes has rapidly increased 4.15 times from 200 million in 1990 to 830 million in 2022¹. Diabetes is mainly classified into two main categories: (a) Type-1 Diabetes (T1D) and (b) Type-2 Diabetes (T2D)^{3,4}. But sometimes diabetes has been classified into three major types, including Gestational Diabetes (GD) as the third category^{5,6}. If the body develops an autoimmune response and damages the pancreatic islet β -cells, leading to little or no insulin production, resulting in T1D⁷. However, T2D is a long-term metabolic condition in which the body either resists the effects of insulin or does not produce enough, which interferes with the metabolism of carbohydrates, fats, and proteins, ultimately causing chronically high blood sugar levels⁸. According to the Global Burden of Disease Study 2021, the rate of occurrence of T2D is more often compared to T1D, as well as for the death rate, shown in Fig. 1⁹. In the year 2021, the total mortality associated with

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diabetes was 1656634, where T2D had 1608123 and T1D had 48511. And the graph indicates that the T1D mortality rate over the year is almost constant, but the T2D is increasing daily.

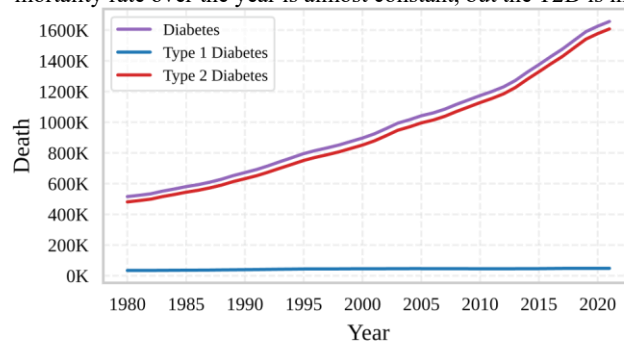


Fig. 1. Incidence of diabetes mortality from 1980–2021.

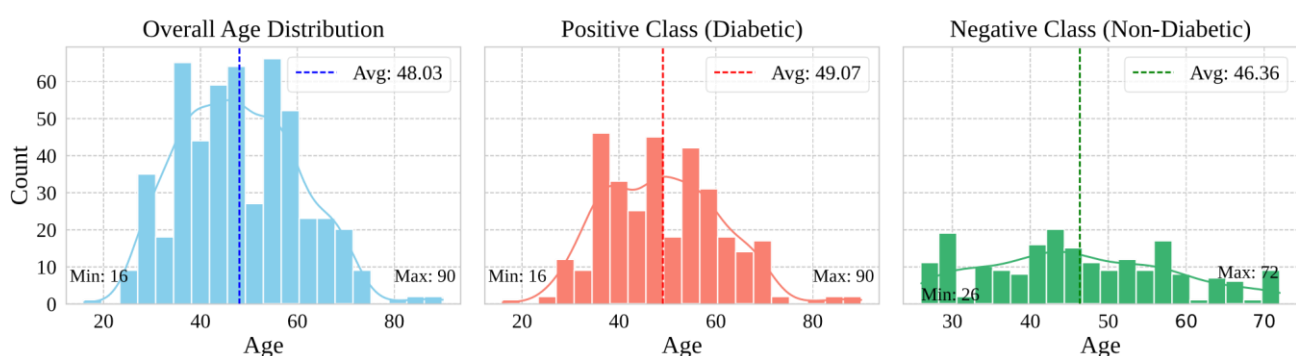


Fig. 2. Age distribution of the dataset.

Early-stage detection and prevention of diabetes are essential in mitigating the progression of the disease, reducing the risk of severe complications (e.g., vascular complications), and minimizing the patient's burden¹⁰. Recent advancements in Machine Learning (ML) have proven its utilization in different sectors, such as the medical sector and engineering sector^{11–14}. In the context of early diabetes identification, ML offers an effective and reliable approach for detecting the presence of diabetes in the human body by analyzing clinical, physical, and questionnaire-based data. Several studies have already proven the usage of ML in diabetes prediction based on different datasets^{15–17}. Among them, the Pima Indians Diabetes Database (PIDD), the CDC Diabetes Health Indicators (CDC-DHI), and the Early Stage Diabetes Risk Prediction (ESDRP) are the well-known datasets. This study has utilized the ESDRP dataset to develop an efficient, accurate, and reliable machine learning-based model for predicting diabetes. The ESDRP dataset has 16 features with a binary target class, including 520 instances (320 diabetes positive and 200 diabetes negative individuals). The features are: Age, Gender, Polyuria, Polydipsia, Sudden weight loss, Weakness, Polyphagia, Genital thrush, Visual blurring, Itching, Irritability, Delayed healing, Partial paresis, Muscle stiffness, Alopecia, and Obesity. Among the features, all of them are binary categorical features except age. The average age is 48.03, 49.07, and 46.36 years old for the overall dataset, diabetic positive, and diabetic negative individuals, as shown in Fig. 2. The minimum and maximum ages of the overall dataset are 16 and 90 years old, respectively, and the same as the positive class. However, for the negative class, the minimum age is 26, and the maximum age is 72 years old. The dataset has been described in Section 2.1.

Recent studies have shown significant improvements in diabetes prediction using ML and the ESDRP dataset. Islam et al. introduced the dataset and demonstrated an impressive classification accuracy of 97% using a Random Forest (RF) classifier with 10-fold cross-validation (CV)¹⁸. For better understanding, previous research works have been discussed in two groups: No Cross-Validation (NCV) Models and with Cross-Validation (WCV) Models.

Firstly, we have summarized the results of the NCV models. Nagpal et al. have implemented an Ensemble Bagging Decision Tree (EBDT) classifier attached with Principal Component Analysis (PCA) Optimizer and achieved an accuracy of 93.26% without CV³. Another research work has gained 94% accuracy with the RF classifier using 16 features¹⁹. Le et al. have used Multilayer Perceptron (MLP) wrapped with Adaptive Particle - Grey Wolf Optimization (APGWO) and been able to attain 97% accuracy⁴. Another study on the ESDRP dataset has used the Extra Trees (ET) classifier and has achieved an accuracy of 97.11%, identifying polyuria, polydipsia, gender, and itching as the most influential predictive features⁵. Another article based on

the K Nearest Neighbor (KNN) classifier has outperformed the few previous works with an accuracy of 97.36%²⁰. The Gradient Boosting (GB) classifier has achieved 98.07% accuracy on this dataset²¹. However, Gundougdu et al. have shown the superiority and been able to reach 99.23% accuracy using the XGBoost classifier, integrated with RF Feature Selector (RFFS), while they have only used 9 features²².

Secondly, we have summarized the findings of the WCV models on the ESDRP dataset. Navazi et al. have used the Particle Swarm Optimizer (PSO) wrapped with a Genetic Algorithmic Support Vector Machine (GASVM) and reduced the number of features from 16 to 9²³. They achieved an accuracy of 93.42% using 10-fold cross-validation. Rahman et al. used the RF classifier with SMOTE to achieve 94.87% accuracy using 5-fold CV²⁴. Two different research works using the RF classifier and 10-fold CV achieved accuracies of 95.03% and 96.58%^{6,25}. Another study reported a 10-fold mean accuracy of 96.81% using an optimized decision forest (ODF) with the random forest ensemble (RFE) technique assisted by binary multineighborhood artificial bee colony (BMNABC)²⁶. Saxena et al. have implemented the Stacked Ensemble (SE) classifier integrated with the Impurity method by Extra Tree Classifier (IMETC) feature selector, achieving 97% prediction accuracy while ensuring 10-fold CV²⁷. Two different research works,^{28,29} used 10-fold CV and achieved accuracies of 98.08% each, while using two different ML methods, a Neural Network (NN) and an RF classifier, respectively. Table 1 presents the overall findings of the recent works. Based on the reviewed literature, the research gaps can be summarized as follows:

- Existing high-accuracy models, such as Gundougdu et al., have not incorporated cross-validation (CV), limiting their generalizability and reliability on unseen data.
- Studies that have employed CV, including Raju et al. and Chaves et al., have not applied feature-reduction techniques, resulting in increased computational cost and model complexity.
- None of the existing models—whether NCV or WCV—has provided a comprehensive explainability analysis to interpret model behavior and feature importance.
- Prior work lacks a detailed misclassification analysis, leaving error patterns and model weaknesses unexplored.
- No study has presented strong evidence validating the significance and contribution of selected features to overall model performance.

Building on these limitations, we have designed our model to address these gaps, with the following key contributions or the research objectives can be written as follows:

- **Feature Reduction and Parameter Tuning:** This study has employed three Swarm-Based Metaheuristic Algorithms (SBMHAs)—Fox Optimizer (FOX), Honey Badger Algorithm (HBA), and Tuna Swarm Optimization (TSO)—for performing feature reduction and hyperparameter tuning.
- **Cross-Validated Classification:** This work has applied a Random Forest (RF) classifier with a 10-fold cross-validation strategy to ensure high classification accuracy, as well as robustness, generalizability, and reliability in distinguishing diabetic and non-diabetic individuals.
- **Performance Evaluation:** This study has evaluated model performance using standard metrics such as Accuracy, F-score, Precision, and Recall. Additionally, ROC curves and computational time complexity analyses have been presented to provide further insights into the model’s efficiency.
- **Feature Importance Analysis:** We have interpreted the contribution of each feature using Shapley Additive Explanations (SHAP), enabling a deeper understanding of feature influence at both the global and individual prediction levels.
- **Misclassification Analysis:** This work has examined misclassified instances and analyzed potential causes of misclassification to identify areas for model improvement.

SL No	Reference	Year	Method	Accuracy (%)	Features
Without Cross-Validation					
01	3	2024	EBDT + PCA	93.26	16
02	19	2021	RF	94.00	16
03	4	2020	MLP + APGWO	97.00	16
04	5	2024	ET	97.11	16
05	20	2022	KNN	97.36	16
06	21	2025	GB	98.07	16
07	22	2023	XGBoost + RFFS	99.23	09
With Cross-Validation					
01	23	2023	GA-SVM + PSO	93.42	09
02	24	2023	RF + SMOTE	94.87	16
03	25	2024	RF	95.03	16
04	6	2024	RF	96.58	16
05	26	2024	ODF(RFE) + BMNABC	96.81	16
06	27	2021	ES+IMETC	97.00	12
07	28	2021	RF	98.08	16
08	29	2024	RF	98.08	16

The remainder of this research paper is structured into three main sections: Methodology, Results, and Conclusion. In Section Methodology, we have presented a comprehensive overview of the research approach, including a description of the dataset, the ML algorithms employed, and the proposed model framework. Section Results presents the overall findings of this research study and describes the use of SHAP for introducing the model's trends and feature importance. Finally, Section Conclusion concludes the work, highlighting the key findings and future work plan.

Methodology

In this scientific study, diabetes affection has been classified using the ESDRP dataset, which includes data from 502 individuals. The dataset has 17 columns, including 16 predictor features as predictors and one target class. Only the age is numerical (Integer), and other features are categorical binary features (e.g., Gender: Male or Female, Polyuria: Yes or No, Weakness: Yes or No). First, we have prepared a 10-fold cross-validation dataset from the original dataset, ensuring proper shuffling to maintain randomness and reduce bias. We have designed our framework by integrating SBMA with a classifier to perform feature selection and hyperparameter tuning, ensuring optimal performance across each fold of the 10-fold cross-validation dataset. And then, an optimized model has been developed using the voting method, and SHAP has been introduced to explain the models as shown in Fig. 3.

Dataset

The Early Stage Diabetes Risk Prediction (ESDRP) dataset is derived primarily from a set of direct questionnaires conducted by Sylhet Diabetes Hospital in Sylhet, Bangladesh, which has been approved and used by specialists and doctors. The dataset is available at the UCI Machine Learning Repository: <https://archive.ics.uci.edu/dataset/529/early+stage+diabetes+risk+prediction+dataset>, accessed on: 13th of April, 2025. As we previously

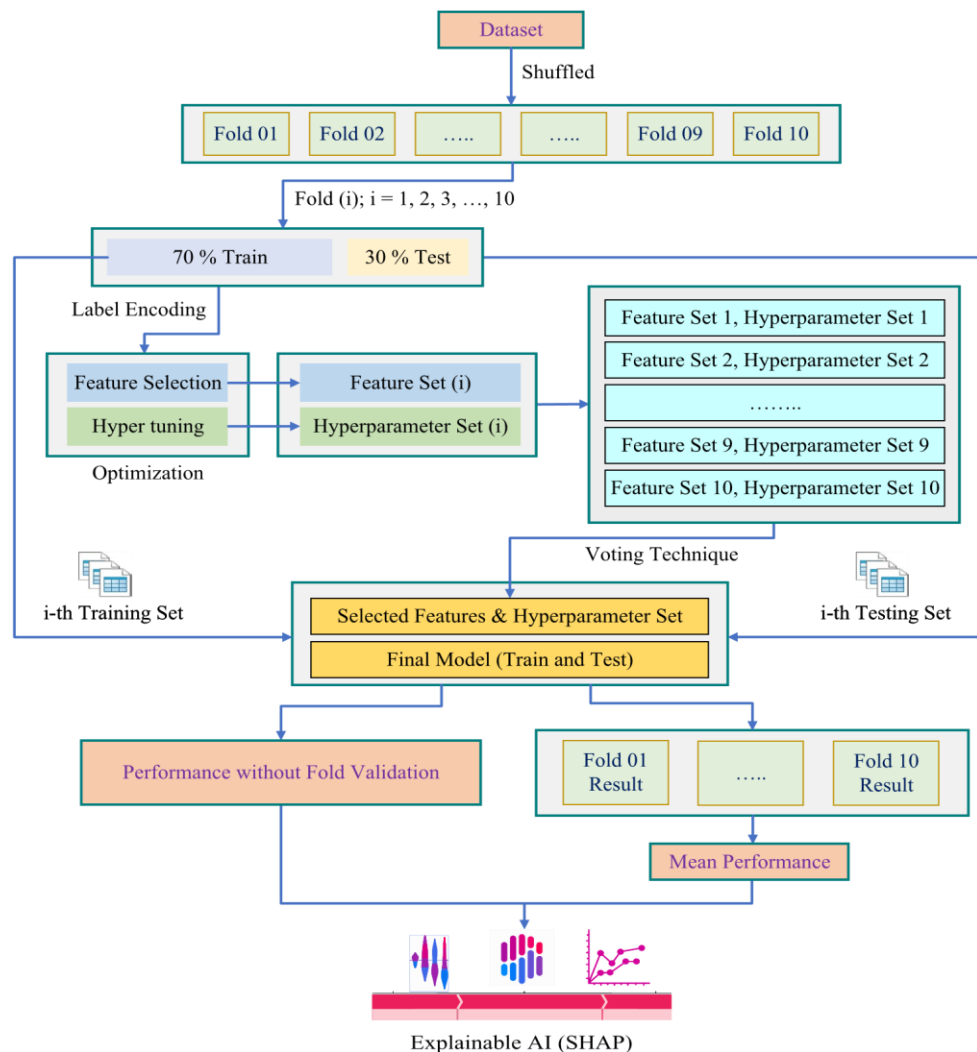


Fig. 3. Methodology of the proposed framework.

mentioned, the features are: Age, Gender, Polyuria, Polydipsia, Sudden weight loss, Weakness, Polyphagia, Genital thrush, Visual blurring, Itching, Irritability, Delayed healing, Partial paresis, Muscle stiffness, Alopecia, and Obesity as shown in Table 2 including their value and encoded value in the column Value (Enc. Value). In the dataset, the ratio of the Positive and Negative classes is 320:200.

Figure 4 visualizes the correlation among the features, where the large yellow diagonal circle represents the highest correlation between a feature and itself. The target class has the highest correlation with Polyuria (V_2) and Polydipsia (V_3). However, Itching (V_9) has the lowest correlation with the target class. According to Fig. 4, the features Age (V_0), Gender (V_1), Polyuria, Polydipsia, Sudden weight loss (V_4), and Partial paresis (V_{12}) exhibit a strong correlation with the target class, suggesting that these variables are among the most influential predictors for diabetes classification. **Random forest classifier**

RF Classifier is an ensemble ML algorithm that combines multiple randomly constructed decision tree classifiers to improve classification accuracy and reduce overfitting, proposed by Breiman³⁰. RF includes the bagging sampling approach with the random selection of features process³¹. The smaller tree classifiers are represented in Eq. 1, as described by Breiman³⁰.

$$\{h_{(x'\Theta_k)}\} \tag{1}$$

Here, $k = 1, 2, 3, \dots$, and Θ_k denotes an independently distributed random vector identifying the most significant class for input x . The Gini index is used to evaluate each tree and its decision. The Gini index is

Feature	Notation	Datatype	Value (Enc. Value)	Count (%)	Missing Values	Role
Age	V_0	Integer	—	—	No	Feature
Gender	V_1	Categorical	Male (1)	328 (63.08%)	No	Feature
			Female (0)	192 (36.92%)		
Polyuria	V_2	Binary	$X(0)$	262 (50.38%)	No	Feature
			$\sqrt{(1)}$	258 (49.62%)		
Polydipsia	V_3	Binary	$X(0)$	287 (55.19%)	No	Feature
			$\sqrt{(1)}$	233 (44.81%)		
Sudden weight loss	V_4	Binary	$X(0)$	303 (58.27%)	No	Feature
			$\sqrt{(1)}$	217 (41.73%)		
Weakness	V_5	Binary	$X(0)$	215 (41.35%)	No	Feature
			$\sqrt{(1)}$	305 (58.65%)		
Polyphagia	V_6	Binary	$X(0)$	283 (54.42%)	No	Feature
			$\sqrt{(1)}$	237 (45.58%)		
Genital thrush	V_7	Binary	$X(0)$	404 (77.69%)	No	Feature
			$\sqrt{(1)}$	116 (22.31%)		
Visual blurring	V_8	Binary	$X(0)$	287 (55.19%)	No	Feature
			$\sqrt{(1)}$	233 (44.81%)		
Itching	V_9	Binary	$X(0)$	267 (51.35%)	No	Feature
			$\sqrt{(1)}$	253 (48.65%)		
Irritability	V_{10}	Binary	$X(0)$	394 (75.77%)	No	Feature
			$\sqrt{(1)}$	126 (24.23%)		
Delayed healing	V_{11}	Binary	$X(0)$	281 (54.04%)	No	Feature
			$\sqrt{(1)}$	239 (45.96%)		
Partial paresis	V_{12}	Binary	$X(0)$	296 (56.92%)	No	Feature
			$\sqrt{(1)}$	224 (43.08%)		
Muscle stiffness	V_{13}	Binary	$X(0)$	325 (62.50%)	No	Feature
			$\sqrt{(1)}$	195 (37.50%)		
Alopecia	V_{14}	Binary	$X(0)$	341 (65.58%)	No	Feature
			$\sqrt{(1)}$	179 (34.42%)		
Obesity	V_{15}	Binary	$X(0)$	432 (83.08%)	No	Feature

			$\sqrt{1}$	88 (16.92%)		
Class	–	Binary	Positive (1)	320 (61.54%)	No	Target
			Negative (0)	200 (38.46%)		

Table 2. The ESDRP dataset description.

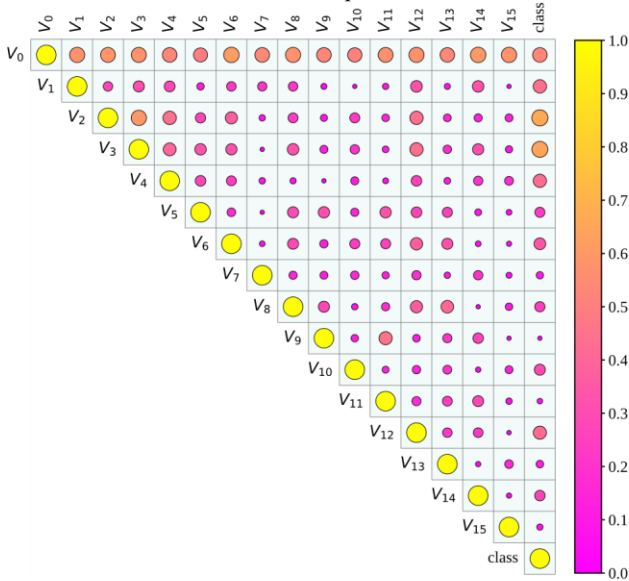


Fig. 4. The ESDRP dataset Feature correlation.

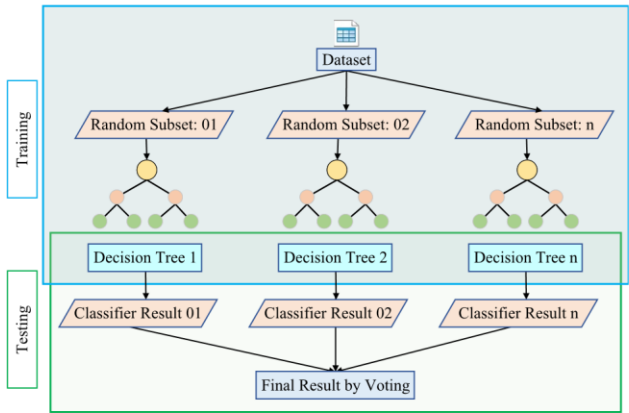


Fig. 5. Structure of the RF classifier³³.

calculated using Eq. 2 where t , N are the conditions, number of classes, and C_i is the i th class label in the data set³¹.

$$\text{Gini}(t) = 1 - \sum_{i \in N} P(C_i | t)^2 \tag{2}$$

=1

The final result has been determined through majority voting based on the outputs of individual smaller classifiers, each built using decision trees, as shown in Fig. 5. The generalization error of the RF, denoted as PE^* , is defined in Eq. 3³⁰.

$$PE^* = P_{X,Y}(f_m(X,Y) < 0) \tag{3}$$

where PE^* represents the probability of misclassification over the joint distribution of X and Y , $f_m(X, Y)$ is the margin function defined in Eq. 4³⁰.

$$f_m(X, Y) = \text{av}_k I(h_k(X) = Y) - \max_{j \neq Y} \text{av}_k I(h_k(X) = j) \quad (4)$$

where $f_m(X, Y)$ is the margin for input X and true class Y , X represents the input feature vector, Y is the true class label, av_k denotes the average over all trees $k = 1, \dots, K$ in the forest, I is the indicator function (1 if the argument is true, 0 otherwise), $h_k(X)$ is the prediction of the k -th decision tree for input X , and $j \neq Y$ refers to competing class labels ($j \in \text{classes}, j \neq Y$).

RF error rate is affected by tree correlation and strength. A higher correlation between two trees increases the error, while strong individual trees reduce it³¹. However, RF is popular for its clear decisions, strong accuracy, outlier robustness, high-dimensional handling, and efficient, parallel-friendly computation³².

Feature selection algorithms

Feature Selection (FS) is a mathematical approach that selects a subset of the most influential features from all available features³⁴. The classification capability of classifier algorithms is upheld and strengthened through FS algorithms (FSA)³⁵. While various types of Feature Selection Algorithms (FSAs) exist, this research has specifically focused on Swarm Intelligence Algorithms (SIAs), a subset of metaheuristic algorithms, first introduced by Beni et al.³⁶. SIAs are bio-inspired optimization algorithms that follow the behavior of social organisms, such as animals, birds, insects, fish, or humans, to solve complex problems³⁷.

Choosing feature selection algorithms and justification

Rajwar et al. have listed 540 metaheuristic algorithms (MHAs) in their review paper and categorized them based on different benchmarks³⁷. However, in this research, Fox Optimizer (FOX), Honey Badger Algorithm (HBA), and Tuna Swarm Optimization (TSO) have been used as SBMHAs to reduce the feature number while ensuring hyper-tuning of the classifier's parameters.

FOX compatibility has been evaluated on five classical benchmark functions (CBFs) and the CEC2019 benchmark test functions³⁸. Table 3 lists the CBFs and indicates their two types: Unimodal and Multimodal. FOX has demonstrated superior performance compared to several other algorithms, including the Grey Wolf Optimizer (GWO), Chimp Optimization Algorithm (ChOA), Genetic Algorithm (GA), Particle Swarm Optimization (PSO), and Butterfly Optimization Algorithm (BOA). The jump technique in FOX has enhanced the performance.

The HBA model's efficiency evaluation has been tested on 24 standard benchmark functions (SBFs) (Unimodal, Multimodal, and Fixed-dimensional), the CEC'17 test suite, and four engineering designs. Ten well-known MHAs, including Simulated annealing (SA), PSO, Moth-flame Optimization (MFO), Whale Optimization Algorithm (WOA), have been compared to HBA. HBA has shown superiority in converging speed and maintaining the balance between exploration and exploitation.

The TSO algorithm has also been tested on 23 test functions, including 7 unimodal, 6 multimodal, and 10 multimodal with fixed dimension³⁹. The outcomes of the functions have been compared with the outcomes of seven well-accepted MHAs, such as the tunicate swarm algorithm (TSA), PSO, GWO, WOA, equilibrium optimizer (EO), Salp swarm algorithm (SSA), and Harris Hawks optimization (HHO), where TSO has shown its superiority over the algorithms.

Fox optimizer

FOX is a type of SBMHA, developed by Mohammed et al. in 2022, mimics the hunting behavior of red foxes³⁸. The red foxes go hunting when the surface is covered by snow. They use ultrasound to find and trace the prey. When they get close to the prey, they use a special jump technique to complete the hunt in the minimum time and from the best position. This hunting behavior has been divided into two main phases: Exploration and Exploitation.

During the exploitation period, the total distance traveled by the ultrasound produced by the fox is denoted by $Dist_S_T_{it}$ as shown in Eq. 5, where Sp_S is the speed of sound in air (343 ms^{-1}), and $Times_S_T_{it}$ is the time traveled by the sound³⁸. The relation between the best position ($BestPosition_{it}$) of exploitation and the $Times_S_T_{it}$ has been shown in Eq. 6³⁸.

$$Dist_S_T_{it} = Sp_S \times Times_S_T_{it} \quad (5)$$

$$Sp_S = \frac{BestPosition_{it}}{Times_S_T_{it}} \quad (6)$$

The distance between the red fox and the prey, $Dist_Fox_Prey_{it}$, is calculated by Eq. 7 and the height of the jump is denoted by $Jump_{(it)}$ as shown in Eq. 8³⁸.

SL no	Expression	Type
1	$\sum_{i=1}^n x_i^2$	Unimodal
2	$\sum_{i=1}^n [100(x_{i+1} - x_i^2)^2 + (x_i - 1)^2]$	Unimodal
3	$\sum_{i=1}^n -x_i \sin(\sqrt{ x_i })$	Multimodal
4	$\sum_{i=1}^n [x_i^2 - 10 \cos(2\pi x_i) + 10]$	Multimodal
5	$\frac{1}{4000000} \sum_{i=1}^n x_i^2 - \sum_{i=1}^n \cos \sqrt{ x_i } + 1$	Multimodal

$$\sum \prod ()$$

Table 3. Five classical benchmark functions used in FOX³⁸.

$$Dist_Fox_Prey_{it} = 0.5 \times Dist_S_T_{it} + 12 \times Dist_S_T_{it} \quad (7)$$

$$Jump_{(it)} = 0.5 \times 9.81 \times t^2; t = \text{Average time of sound to travel.} \quad (8)$$

A random number p is generated in the range $[0, 1]$. If $p > 0.18$, the next position of the fox is calculated using

Eq. Eq. 109, where the parameter , with c_2 ranging from 0.19 to 1 c_1 varies between 0 and 0.18. Conversely, if $p \leq 0.18$, the position is updated using

$$X_{(it+1)} = Dist_Fox_Prey_{it} \times Jump_{it} \times c_1 \quad (9)$$

$$X_{(it+1)} = Dist_Fox_Prey_{it} \times Jump_{it} \times c_2 \quad (10)$$

During the exploration phase, the position of the fox is updated using Eq. 11, where $MinT$ denotes the minimum time variable and a represents the search variable³⁸. Figure 6 shows the overall process of FOX³⁸.

$$X_{(it+1)} = BestX_{it} \times rand(1, dimension) \times MinT \times a \quad (11)$$

Honey badger algorithm

HBA was proposed by Hashim et al. in 2021 that mimics the intelligent foraging behavior of the honey badger⁴⁰. Honey badger uses two techniques to locate the prey: (i) Smelling and Digging, and (ii) Following honey-birds. In the first technique, the honey badger smells the prey and steps forward to estimate the location. When the position is identified, the honey badger digs the place. In the second technique, the honey badger follows the honey birds directly to the prey (beehive). The whole hunting process has been mathematically modeled in five steps. Step 01 represents the initialization phase of the hunting process, showing the i_{th} position of the honey badger using Eq. 12, where lb_{ui} , ub_i , and r_1 are lower bound, upper bound, and random number between 0 and 1, respectively⁴⁰.

$$x_i = lb_i + r_1 \times (ub_i - lb_i) \quad (12)$$

Step 02 is associated with the intensity of the smell. Position of prey is X_{prey} , and the distance of the prey in i_{th} iteration is $th\ di = X_{prey} - x_i$. The source of strength to the position of the prey is 13, where r_2 varies between 0 and 1, and $S_{40} = (x_i - x_{i+1})^2$, then the i intensity of the smell can be represented by Eq.

$$I_i = r_2 \times \frac{1}{4\pi d S_{2i}} \quad (13)$$

Step 03 controls the switching between the exploration and exploitation states using a density factor α , as defined by Eq. 14, with $C \geq 1$ (default 2)⁴⁰.

$$\alpha = C \times e^{t_{max} - t} \quad (14)$$

Here, t_{max} denotes the maximum number of iterations. In step 04, a flag, F , that controls the search direction and avoids a local minimum. Step 05 concerns the honey badger's new position. For the Smelling and Digging technique, the new position is denoted by x_{new} in Eq. 15⁴⁰ as shown in Fig. 7.

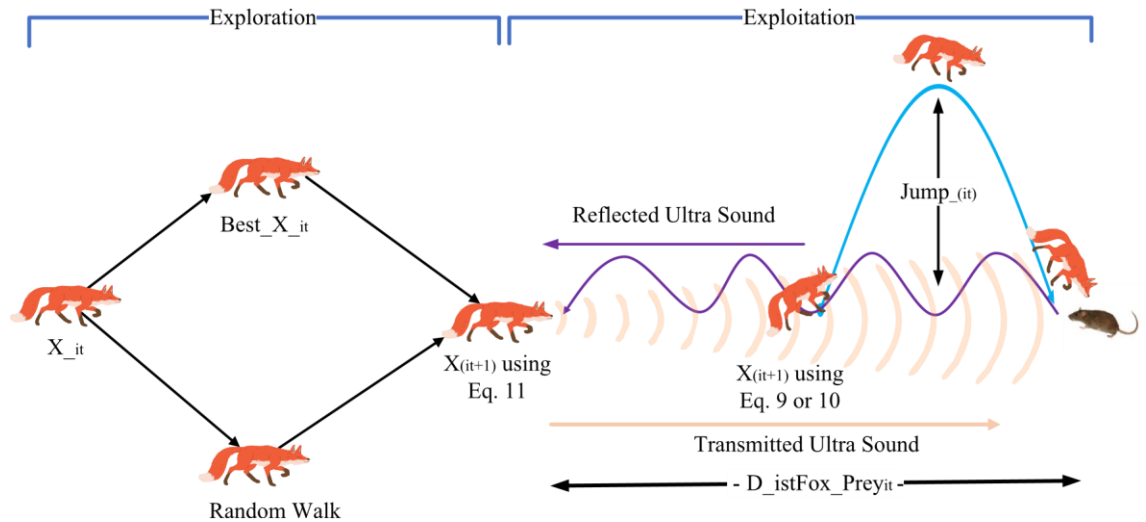


Fig. 6. Red Fox hunting behavior and mathematical model development³⁸.

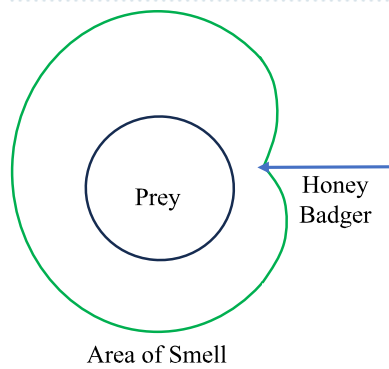


Fig. 7. Honey badger hunting behavior using smell⁴⁰.

$$X_{new} = x_{prey} + F \times \beta \times I \times x_{prey} + F \times r_3 \times \alpha \times d_i |\cos(2\pi r_4) \times [1 - \cos(2\pi r_5)]| \quad (15)$$

where random variables that range between 0 and 1. Following the honey-birds technique, the new position is updated $\beta \geq 1$ is the honey badger's capability to catch its prey, with a default value of 6. And r_3 , r_4 , and r_5 are using Eq. 16 where r_7 varies between 0 and 1⁴⁰.

$$X_{new} = x_{prey} + F \times r_7 \times \alpha \times d_i \quad (16)$$

Tuna swarm optimizer

TSO is proposed by Xie et al. in 2021 that follows the foraging pattern of Tuna Swarm³⁹. Tuna uses two approaches to get the prey: (i) Spiral Foraging, and (ii) Parabolic Foraging. The initialization of the population is mathematically shown in Eq. 17³⁹.

$$X_i^{int} = \text{rand} \cdot (\text{ub} - \text{lb}) + \text{lb}, \quad i = 1, 2, \dots, NP \quad (17)$$

Here, X_i^{int} is the starting position of the i^{th} individual. It's generated randomly between the lower bound (lb) and upper bound (ub) of the search space. The random value helps to spread out the initial solutions across the space.

$$X_{it} = \begin{cases} \alpha_1 X_{bestit} & ++ \beta \beta |XX_{bestit} - XX_{iit}| ++ \alpha_2 XX_{iit}, \quad i = 1 = \\ 2, 3, \dots, NP & (18) \end{cases} \quad +1 \alpha_1 X_{best}$$

In spiral foraging, when the prey is too close to hunt, the Eq. 18 updates the position of each individual, where X_t^i and X_{t+1}^i denote the i^{th} candidate at t and $t + 1$ iterations, respectively. X_{best}^t represents the optimal solution (food), and α_1 , and α_1 are the coefficients of weights. β is a cosine function.

But if the best candidate fails to hunt the prey, then the position of the individuals is updated based on a random coordinate by Eq. 19, where X_{rand}^t is a random reference³⁹.

$$X_{it+1}^{rand} = \begin{cases} \alpha_1 X_{best}^t + \alpha_1 X_{rand}^t \cos(\beta) & \text{if } rand \leq \frac{t}{t_{max}} \\ X_{it} & \text{otherwise} \end{cases} \quad (19)$$

Therefore, the Eq. 18 and 19 are selected based on the value of rand. If $rand \leq \frac{t}{t_{max}}$, the Eq. 18 will be selected, otherwise Eq. 19 as shown in 20, where t and t_{max} are the current and max iteration number

$$X_{it+1} = \begin{cases} \text{Eq. (18)} & \text{if } rand \leq \frac{t}{t_{max}} \\ \text{Eq. (19)} & \text{otherwise} \end{cases} \quad (20)$$

But for parabolic foraging, the individual is updated with Eq. 21, where p^{max} and TF is a random value between 1 and -1³⁹.

$$X_{it+1}^{parabolic} = \begin{cases} X_{best}^t + TF \cdot p^{max} \cdot (X_{best}^t - X_{it}^t) & \text{if } rand \leq 0.55 \\ X_{it}^t & \text{otherwise} \end{cases} \quad (21)$$

Framework development

$$X_{it+1}^{parabolic} = \begin{cases} X_{best}^t + TF \cdot p^{max} \cdot (X_{best}^t - X_{it}^t) & \text{if } rand \leq 0.55 \\ X_{it}^t & \text{otherwise} \end{cases} \quad (21)$$

The RF classifier has been integrated with every SBMHA to ensure hyper-tuning and feature extraction for each fold of the dataset. The proposed framework comprises five key stages: (1) Preparing the Custom Fold Dataset, (2) Initialization of Boundary Conditions, (3) Developing Optimization Problem (4) Application of

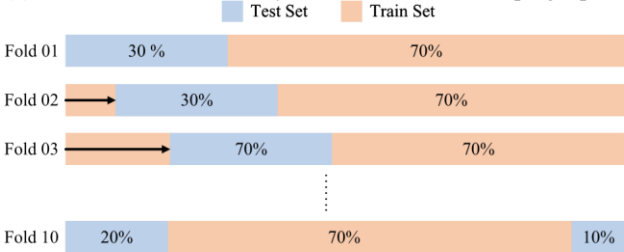


Fig. 8. Fold-wise train and test splitting.

Hyperparameter	Type	Range
Number of Estimators (n_estimators)	Integer	100 to 300
Maximum Depth (max_depth)	Integer	3 to 10
Minimum samples split (min_samples_split)	Integer	2 to 20
Minimum samples leaf (min_samples_leaf)	Integer	1 to 20
Feature Selection (features)	Binary Vector (F _{all} =16)	Binary (0 or 1)

Table 4. Boundary Conditions for the RF Hyperparameters.

Voting Method, and (5) Final Model Development, and its performance has been evaluated based on the mean CV results.

Preparing the custom fold dataset

For preparing the 10-fold CV dataset with a 70:30 ratio between train and test fold sets, the dataset has been shuffled and partitioned into 10 overlapping folds using a sliding-window strategy. For each fold i ($i = 1, 2, \dots, 10$), the test set has comprised a contiguous block of $N \times 30\%$ of the data, starting at position $(i - 1) \times 0.1N$, where N denotes the total dataset size. The remaining of the data (non-overlapping with the test block) has formed the training set. This process has iteratively shifted the test window forward by 10% for subsequent folds, creating a cyclic progression that has ensured comprehensive coverage of the dataset while preserving class distribution integrity as shown in Eq. 22, and 22 respectively for test and train fold sets. Figure 8 presents the train-test splitting of the shuffled dataset for the folds.

$$FTest_i = D_{\text{shuff}}[\lfloor 0.1N(i-1) \rfloor : \lfloor 0.1N(i-1) + 0.3N \rfloor \bmod N], \quad i = 1, 2, \dots, 10 \quad (22)$$

$$FTrain_i = D_{\text{shuff}} \setminus FTest_i \quad (23)$$

Initialization of boundary conditions

As this research has listed the RF classifier integrated with three different SBMHAs, a set of hyperparameters has been used: number of estimators, maximum depth, min sample split, and min sample leaf. This hybrid model (combination of Classifier and SBMHAs) finds the best value of the parameters and extracts the most appropriate features. Therefore, the boundary condition has been deployed for the hyperparameters, including a range of features as shown in Table 4.

Developing optimization problem

According to the Fig. 3, each train-fold set (Ftrain) has been optimized considering F-score as the cost function. During optimization, 100 epochs (Ep) and 50 populations (Pop) have been used for the SBMHA. Based on the best cost function value or the global best value over the iteration (G_{best}), the hyperparameters (M_{best}^i) and an optimized set of features (F_{best}^i) have been selected for each fold, where i is the fold number. Each M_{best}^i consists of the value of the number of estimators, maximum depth, min sample split, and min sample leaf, respectively represented by n_{best}^i , m_{best}^i , $split_{\text{best}}^i$, and $leaf_{\text{best}}^i$ for the i^{th} fold. The algorithm 1 has visualized the optimization process.

```

1: Start Process
2: (i) Initialization of the Problem with Boundary Conditions (B):
3:   Define boundaries for the RF hyperparameters:
4:     Number of Estimators ( $n$ ):  $100 \leq d \leq 300$ 
5:     Maximum Depth ( $m$ ):  $3 \leq d \leq 10$ 
6:     Minimum samples split ( $split$ ):  $2 \leq \eta \leq 20$ 
7:     Minimum samples leaf ( $leaf$ ):  $1 \leq t \leq 20$ 
8:     Feature Selection ( $V \in V_{\text{all}}$ ): Binary vector of length  $\leq V_{\text{all}}(16)$ 
9:   Initialize MHA (epoch  $Ep = 100$ , population size  $Pop = 50$ , seed)
10: (ii) Performing the RF Classification for  $i^{\text{th}}$  fold, where  $i = 1, 2, \dots, 10$ 
11:   for Current Epoch of  $i^{\text{th}}$  fold,  $t^i = 1$  to 100 do:
12:     M Features at  $t$ ,  $V_t^i = \{V_1, V_2, \dots, V_M\} \in V_{\text{all}}$ 
13:     Train RF model with the hyperparameters:  $M_t^i(n_t^i, m_t^i, split_t^i, leaf_t^i) \in B$ 
14:     Compute the cost function metric  $FS_t^i$  (F-score) using  $FTrain_i$ 
15:      $FS^i$  Stores all the F-scores,  $FS^i = [FS_1^i, FS_2^i, \dots, FS_{100}^i]$ 
16:      $V^i$  Stores all the features indices,  $V^i = [V_1^i, V_2^i, \dots, V_{100}^i]$ 
17:      $M^i$  Stores all the hyperparameters,  $M^i = [M_1^i, M_2^i, \dots, M_{100}^i]$ 
18: (iii) Tuned Parameters and Extracted Features:
19:   Decode all solutions, and get the best F-score,  $G_{\text{best}}^i$ , at the epoch  $t_{\text{best}}^i$ 
20:   If  $t^i = t_{\text{best}}^i$ :
21:     Extract  $M_{\text{best}}^i(n_{\text{best}}^i, m_{\text{best}}^i, split_{\text{best}}^i, leaf_{\text{best}}^i)$  and  $V_{\text{best}}^i$ 
22: End Process

```

Algorithm 1. Optimization Process using SBMHAs for each fold.

Application of voting method

After completing optimization for 10 folds, ten sets of hyperparameters and feature sets have been extracted. The final model has been developed using the majority voting (MV) technique, where the optimal values for the

number of estimators (n_f), maximum depth (m_f), minimum sample split ($split_f$), and minimum sample leaf ($leaf_f$) have been chosen based on their frequency of appearance across the folds as shown in Eq. 24, where $\theta_{best} \in \{n_{ibest}, m_{ibest}, split_{ibest}, leaf_{ibest}\}$. However, for selecting the final set of features (F_f), those 25. consistently appearing in the majority of the folds, at least six out of ten, have been chosen as described in Eq.

$$\theta_f(n_f, m_f, split_f, leaf_f) = MV(\theta_{best}^1, \theta_{best}^2, \dots, \theta_{best}^{10}) \quad (24)$$

$$\bigcup_{i=1}^{10} \left\{ V \in V_{best}^i \right\} \geq 6$$

$$V_f = V \in V_{best}^i \quad \text{and } i$$

$$=1 \quad (25)$$

Final model development

Each fold test set (F_{Test_i}) has been evaluated using the classifier model developed using the voted hyperparameters and features based on the MV. The final classifier model has the hyperparameters $\theta_f(n_f, m_f, split_f, leaf_f)$ and feature set V_f as shown in Eq. 26.

$$Final_Model = RF_{classifier}\{\theta_f, V_f\} \quad (26)$$

Performance analysis

The models have been evaluated using well-known metrics such as Accuracy (27), Precision (28), Recall (29), F-score (30), ROC curve, and AUC. True Positive (TP) refers to cases where the model correctly predicts the positive class for actual positive cases. True Negative (TN) denotes correct predictions of the negative class for actual negative cases. In contrast, a False Positive (FP) occurs when the model makes a wrong prediction of the positive class for an actual negative case; however, a False Negative (FN) represents the incorrect prediction of the negative class for an actual positive case.

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \quad (27)$$

$$Precision = \frac{TP}{TP + FP} \quad (28)$$

$$Recall = \frac{TP}{TP + FN} \quad (29)$$

$$F - score = \frac{2 \times Precision \times Recall}{Precision + Recall} \quad (30)$$

SHAP Analysis

Finally, SHapley Additive ExPlanations (SHAP), an explainable artificial intelligence, has been used to demonstrate the feature's importance and the model's predictive nature. The model interpretation technique of SHAP has been introduced by Lundberg et al.⁴¹. The main base of the SHAP is built using the Shapley values, introduced by Lloyd Shapley⁴². The feature significance is measured by a SHAP value of each feature calculated by Eq. 31.

$$\phi_i = \sum_{S \in \mathcal{F}_{-i}} \frac{|S|!(|F| - |S| - 1)!}{|F|!} [f_{S \cup \{i\}}(x_{S \cup \{i\}}) - f_S(x_S)] \quad (31)$$

Where subset ϕ_i is the SHAP value for feature s_i (s_i, F is the full feature set, $S \subseteq SF$ and $\setminus \{i\}$), $f_{f|S}$ denote the sizes of (x_S) is the model output for $f(x)$ according S and F , respectively. Individual prediction in SHAP is based on the cumulative SHAP value denoted by $v_i(x, v_i)$ is the output when feature i is added. |

S , and f

to Eq. 32, where $E[f(x)]$ is the base value, n is the number of features, as shown in Fig. 9. According to the SHAP explanation, if the value of $f(x)$ has been higher than 0.5, the model has predicted the current instance as Class A—specifically, Diabetic Positive in this research—otherwise, it has classified it as Class B.

$$f(x) = E[f(x)] + \sum_0^n \phi_i \quad (32)$$

Results

The Results section presents the findings of this research, structured into four main distinct parts for enhanced clarity and comprehension: (1) Outcomes of the Optimizer, (2) Classifier Performance, (3) SHAP-Based Interpretability, and (5) Discussion. In the “Outcomes of the Optimizer” section, we have discussed the convergence behavior of the optimization algorithms, the relationship between epochs and F-score, the selected hyperparameters, and the features that have been chosen through the optimization process. In the “Classifier Performance” section, mainly the prediction metrics and the misclassification entities have been discussed. In the section, SHAP-Based Interpretability, models’ nature, feature importance, and the relation between misclassification and the dataset have been upheld. Finally, in the section “Discussion”, we have discussed our findings and compared them with recent publications.

Outcomes of the optimizer

During the optimization session, the FOX has reached the highest cost function or the fitness (F-score = 1) while optimizing all the train-fold sets (Ftrain), except in fold no 9. Here, in fold 5, the highest F-score has been attained at epoch 7. The HBA Optimization has acquired the maximum cost function (F-score = 1) in four different folds: Fold 1 at Epoch 20, Fold 2 at Epoch 4, Fold 3 at Epoch 4, and Fold 4 at Epoch 6. However, other folds (5, 6, 7, 8, 9, and 10) have not attained the highest F-score during optimization. In contrast, the TSO has shown the best performance in optimization. TSO has got the F-score = 1, in all folds, even this time it has happened so fast that the highest fitness has been gained at epoch 3 in the 2nd and 9th fold. Not only that, but TSO has also shown a more sustainable convergence to the optimal solution. Figure 10 presents the cost function over epochs for the optimizer.

During optimization, each algorithm has optimized each fold separately and returned the best value of hyperparameters (n_{best}^i , m_{best}^i , $split_{best}^i$, and $leaf_{best}^i$) and optimal feature sets (F_{best}^i) as shown in Table 5. While HBA and TSO have exhibited a consistent pattern in hyperparameter selection, FOX has demonstrated greater variability across folds. Nevertheless, the application of majority voting (MV) has facilitated the selection of an optimal set of hyperparameters, $\theta(n_f, m_f, split_f, leaf_f)$, along with the corresponding feature

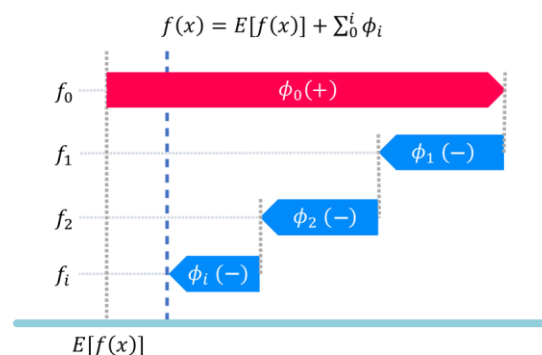


Fig. 9. SHAP value determination.⁴¹

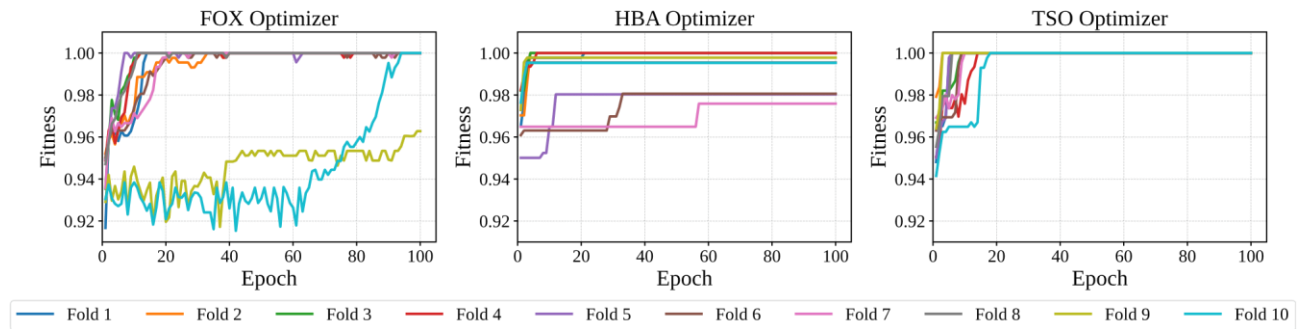


Fig. 10. Epoch vs. Cost function (Fitness).

Opti.	Fold (i)	<i>nibest</i>	<i>mibest</i>	<i>splitibest</i>	<i>leafbesti</i>	<i>Fbesti</i>
FOX	1	143	8	2	1	$V_0, V_1, V_3, V_4, V_5, V_7, V_8, V_{10}, V_{12}, V_{13}, V_{14}$
	2	150	8	2	1	$V_0, V_1, V_2, V_3, V_5, V_7, V_9, V_{11}, V_{12}, V_{13}, V_{14}$
	3	139	9	2	1	$V_0, V_1, V_2, V_3, V_4, V_5, V_7, V_8, V_{10}, V_{12}, V_{13}, V_{14}$
	4	139	10	2	1	$V_0, V_1, V_2, V_3, V_4, V_7, V_8, V_9, V_{10}, V_{13}, V_{14}$
	5	137	10	3	1	$V_0, V_1, V_2, V_3, V_4, V_8, V_9, V_{11}, V_{12}, V_{13}, V_{14}, V_{15}$
	6	143	10	5	1	$V_0, V_1, V_2, V_3, V_4, V_5, V_6, V_7, V_9, V_{10}, V_{11}, V_{13}, V_{14}$
	7	142	9	3	1	$V_0, V_1, V_2, V_3, V_4, V_5, V_7, V_8, V_9, V_{12}, V_{13}, V_{14}$
	8	136	9	2	1	$V_0, V_1, V_2, V_3, V_4, V_5, V_7, V_8, V_9, V_{10}, V_{11}, V_{13}$
	9	191	10	11	5	$V_0, V_1, V_2, V_3, V_6, V_7, V_8, V_9, V_{10}, V_{11}, V_{13}, V_{14}$
	10	193	8	2	1	$V_0, V_1, V_2, V_3, V_4, V_5, V_6, V_7, V_8, V_9, V_{11}, V_{13}, V_{14}$
HBA	1	100	10	2	1	$V_0, V_1, V_2, V_3, V_8, V_{10}, V_{11}, V_{12}$
	2	300	10	2	1	$V_0, V_1, V_2, V_3, V_8, V_{10}, V_{11}, V_{12}, V_{13}$
	3	300	10	2	1	$V_0, V_1, V_2, V_3, V_8, V_{10}, V_{11}, V_{12}, V_{13}$
	4	300	10	5	1	$V_0, V_1, V_2, V_3, V_8, V_{10}, V_{11}, V_{12}, V_{13}$
	5	300	10	2	1	$V_1, V_2, V_3, V_4, V_7, V_8, V_{11}, V_{14}$
	6	300	10	9	1	$V_1, V_2, V_3, V_7, V_8, V_{11}, V_{14}$
	7	300	10	2	1	$V_1, V_2, V_3, V_7, V_8, V_{11}, V_{14}$
	8	300	10	2	1	$V_0, V_3, V_4, V_6, V_8, V_{10}, V_{11}, V_{12}, V_{13}, V_{14}, V_{15}$
	9	300	10	2	1	$V_0, V_3, V_4, V_6, V_8, V_{10}, V_{11}, V_{12}, V_{13}, V_{14}, V_{15}$
	10	300	10	2	1	$V_0, V_3, V_4, V_6, V_8, V_{10}, V_{11}, V_{12}, V_{13}, V_{14}, V_{15}$
TSO	1	100	10	2	1	$V_0, V_1, V_2, V_3, V_4, V_5, V_7, V_8, V_9, V_{10}, V_{11}, V_{13}, V_{14}, V_{15}$
	2	300	10	2	1	$V_0, V_1, V_2, V_3, V_4, V_5, V_7, V_8, V_9, V_{10}, V_{11}, V_{13}, V_{14}, V_{15}$
	3	300	10	2	1	$V_0, V_1, V_2, V_3, V_4, V_5, V_7, V_8, V_9, V_{10}, V_{11}, V_{13}, V_{14}, V_{15}$
	4	300	10	2	1	$V_0, V_1, V_2, V_3, V_4, V_5, V_6, V_7, V_8, V_{10}, V_{11}, V_{12}, V_{13}, V_{14}$
	5	100	10	3	1	$V_0, V_1, V_2, V_3, V_4, V_5, V_7, V_8, V_9, V_{10}, V_{11}, V_{13}, V_{14}, V_{15}$
	6	300	10	2	1	$V_0, V_1, V_2, V_3, V_4, V_5, V_7, V_8, V_9, V_{10}, V_{11}, V_{13}, V_{14}, V_{15}$
	7	100	10	2	1	$V_0, V_1, V_2, V_3, V_4, V_5, V_8, V_9, V_{11}, V_{13}, V_{14}, V_{15}$
	8	100	10	2	1	$V_0, V_1, V_3, V_4, V_5, V_7, V_8, V_9, V_{10}, V_{13}, V_{14}, V_{15}$
	9	100	10	2	1	$V_0, V_1, V_2, V_3, V_4, V_5, V_7, V_8, V_9, V_{10}, V_{11}, V_{13}, V_{14}, V_{15}$
	10	300	10	2	1	$V_0, V_1, V_2, V_3, V_7, V_9, V_{10}, V_{11}, V_{12}, V_{13}, V_{14}, V_{15}$

Table 5. Fold-wise optimized hyperparameters and selected features.

subset, F_f . The finalized values have been presented in Table 6. In the FOX optimizer, the finalized 13 features are $F_{FOX} = \{V_0, V_1, V_2, V_3, V_4, V_5, V_7, V_8, V_9, V_{10}, V_{11}, V_{13}, V_{14}\}$.

However, the HBA has returned the fewest number of features over the folds, as well as after voting, it has been finalized with only 10 features (other features (V_4, V_5, V_6, V_7, V_9 , and V_{F15HBA}). TSO has only trashed $\{V_0, V_1, V_2, V_3, V_8, V_6$ and $10, V_{11,12}, V$ and made a set of 14 features. $12, V_{13}, V_{14}\}$ by excluding 6

Optimizer	n_f	m_f	$split_f$	$leaf_f$	F_f
FOX	143	10	2	1	$F_{FOX} = \{V_0, V_1, V_2, V_3, V_4, V_5, V_7, V_8, V_9, V_{10}, V_{11}, V_{13}, V_{14}\}$
HBA	300	10	2	1	$F_{HBA} = \{V_0, V_1, V_2, V_3, V_8, V_{10}, V_{11}, V_{12}, V_{13}, V_{14}\}$
TSO	100	10	2	1	$F_{TSO} = \{V_0, V_1, V_2, V_3, V_4, V_5, V_7, V_8, V_9, V_{10}, V_{11}, V_{13}, V_{14}, V_{15}\}$

Table 6. Finalized hyperparameters and features using MV.

Models	Feat.	Train				Test			
		Accu.	F-sc.	Prec.	Reca.	Accu.	F-sc.	Prec.	Reca.
FOX_RF	13	100%	100%	100%	100%	99.36%	99.51%	100%	99.02%
HBA_RF	10	100%	100%	100%	100%	99.36%	99.51%	100%	99.02%
TSO_RF	14	100%	100%	100%	100%	100%	100%	100%	100%

Table 7. Model's result without Fold Validation.

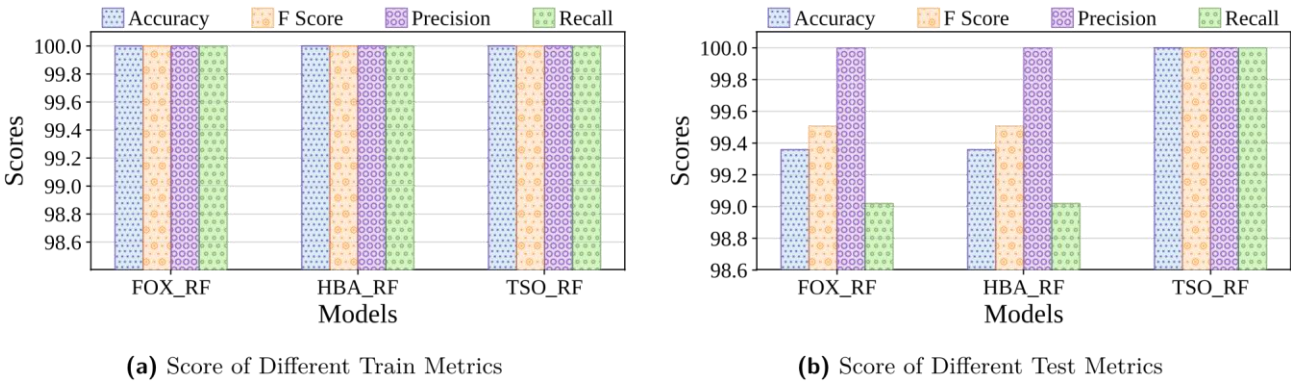


Fig. 11. Result of different metrics without Fold Validation.

FOX, HBA, and TSO have shown the same selections for $m_f=10$, $split_f=2$, and $leaf_f=1$, whereas n_f has differed across them—143, 300, and 100 for FOX, HBA, and TSO, respectively.

Classifier performance

In this section, the final RF classifier has been developed using the finalized parameters and trained and tested on the finalized feature sets. The classifier performance has been discussed in two aspects: Performance without Fold Validation and Performance using Fold Validation.

Performance without fold validation

Table 7 visualizes the outcomes for the first scenario, where the RF classifier optimized with TSO (TSO_RF) has shown superior performance, having 100% of all evaluation categories in both training and testing periods, surpassing the other two algorithms. However, the RF classifier optimized with FOX (FOX_RF) and HBA (HBA_RF) has shown a similar result with 99.36%, 99.51%, 99.02%, and 100%, respectively, in Accuracy, F-score, Precision, and Recall during testing as shown in Fig. 11. TSO_RF with no fold validation has no misclassification, but FOX_RF and HBA_RF have a wrong prediction (Index 68) as FN presented in Fig. 12 via confusion matrix.

Performance using fold validation

The finalized model has also been evaluated for each fold. Each model (FOX_RF, HBA_RF, and TSO_RF) has shown compatibility over the fold sets. Table 8 presents the fold-wise performance of the model. Figure 13 highlights the distribution of scores across the folds for both training and testing. All the models have gained the maximum value (100%) in all evaluation criteria during training. However, the test performances have been found to vary slightly across the folds, but the deviation has not been significantly high, as described in the Table 9.

The mean value of all evaluation metrics has been evaluated and listed in Table 10. The mean accuracies for FOX_RF, HBA_RF, and TSO_RF have been measured as 98.01%, 97.24%, and 98.14%, respectively, indicating that TSO_RF has achieved the best performance among the models. Even, TSO_RF has gained the highest F-score (98.47%), and Recall (98.54%). However, FOX_RF has surprisingly attained the highest Precision of

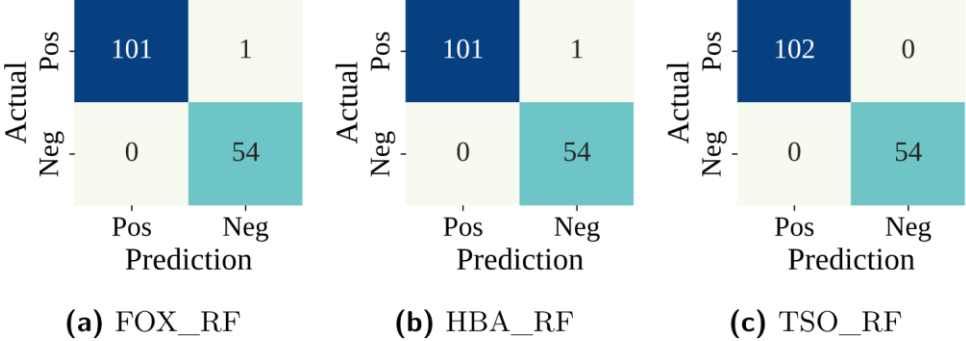


Fig. 12. Confusion matrix on the test dataset without Fold Validation.

Models	Fold	Feat.	Train				Test			
			Accuracy	F-score	Precision	Recall	Accuracy	F-score	Precision	Recall
FOX_RF	1	13	100%	100%	100%	100%	99.36%	99.51%	100%	99.02%
	2		100%	100%	100%	100%	98.08%	98.49%	100%	97.03%
	3		100%	100%	100%	100%	96.79%	97.30%	100%	94.74%
	4		100%	100%	100%	100%	98.08%	98.32%	98.88%	97.78%
	5		100%	100%	100%	100%	98.08%	98.36%	97.83%	98.90%
	6		100%	100%	100%	100%	97.44%	97.80%	96.74%	98.89%
	7		100%	100%	100%	100%	96.15%	96.84%	94.85%	98.92%
	8		100%	100%	100%	100%	96.79%	97.46%	96.00%	98.97%
	9		100%	100%	100%	100%	99.36%	99.47%	100%	98.95%
	10		100%	100%	100%	100%	100%	100%	100%	100%
HBA_RF	1	10	100%	100%	100%	100%	99.36%	99.51%	100%	99.02%
	2		100%	100%	100%	100%	97.44%	97.98%	100%	96.04%
	3		100%	100%	100%	100%	96.79%	97.30%	100%	94.74%
	4		100%	100%	100%	100%	96.79%	97.21%	97.75%	96.67%
	5		100%	100%	100%	100%	96.79%	97.27%	96.74%	97.80%
	6		100%	100%	100%	100%	96.15%	96.70%	95.65%	97.78%
	7		100%	100%	100%	100%	96.15%	96.84%	94.85%	98.92%
	8		100%	100%	100%	100%	95.51%	96.41%	95.92%	96.91%
	9		100%	100%	100%	100%	98.08%	98.40%	100%	96.84%
	10		100%	100%	100%	100%	99.36%	99.53%	100%	99.06%
TSO_RF	1	14	100%	100%	100%	100%	100%	100%	100%	100%
	2		100%	100%	100%	100%	98.08%	98.49%	100%	97.03%
	3		100%	100%	100%	100%	98.08%	98.40%	100%	96.84%
	4		100%	100%	100%	100%	98.08%	98.32%	98.88%	97.78%
	5		100%	100%	100%	100%	98.72%	98.91%	97.85%	100%

6	100%	100%	100%	100%	97.44%	97.80%	96.74%	98.89%
7	100%	100%	100%	100%	96.15%	96.84%	94.85%	98.92%
8	100%	100%	100%	100%	98.72%	98.97%	98.97%	98.97%
9	100%	100%	100%	100%	97.44%	97.89%	97.89%	97.89%
10	100%	100%	100%	100%	98.72%	99.06%	99.06%	99.06%

Table 8. Model’s result using Fold Validation.

98.43%. But, HBA_RF has shown the highest feature reduction while maintaining a moderate performance. Therefore, HBA_RF has the highest number of wrong predictions, while TSO_RF has shown minimal wrong predictions.

Figure 14 shows the 10-fold confusion matrices of FOX_RF, where at 10th fold the model has made no wrong predictions, but at the 7th fold the model has made the maximum number of misclassifications (FP: 5, FN: 1) among the 10 folds. By aggregating predictions across all folds, FOX_RF has produced a total of 31 misclassifications out of 1560 total test predictions (Size of each test set: 156 × number of folds: 10).

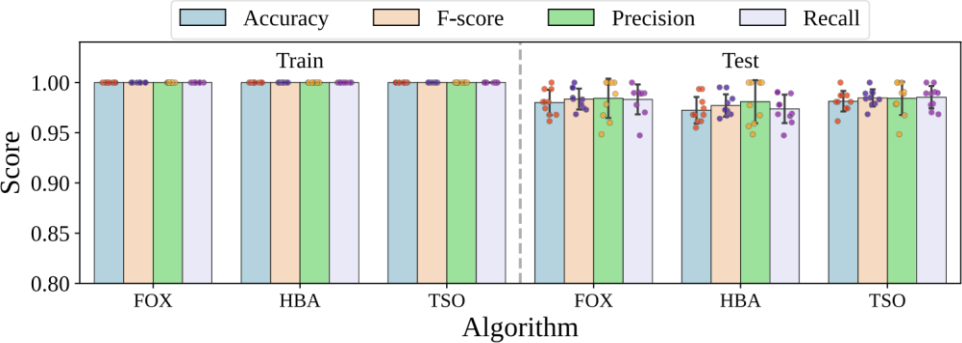


Fig. 13. Distribution of Different Performance Metrics of Models.

Models	Accuracy	F-score	Precision	Recall
FOX_RF	1.26	1.04	1.96	1.49
HBA_RF	1.32	1.11	2.14	1.42
TSO_RF	1.02	0.86	1.65	1.11

Table 9. Standard Deviation of model’s test performance.

Models	Features	Train				Test			
		Accuracy	F-score	Precision	Recall	Accuracy	F-score	Precision	Recall
FOX_RF	13	100%	100%	100%	100%	98.01%	98.36%	98.43%	98.32%
HBA_RF	10	100%	100%	100%	100%	97.24%	97.71%	98.09%	97.38%
TSO_RF	14	100%	100%	100%	100%	98.14%	98.47%	98.42%	98.54%

Table 10. Model’s mean result using Fold Validation.

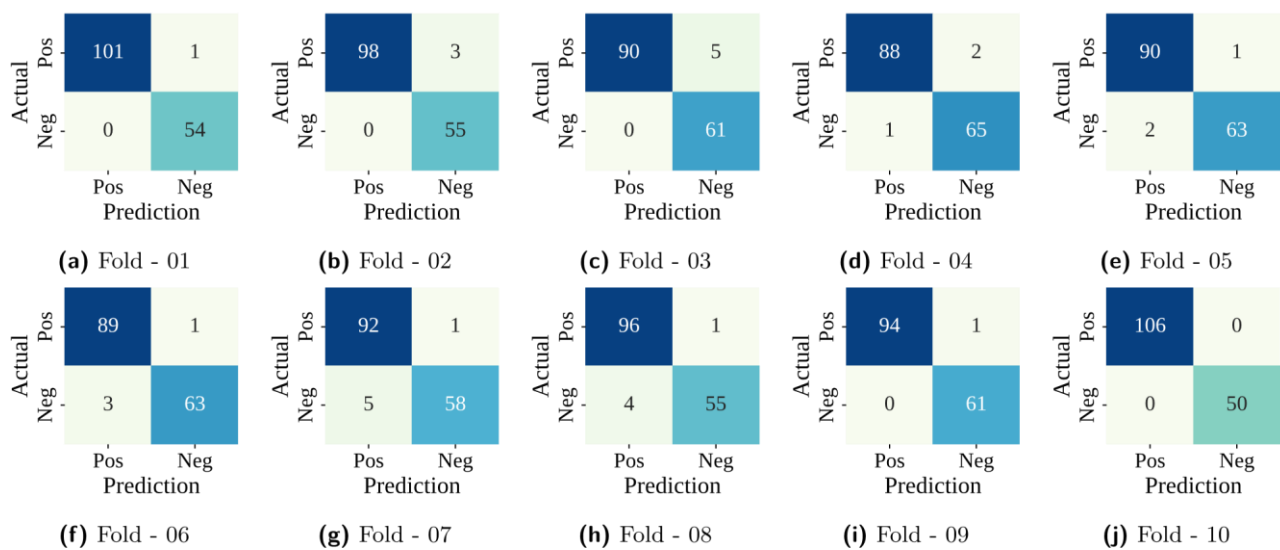


Fig. 14. FOX_RF confusion matrices for test fold sets.

Figure 15 displays all the confusion matrices for HBA_RF. The 8th fold has encountered the highest number of incorrect predictions, whereas the 1st and 10th folds have shown the lowest number of misclassifications during the evaluation of their respective test sets. Therefore, combining all fold predictions, the total wrong prediction numbers are 43 for HBA_RF out of 1560.

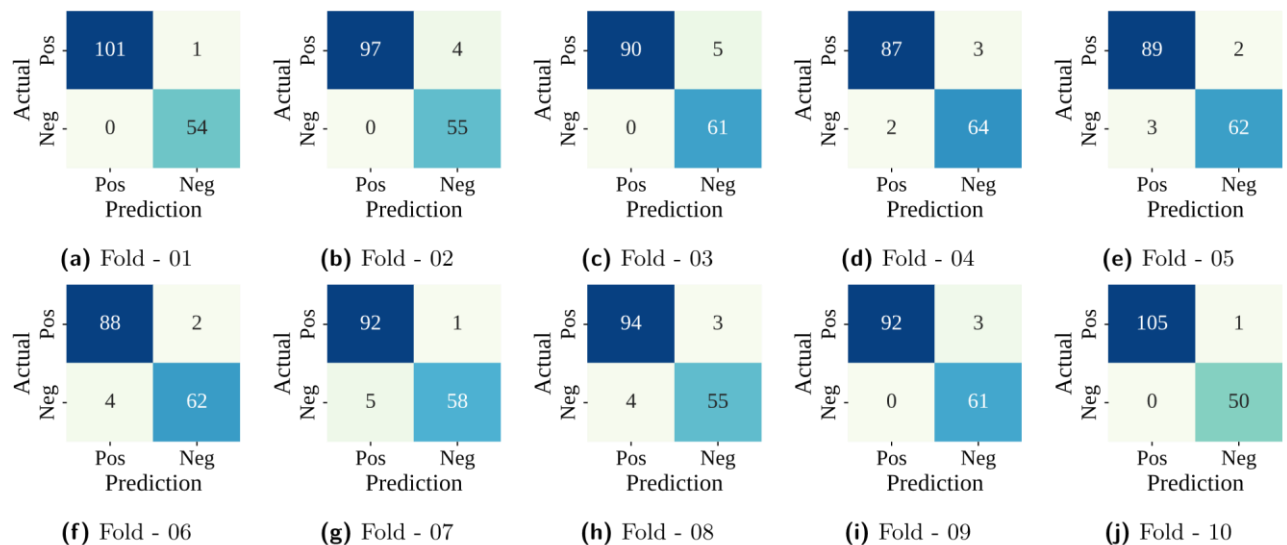


Fig. 15. HBA_RF confusion matrices for test fold sets.

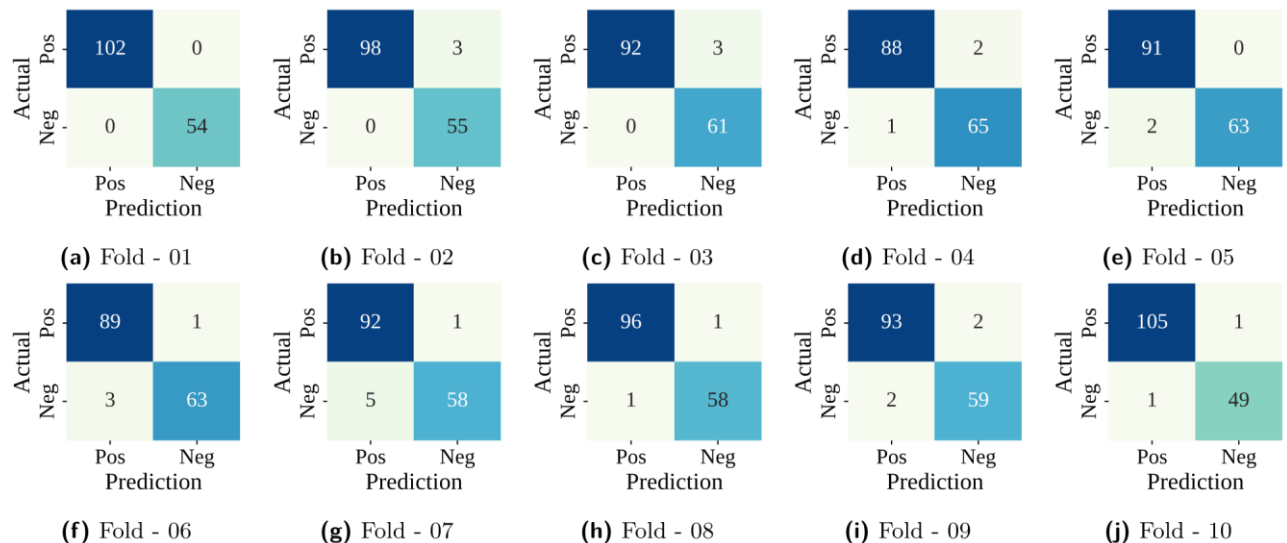
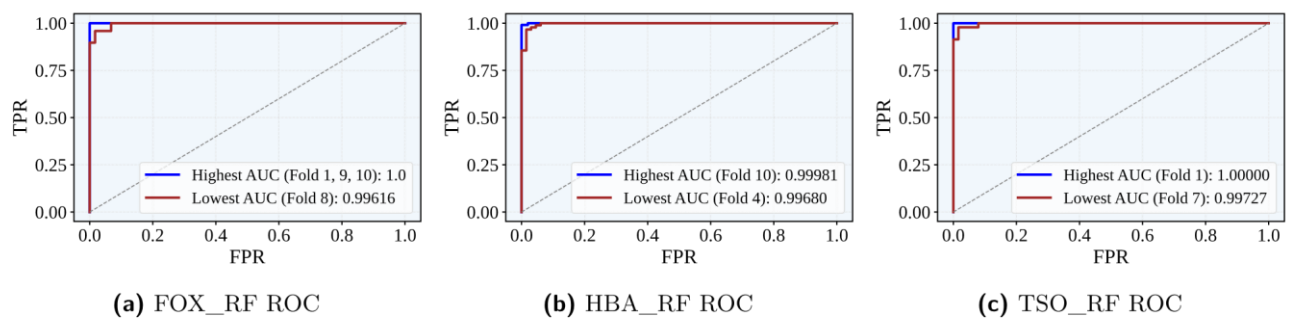


Fig. 16. TSO_RF confusion matrices for test fold sets.

Figure 16 presents the confusion matrices for TSO_RF, where the model has made no misclassifications in test fold set 01. However, test fold set 07 has exhibited the highest number of incorrect predictions, which has been consistent with the pattern observed in the other two models. Finally, TSO_RF has made a total of wrong predictions of only 29 out of 1560 across all folds, which is the lowest among the models.

To further analyze misclassification, the indices of misclassified entities over the folds of each model have been gathered in one place. Table 11 presents the eleven common indices that have been misclassified across the three proposed models, considering the total misclassification across all the folds. Eleven indices are misclassified (5 FN and 6 FP), common in all models. The best and worst true positive rate (TPR) against the false positive rate (FPR), Receiver Operating Characteristic (ROC) among the folds have been reported in Fig. 17 for each model. For FOX_RF, the best Area Under the Curve (AUC) has been reported in multiple folds (1, 9, and 10), and the lowest AUC has been seen in fold 8, during evaluation. But, the highest and lowest AUC for HBA_RF have been seen in fold 10 and 4, respectively. However, TSO_RF has gained the highest and lowest AUC in folds 1 and 2, respectively. Table 12 shows all AUC scores over the folds of all models, including the mean AUC. The highest and lowest AUC in FOX_RF are 1 and 0.9962, respectively, and the mean AUC is reported as 0.9987. In HBA_RF, the model has failed to attain the AUC of 1 in any fold while testing. The highest and lowest AUC values are 0.9998 and 0.9968, respectively, with a mean AUC of 0.9983. However, the TSO_RF has gained the highest mean AUC of 0.9988, including the highest and lowest AUC of 1 and 0.9973.

Index	V ₀	V ₁	V ₂	V ₃	V ₄	V ₅	V ₆	V ₇	V ₈	V ₉	V ₁₀	V ₁₁	V ₁₂	V ₁₃	V ₁₄	V ₁₅	Class	Pred
3	45	M	X	X	✓	✓	✓	✓	X	✓	X	✓	X	X	X	X	Pos	Neg
26	42	M	X	X	X	✓	✓	X	X	X	✓	X	X	✓	X	X	Pos	Neg
68	25	F	X	X	X	✓	✓	X	✓	X	X	X	X	X	✓	X	Pos	Neg
136	54	M	X	X	X	X	X	✓	X	✓	✓	X	X	✓	✓	✓	Pos	Neg
213	49	M	X	✓	X	X	X	X	X	✓	X	X	✓	X	X	✓	Neg	Pos
232	56	M	X	✓	X	✓	✓	X	✓	✓	X	✓	X	✓	✓	X	Neg	Pos
317	36	F	X	X	X	✓	X	X	X	X	✓	X	X	X	X	X	Neg	Pos
328	58	M	X	✓	X	X	X	X	✓	✓	X	X	X	✓	X	X	Neg	Pos
391	58	M	X	✓	X	X	X	X	✓	✓	X	X	X	✓	X	X	Neg	Pos
405	58	M	X	✓	X	X	X	X	✓	✓	X	X	X	✓	X	X	Neg	Pos
431	42	M	X	X	X	✓	✓	X	X	X	✓	X	X	✓	X	X	Pos	Neg

Table 11. Indices of misclassified samples across different folds in all models.**Fig. 17.** ROC Curves for all Models.

Model	Fold 1	Fold 2	Fold 3	Fold 4	Fold 5	Fold 6	Fold 7	Fold 8	Fold 9	Fold 10	Mean
FOX_RF	1.0000	0.9988	0.9995	0.9982	0.9986	0.9986	0.9974	0.9962	1.0000	1.0000	0.9987
HBA_RF	0.9993	0.9995	0.9995	0.9968	0.9981	0.9972	0.9973	0.9969	0.9990	0.9998	0.9983
TSO_RF	1.0000	0.9984	0.9998	0.9975	0.9989	0.9983	0.9973	0.9990	0.9993	0.9998	0.9988

Table 12. Fold-wise AUC scores and Mean AUC for FOX, HBA, and TSO models.

Fold	FOX_RF			HBA_RF			TSO_RF		
	TrT (ms)	TsT (ms)	OpT (s)	TrT (ms)	TsT (ms)	OpT (s)	TrT (ms)	TsT (ms)	OpT (s)
1	630.71	33.27	664	860.47	8.26	1677	390.48	12.73	903
2	380.36	9.60	659	487.23	7.99	2224	432.05	12.09	1082
3	413.16	13.04	654	493.07	8.07	2196	366.41	20.63	1538
4	477.55	11.94	638	510.16	8.46	1990	180.65	7.25	1527
5	516.90	18.18	642	510.74	7.75	2048	177.51	10.34	866
6	633.38	10.43	663	514.44	7.99	1919	262.53	11.26	1587
7	392.05	12.86	661	837.27	14.45	2047	284.65	7.75	879
8	249.93	8.15	638	525.29	7.89	2253	163.13	9.29	1090
9	247.00	8.99	859	516.71	11.11	2313	163.13	7.74	862
10	247.92	8.00	866	508.27	8.38	2264	166.41	8.33	1535
Mean	418.90	13.45	694.40	576.36	9.04	2093.10	258.67	10.74	1186.90

Table 13. Training, Testing, and Optimization Times for Each Fold.

In this research, the complexity has been evaluated based on the execution time, which includes both the training and testing phases of the models. Each model requires one-time optimization to generate the optimal set of features and the hyperparameters. As soon as they are found, the model can be used anytime, only via training and testing. The optimization is quite high as shown in the Table 13. The FOX_RF has taken almost 694.40 seconds on average, the lowest mean time among the models. However, HBA_RF and TSO_RF have taken the mean optimization time, respectively, 2093.10 and 1186.90 seconds. However, TSO_RF has taken the lowest mean training time of 258.67 milliseconds. In contrast, FOX_RF and HBA_RF have required mean training times of 418.90 milliseconds and 576.36 milliseconds, respectively. HBA_RF has outperformed the other models during the testing session, having a mean testing time of 9.04 milliseconds, whereas FOX_RF and TSO_RF have taken a mean testing time of 13.45 and 10.74 milliseconds, respectively. Although FOX_RF has demonstrated lower execution time complexity during optimization, TSO_RF has outperformed in testing execution time, while HBA_RF has excelled during training.

SHAP analysis

SHAP analysis has been designed in three sub-sections: SHAP evaluation without optimization, Optimized Model without Fold Validation, and Optimized Model using Fold Validation. The Section, Optimized Model using Fold Validation, broadly explains the misclassification and individual prediction of the models.

SHAP evaluation without optimization

We have initially determined the feature importance without applying optimization to understand the dataset and relate it to the optimized models. Figure 18 shows the top seven features, showing their mean SHAP value and overall summary plot. Polyuria, Polydipsia, Gender, Sudden weight loss, and partial paresis are the top five features while performing the classification task without optimization. Polyuria has attained a mean SHAP value of +0.15 and conquered the top position. As mentioned in Table 2, Polyuria has been encoded as Yes, $\sqrt{}$:1 and No:0. The higher value of Polyuria (1) has influenced the model with a positive SHAP value, tending the decision towards the positive class of diabetes. Similarly, lower Polyuria has a negative SHAP value. And this similar trend has been observed for Polydipsia, Sudden weight loss, partial paresis, and Irritability. However, the lower value of Gender (Female: 0) and Alopecia (No: 0) has a higher SHAP value, leading the model predictions toward the diabetes positive class.

SHAP evaluation of optimized model without fold validation

Each optimized model has a distinct set of hyperparameters and feature set, previously mentioned in Table 7. However, FOX_RF, HBA_RF, and TSO_RF models have identified Polyuria, Polydipsia, and Gender as the top three features separately, which nature has also been noticed while performing classification without optimization. However, the 4th position has been shared by Sudden weight loss and Irritability. In FOX_RF and TSO_RF, Sudden weight loss has secured the 4th position, while in HBA_RF, Irritability has held the same rank.

Figure 19 shows the top seven mean SHAP values and summary plots for each optimized model without using fold validation. The mean SHAP value of Polyuria has been calculated as +0.16, +0.18, and +0.15, respectively, in FOX_RF, HBA_RF, and TSO_RF. Polydipsia has been assigned a mean SHAP value of +0.12, +0.13, and +0.11, respectively, in FOX_RF, HBA_RF, and TSO_RF. The summary plot of each model has confirmed that higher values (Yes, $\sqrt{}$: 1) for Polyuria, Polydipsia, and Irritability positively influence the SHAP value, thereby increasing the value of $f(x)$ for individual predictions. However, Age has played a mixed role, exhibiting a wide range of SHAP value distribution across the model. However, lower values of Gender and Alopecia have negatively impacted the models.

SHAP evaluation of optimized model using fold validation

Our Models have been evaluated using 10-fold validations. Among the folds, FOX_RF has gained the highest 100% of accuracy at fold 10 and the lowest accuracy of 96.15% at fold 7. Therefore, in SHAP analysis, only the Fold 10 and 7 have been presented as mean SHAP value analysis and summary plot.

Figure 20 presents the mean SHAP value of the top seven features while testing using the feature set, F_{FOX} . In folds 10 and 7, the highest-ranked three features are Polyuria, Polydipsia, and Gender. However, the mean

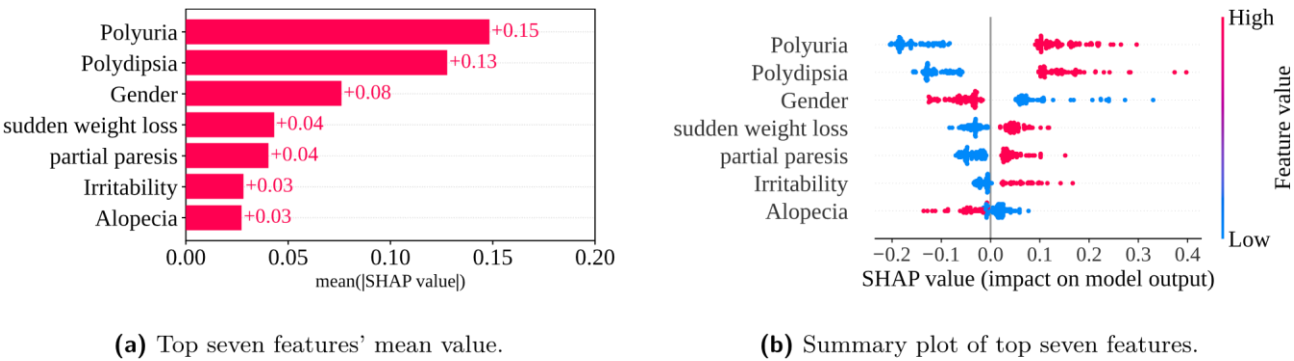


Fig. 18. SHAP evaluation without optimization.

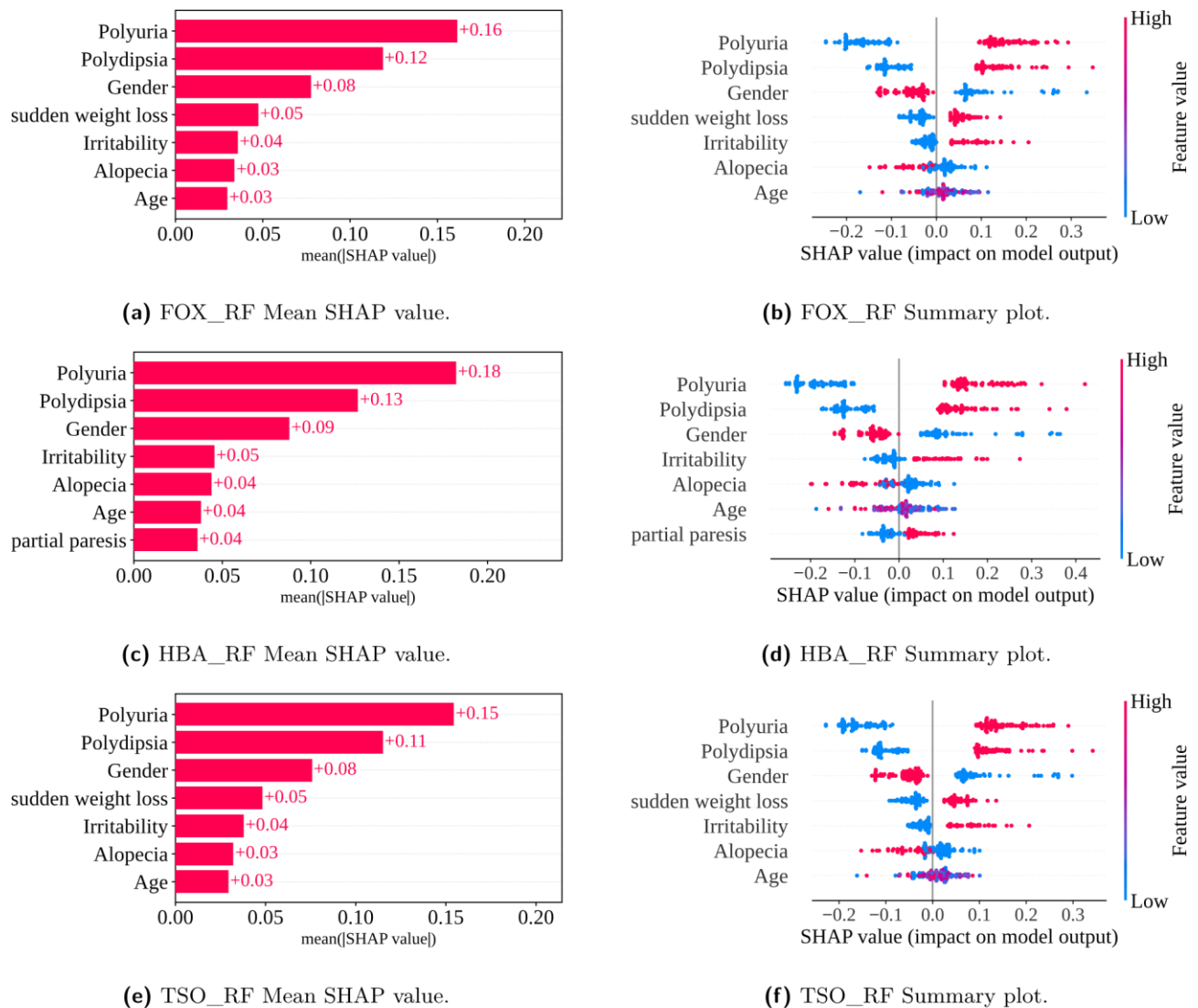


Fig. 19. Mean SHAP value and Summary Plot of optimized models without Fold Validation.

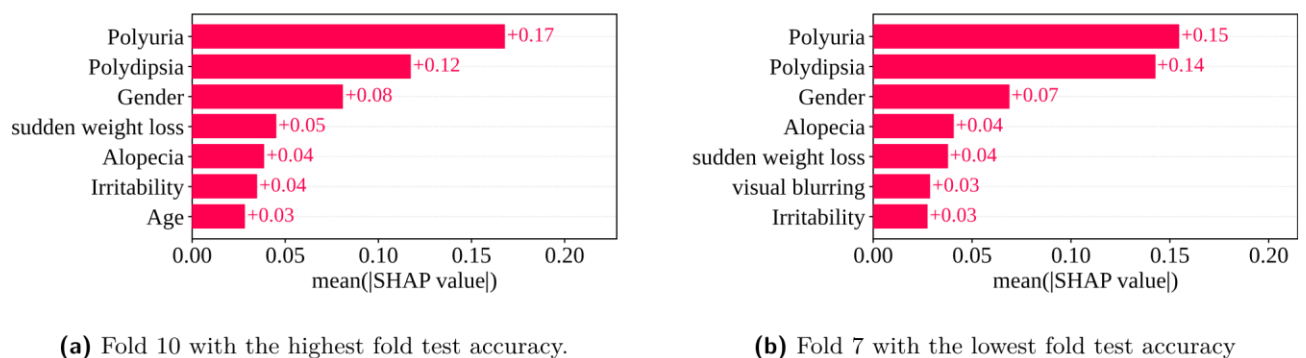


Fig. 20. Mean SHAP value for FOX_RF using Fold Validation.

SHAP value of Polyuria has been calculated as +0.17 and +0.15, at fold 10 and 7, respectively. The mean SHAP value of Polydipsia at fold 10 is +0.12, which is 0.02 units lower than fold 7. Though Gender has a slightly higher mean SHAP value at fold 10 compared to fold 7. Figure 21 summarizes the overall model predictions of folds 10 and 7 for FOX_RF using a summary plot. Both the folds have maintained rigidity in the first three

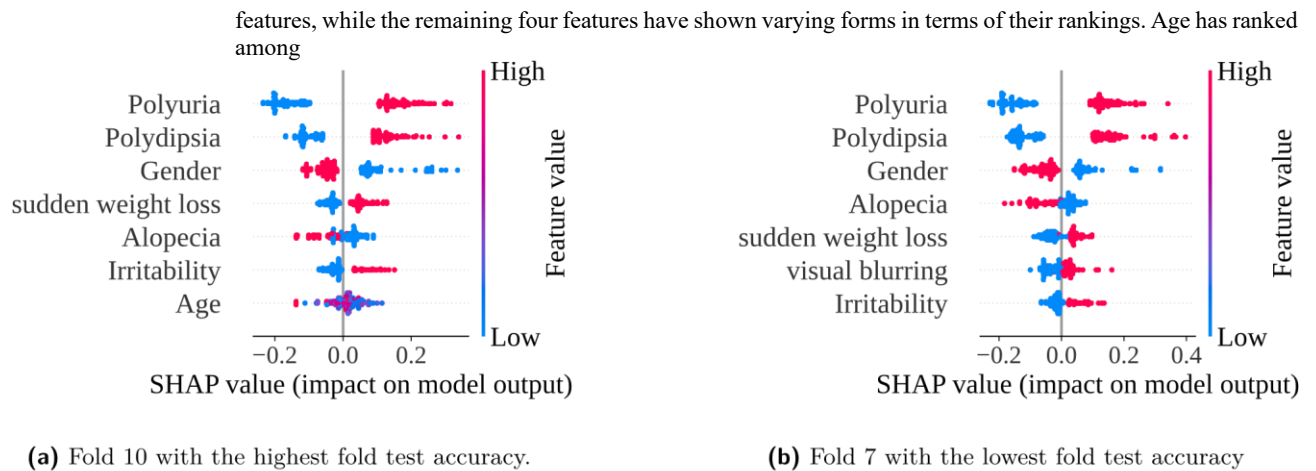


Fig. 21. Summary for FOX_RF using Fold Validation.

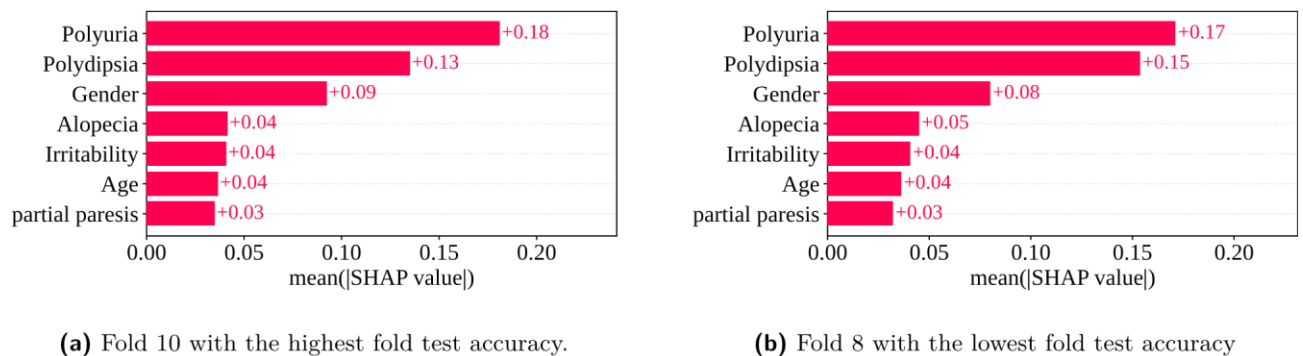


Fig. 22. Mean SHAP value for HBA_RF using Fold Validation.

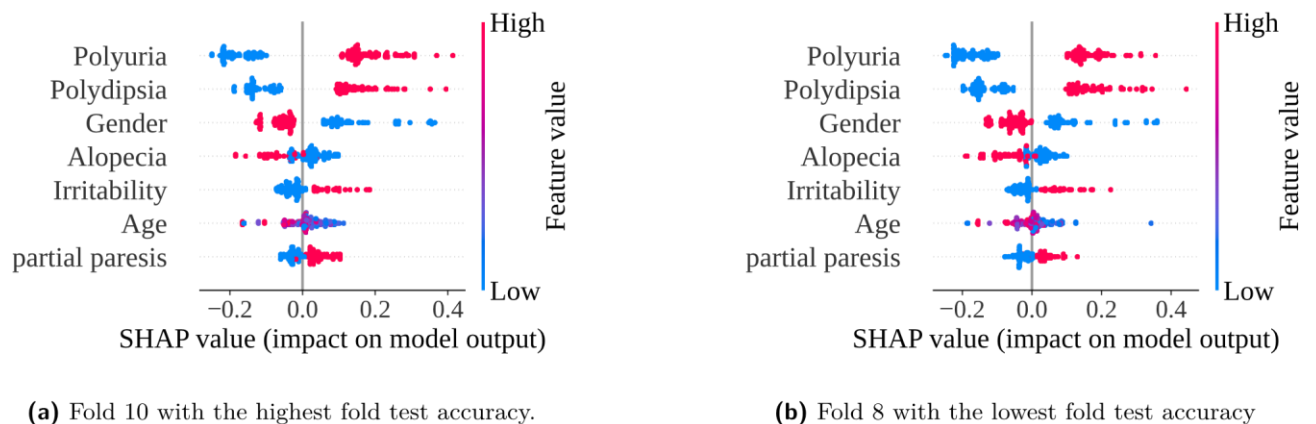


Fig. 23. Summary for HBA_RF using Fold Validation.

the top seven features in fold 10, but has not appeared in the top seven of fold 7. In contrast, Visual Blurring has been included in the top seven features of fold 7 but is absent from fold 10.

In HBA_RF, the highest accuracy of 99.36% has been achieved at folds 1 and 10, while the lowest accuracy of 95.51% has been recorded at fold 8. Figure 22 shows the comparison of the mean SHAP values of the top seven features between fold 10 and 8, which have been identified using the feature set F_{HBA} during testing. The seven highly impacted features are the same in both folds, including their impact ranking. The SHAP values of the top three features are +0.18, +0.13, and +0.09 at fold 10, whereas +0.17, +0.15, and +0.08 at fold 8. Figure

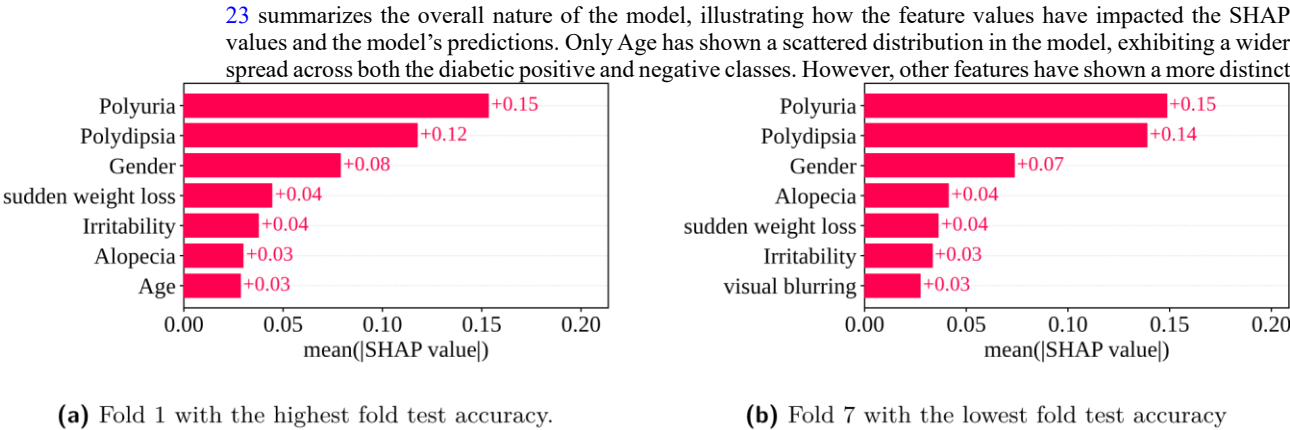


Fig. 24. Mean SHAP value for TSO_RF using Fold Validation.

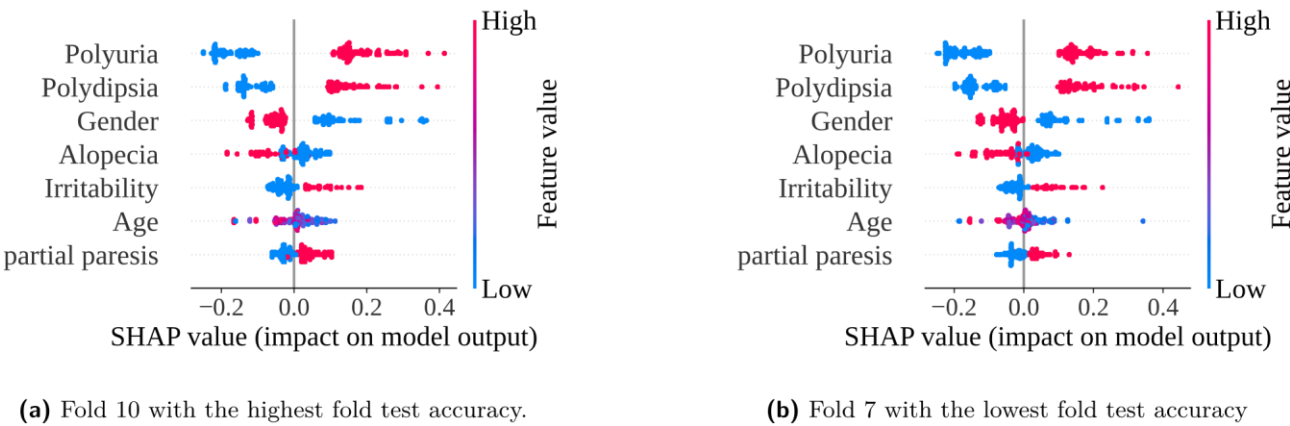


Fig. 25. Summary for TSO_RF using Fold Validation.

Index	V_0	V_1	V_2	V_3	V_4	V_5	V_6	V_7	V_8	V_9	V_{10}	V_{11}	V_{12}	V_{13}	V_{14}	V_{15}	Class
0	40	Male	No	Yes	No	Yes	No	No	No	Yes	No	Yes	No	Yes	Yes	Yes	Positive
332	50	Female	No	No	No	Yes	No	No	Yes	Yes	No	Yes	No	No	Yes	No	Negative
317	36	Female	No	No	No	Yes	No	No	No	No	Yes	No	No	No	No	No	Negative
431	42	Male	No	No	No	Yes	Yes	No	No	No	Yes	No	No	Yes	No	No	Positive
Encoded Value																	
0	40	1	0	1	0	1	0	0	0	1	0	1	0	1	1	1	1
332	50	0	0	0	0	1	0	0	1	1	0	1	0	0	1	0	0
317	36	0	0	0	0	1	0	0	0	0	1	0	0	0	0	0	0
431	42	1	0	0	0	1	1	0	0	0	1	0	0	1	0	0	1

Table 14. Data description of index 0, 332, 317, and 431.

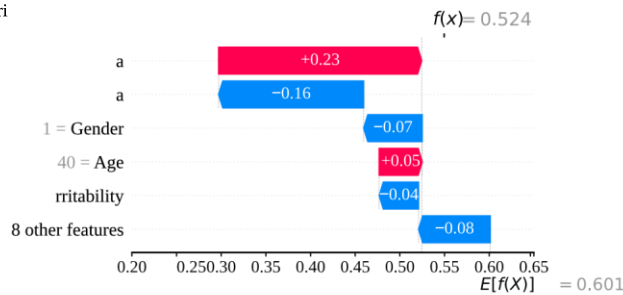
behavior in the model. For instance, a higher value of Gender (Male: 1) has negatively impacted the model in both folds. In contrast, higher values of the top two features have positively influenced the SHAP values.

In TSO_RF, the highest accuracy has been found at fold 1 (100%), while the lowest accuracy has been measured at fold 7 (96.15%). Therefore, presenting the mean SHAP value and summary plot for only these two folds has been more convenient. Figure 24 visualizes the mean SHAP value graph for the highest 7 features that have ranked among the feature set F_{TSO} . In both 1st and 7th fold, Polyuria has been elected as the most impacted feature, with the mean SHAP value of +0.15. And Fig. 25 highlights the feature values and their trend toward the SHAP values.

For further deep analysis, individual predictions have been analyzed using SHAP. This analysis has included both correct and wrong classifications. A TP (Index 0) and A TN (Index 332) have been described from the pull of right predictions. In Table 11, we have reported 11 mutual indices that have been misclassified by all models across multiple folds (10-fold validation). Therefore, among eleven, individual misclassified predictions have been analyzed for an FP (Index 317) and an FN (Index 431) prediction. The indices (TP: 0, TN: 332, FP: 317,

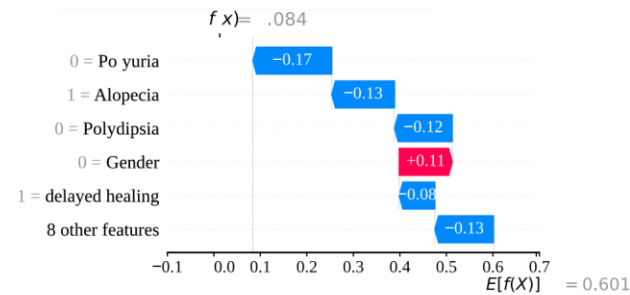
1 = Polydipsia
0 = Polyuria

0 = I

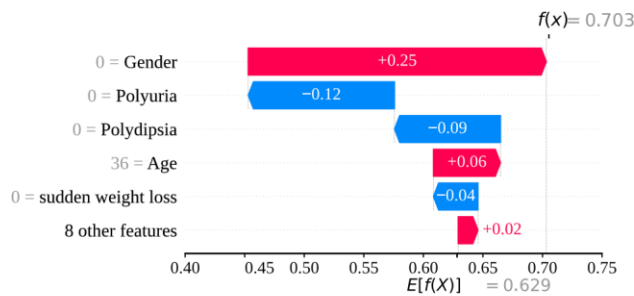


(a) TP prediction (Index 0).

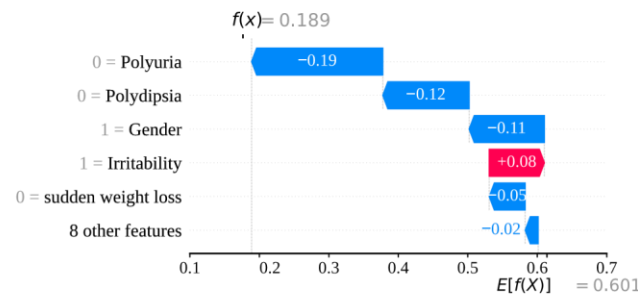
1



(b) TN prediction (Index 332).



(c) FP prediction (Index 317).



(d) FN prediction (Index 431).

Fig. 26. Individual predictions in FOX_RF.

Index	V_0	V_1	V_2	V_3	V_4	V_5	V_7	V_8	V_9	V_{10}	V_{11}	V_{13}	V_{14}
0 (TP)	0.05	— 0.07	— 0.16	0.23	— 0.03	0.00	— 0.02	0.03	— 0.04	— 0.04	0.02	— 0.02	— 0.02
332 (TN)	— 0.01	0.11	— 0.17	— 0.12	— 0.06	— 0.02	— 0.01	0.03	— 0.06	— 0.02	— 0.08	0.02	— 0.13
317 (FP)	0.06	0.25	— 0.12	— 0.09	— 0.04	— 0.01	— 0.01	— 0.04	— 0.03	— 0.04	— 0.00	— 0.02	— 0.00
431 (FN)	— 0.01	— 0.11	— 0.19	— 0.12	— 0.05	0.01	— 0.04	— 0.02	0.02	0.08	— 0.01	0.05	— 0.02

Table 15. SHAP values for the selected predictions in FOX_RF. V_0, V_1 , etc., represent the notation of the actual features listed in Table 2.

FN: 431) have been randomly selected from their respective categories. The description of these four records has been given in the Table 14.

Figure 26 illustrates the SHAP-based feature contributions for four individual predictions using the F_{FOX} feature set in the FOX_RF model. Figure 26a is a TP prediction of the individual indexed as 0, where Polydipsia has contributed the most with a SHAP value of +0.23, having the feature value of 1 (Yes). Age has also positively contributed to this particular prediction with a +0.05 SHAP value. However, Polyuria with the feature value 0 (No), and Gender with 1 (Male) have contributed negatively. Polyuria and Gender have SHAP values of -0.16 and -0.07, respectively.

Notably, Irritability has emerged among the top five influential features, with a SHAP value of -0.04. And the combined contribution of the other 8 features has been calculated as -0.08. Features with negative SHAP values contribute toward the non-diabetic (negative) class, while those with positive SHAP values drive the prediction toward the diabetic (positive) class. Therefore, it has been confirmed that Polydipsia is the key contributor to this particular forecast to lead the cumulative SHAP value, $f(x) = 0.524$, from the base value $E[f(x)] = 0.60$. And finally, as the $f(x) > 0.5$, the model has made its decision as a diabetic positive class (1) for this instance. Table 15 represents all the SHAP values of all the features in the set F_{FOX} .

For the TN prediction at index 332 in Fig. 26b, Polyuria has had the highest SHAP value, at -0.17, due to its lower feature value (No: 0). A similar pattern has been observed for Alopecia and Polydipsia, although their feature values are opposite. For index 332, Alopecia has had a feature value of 1 (Yes) with a negative SHAP value of -0.13, whereas Polydipsia has had a feature value of 0 (No) with a SHAP value of -0.12. Despite their differing feature values, both have contributed to the negative class. Gender, with the feature value of 0 indicating female, has gained a SHAP value of +0.11, which has tried to lead the prediction to the positive class.

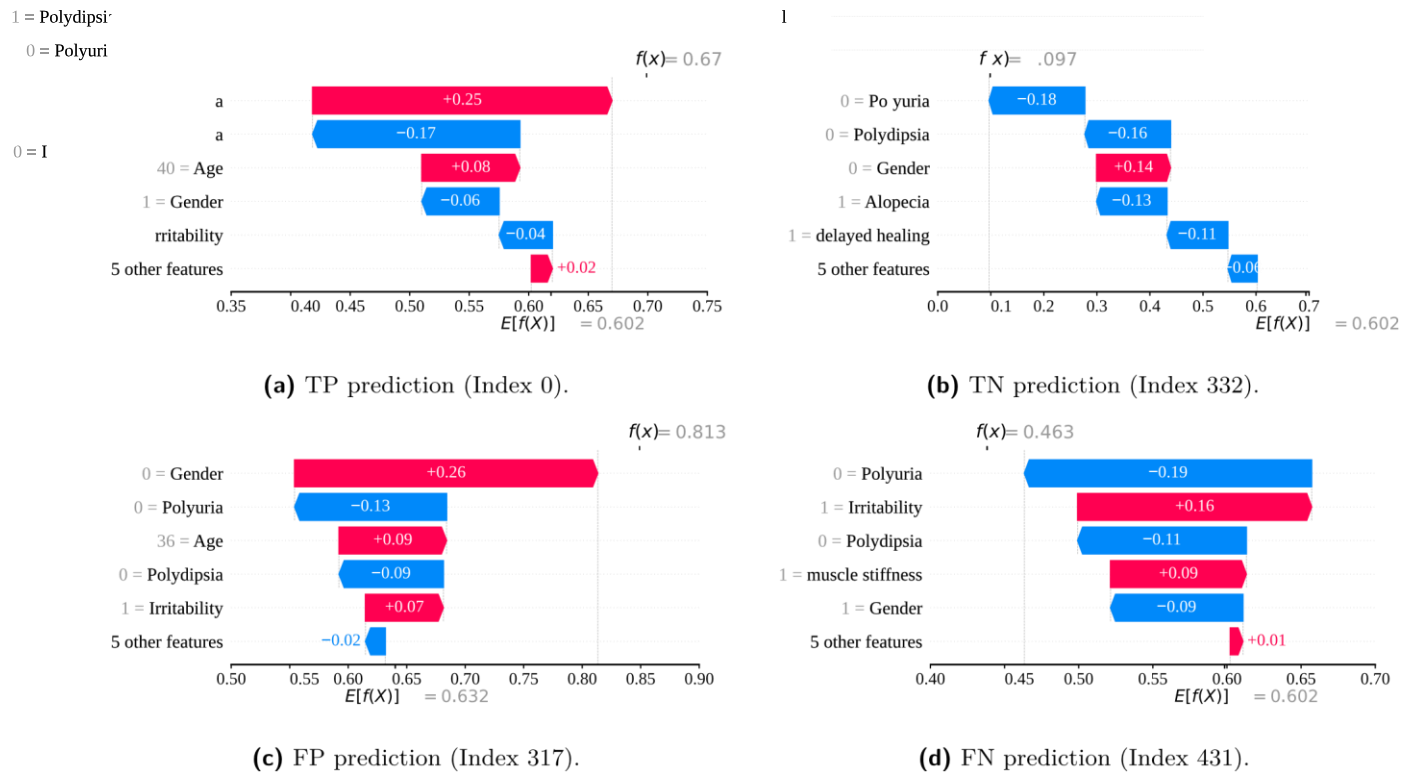


Fig. 27. Individual predictions in HBA_RF.

As a result, the value of $f(x)$ has resulted in 0.084, which is less than 0.5 and leads to the decision as a Non-diabetic class. In comparison with the prediction for index 0, it has been confirmed that Polyuria with a value of 0, Polydipsia with a value of 0, and Gender with a value of 1 have consistently contributed negative SHAP values, steering the prediction toward the non-diabetic class and vice versa.

Index	V_0	V_1	V_2	V_3	V_8	V_{10}	V_{11}	V_{12}	V_{13}	V_{14}
0 (TP)	0.08	0.06	0.17	0.25	0.04	0.04	0.01	0.01	0.04	0.00
332 (TN)	0.02	0.14	0.18	0.16	0.02	0.02	0.11	0.05	0.02	0.13
317 (FP)	0.09	0.26	0.13	0.09	0.01	0.07	0.00	0.01	0.01	0.01
431 (FN)	0.00	0.09	0.19	0.11	0.02	0.16	0.00	0.00	0.09	0.01

Table 16. SHAP values for the selected predictions in HBA Model.

For the FP prediction at index 317, the base has been determined as $E[f(x)] = 0.629$, and the cumulated SHAP value has been measured as $f(x) = 0.703$. As the value of $f(x) > 0.5$, the decision has gone towards the diabetic positive class, though the individual is a non-diabetic. SHAP has found that the main contribution to this FP prediction is made by Gender, having the SHAP value of +0.25, with the feature value of 0. Age also has a positive SHAP value of +0.06. However, Polyuria, Polydipsia, and Sudden weight loss have all had feature values of 0 (No) and have contributed toward the negative class with their negative SHAP values. For the index 431, a FN prediction, only higher Irritability (1: Yes) has a positive SHAP value to prevent this wrong classification, but it has failed to be suppressed by all other features. Where Polyuria is the highest contributor to this misclassification with a SHAP value of -0.19.

Figure 27 shows the individual predictions made by HBA_RF using the feature F_{HBA} feature set. For index 0, Polydipsia is mainly responsible for this right classification with the SHAP of +0.25, assisted by Age having the SHAP value of +0.08. However, Polyuria, Gender, and Irritability have tried to push the prediction to be wrong, but were suppressed by the impact of Polydipsia. HBA_RF has correctly classified the index 332 as non-diabetic. Here, only Gender has a positive SHAP value of +0.14. As a result, the overall SHAP value has been calculated as $f(x) = 0.097$, and the model has made a correct classification, with Polyuria and Polydipsia being the main contributors.

Index 317 has also been misclassified in HBA_RF. Gender has gained a positive SHAP value of +0.26, mainly contributing to this prediction being misclassified, which is similar to the nature observed in FOX_RF.

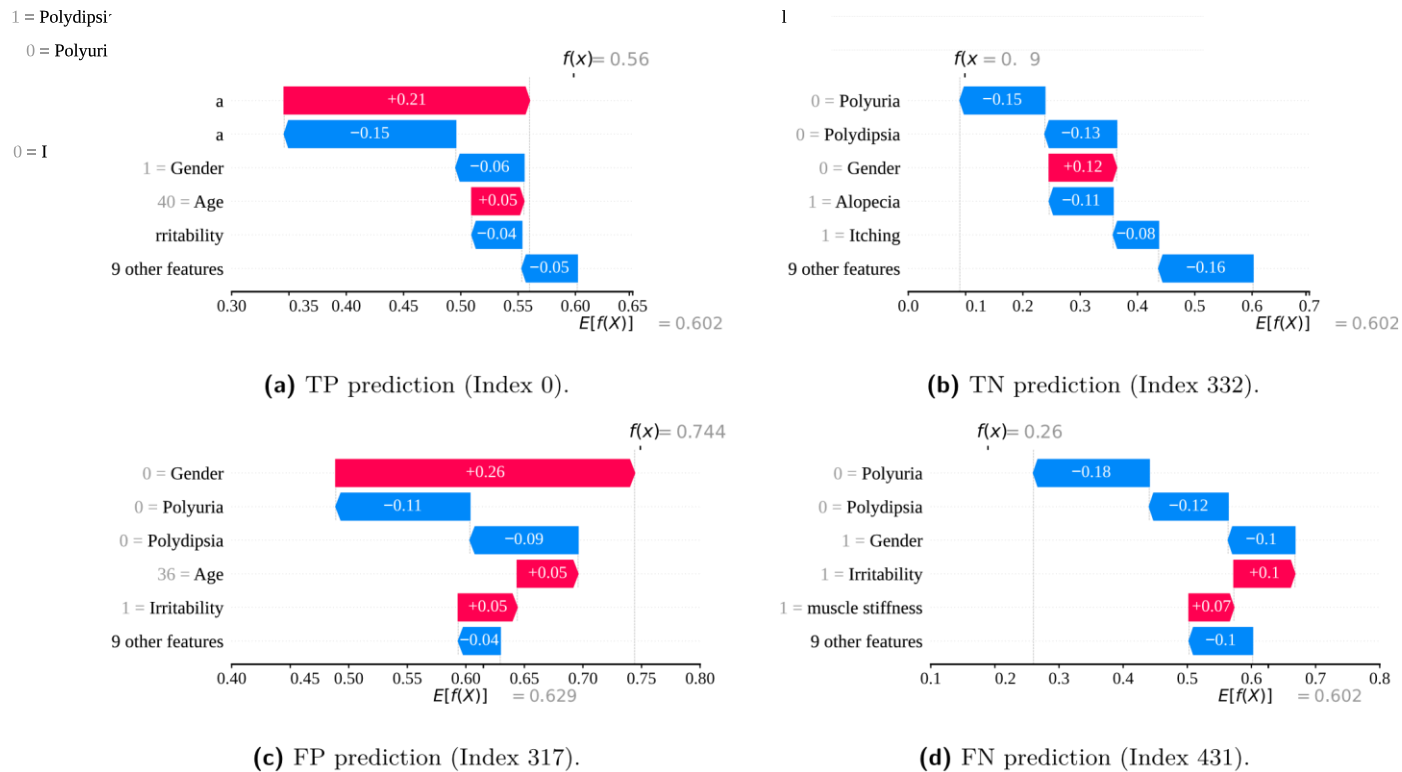


Fig. 28. Individual predictions in TSO_RF.

However, the value of $f(x) = 0.813$ has been higher than the value found in FOX_RF for index 317, primarily due to the cooperative contribution of Irritability. For index 431 in HBA_RF, the model has made an incorrect prediction (FN) with a value of $f(x) = 0.46 < 0.5$. In this prediction, Polyuria has played a key role due to its feature value of 0, supported by Polydipsia and Gender. Although Irritability and Muscle stiffness with feature values of 1 have had positive SHAP values and have tried to prevent misclassification, they were insufficient. Table 16 represents all the SHAP values of all the features in the set F_{HBA} .

Figure 28 presents the forecast for index 0, 332, 317, and 431 as the TP, TN, FP, and FN for TSO_RF based on F_{TSO} feature set. For index 0, Polydipsia is mainly responsible, as noticed in the previous two models.

Index	V_0	V_1	V_2	V_3	V_4	V_5	V_7	V_8	V_9	V_{10}	V_{11}	V_{13}	V_{14}	V_{15}
0 (TP)	0.05	— 0.06	— 0.15	0.21 0.03	— 0.03	0.02 0.01	— 0.01	0.03 0.04	— 0.04	— 0.04	0.02 0.07	— 0.01	— 0.02	0.00
332 (TN)	— 0.01	0.12 0.15	— 0.13	— 0.07	— 0.03	— 0.01	— 0.01	0.04 0.02	— 0.08	— 0.03	— 0.07	0.00 0.02	— 0.11	0.00
317 (FP)	0.05	0.26 0.11	— 0.09	— 0.03	— 0.02	0.02 0.01	— 0.01	— 0.02	0.02 0.05	0.05 0.01	— 0.02	— 0.02	0.00 0.00	— 0.00
431 (FN)	0.01	— 0.10	— 0.18	— 0.12	— 0.05	0.00 0.02	— 0.02	— 0.02	0.02 0.10	— 0.01	— 0.07	— 0.01	— 0.01	0.00

Table 17. SHAP values for the selected predictions in TSO Model.

TSO_RF has maintained a similar trend observed in FOX_RF and HBA_RF regarding Polydipsia, Polyuria, Gender, Irritability, and Age. And the same trend as the previous two models has been found for the index in 332, a TN prediction, where all the features have shown a correct response to the model except the Gender. Index 317 has mainly been misclassified because of a high positive SHAP value of +0.26 for Gender. For index 431, the Polyuria, Polydipsia, Gender, and the combined result of the other 9 features are responsible for this wrong classification. Table 17 represents all the SHAP values of all the features in the set F_{TSO} .

Discussion

A constructive and comparative discussion between our findings and the previous works on the ESDRP dataset has been conducted in this section. Table 18 visualizes the comparison between our proposed frameworks and previous models based on without CV and using CV. Accuracy, F-score, Precision, and Recall have been counted for this comparison. The results of previous research works have been tabulated in ascending order according to accuracy to facilitate a clearer comparison. In the Without Cross-validation section, Nagpal et al. have achieved an accuracy of 93.40%, F-score of 95.40%, Precision of 96.90, and Recall of 94.10% using 16³.

1 = Polydipsi

0 = Polyuri

1

0 = I

Ferdousi et al. have increased the Accuracy up to 94% using 16 features¹⁹. Later,^{4,5,20} have achieved an Accuracy of 97%, 97.11%, 97.36%, respectively, using 16 feature. The GB classifier has achieved 98.07% of Accuracy, F-score, Precision, and Recall²¹. However, Gundougdu et al. have shown a supremacy among the previous works with an Accuracy of 99.23% using only 09 features, including a noticeable F-score, Precision, and Recall of 99%²². However, our proposed frameworks have shown the ability to outperform the previous outcome conducted on the ESDRP dataset. Without CV, FOX_RF (RF+FOX) and HBA_RF (RF+HBA) have achieved an Accuracy of 99.36%, F-score of 9.51%, Precision of 100%, and Recall of 99.02%. Our final model, TSO_RF, which combines the TSO optimizer with the RF classifier, has demonstrated the highest performance, achieving 100% across all metrics with no misclassification when tested without CV using the 14 most valuable features.

In the Cross-validation section, Navazi et al. have used GA-SVM with PSO and successfully reduced the number of features from 16 to 9 while achieving and mean 10-fold cross-validation Accuracy of 93.42%, F-score of 94.51%, Precision of 95.56%, and Recall of 93.48%²³. Rahman et al. have used RF attached with SMOTE, and

SL No	Reference	Method	Accuracy (%)	F-score	Precision	Recall	Features
Without Cross-Validation							
01	³	EBDT + PCA	93.26	95.40	96.90	94.10	16
02	¹⁹	RF	94.00	92.00	93.00	92.00	16
03	⁴	MLP + APGWO	97.00	98.00	99.00	97.00	16
04	⁵	ET	97.11	98.10	98.52		16
05	²⁰	KNN	97.36	–	98.22	99.21	16
06	²¹	GB	98.07	98.07	98.07	98.07	16
07	²²	XGBoost + RFFS	99.23	99.00	99.00	99.00	09
01	Our Model 01	RF + FOX	99.36	99.51	100.0	99.02	13
02	Our Model 02	RF + HBA	99.36	99.51	100.0	99.02	10
03	Our Model 03	RF + TSO	100.0	100.0	100.0	100.0	14
With Cross-Validation (Mean Performance)							
01	²³	GA-SVM + PSO	93.42	94.51	95.56	93.48	09
02	²⁴	RF + SMOTE	94.87	96.08	96.08	96.08	16
03	²⁵	RF	95.03	–	–	–	16
04	⁶	RF	96.58	–	–	98.59	16
05	²⁶	ODF(RFE) + BMNABC	96.81	–	–	98.07	16
06	²⁷	ES+IMETC	97.00	97.00	97.00	97.00	12
07	²⁸	RF	98.08	98.40	98.40	98.40	16
08	²⁹	RF	98.08	–	–	–	16
01	Our Model 01	RF + FOX	98.01	98.36	98.43	98.32	13
02	Our Model 02	RF + HBA	97.24	97.71	98.09	97.38	10
03	Our Model 03	RF + TSO	98.14	98.47	98.42	98.54	14

Table 18. Result Comparison of Previous Methods with Our Method (without CV and with CV).

successfully gained 94.87% of Accuracy and 96.08% of F-score, Precision, and Recall using all 16 features²⁴. Yadu et al. and Addo et al. have used RF and showed that their models are capable of reaching up to 95.03% and 96.58% of Accuracy, respectively^{6,25}. Next, Pradhan et al. have shown that a complex model such as ODF(RFE) + BMNABC is capable of gaining an Accuracy of 96.81% and a Recall of 98.59%²⁶. Saxena et al. have contributed to diabetes classification by achieving 97% Accuracy, F-score, Precision, and recall using ES+IMETC²⁷. Chaves et al. and Raju et al. have independently used Random Forest with cross-validation and have achieved an accuracy of 98.08%^{28,29}. None of the models that have used CV have focused on feature reduction. However, our findings have addressed both aspects by reducing the number of features and fine-tuning the parameters, while maintaining optimal performance. In our proposed frameworks, a 10-fold CV has been introduced, and the mean performances have been displayed in the lower part of the Table 18. Using FOX_RF, we have achieved 98.01% accuracy, 98.36% F-score, 98.43% precision, and 98.32% recall, while reducing the number of features from 16 to 13. Another model, HBA_RF, has shown the maximum feature compression by selecting only the 10 most influential features. This model has also maintained a moderate performance with 97.24%, 97.71%, 98.09%, and 97.38% of Accuracy, F-score, Precision, and Recall, respectively. However, TSO_RF has outperformed all the previous models, including those within our proposed models, and achieved an Accuracy of 98.14%. The F-score, Precision, and Recall of the TSO_RF are 98.47%, 98.42%, and 98.54%, respectively, using only 14 features.

From the SHAP analysis, our frameworks have identified the most impactful features related to diabetes classification. Each proposed framework, whether using CV or not, has confirmed that Polyuria, Polydipsia, and Gender are the most influential features. Not only that, Fig. 4 has also found that Polyuria and Polydipsia are highly correlated with the diabetes target class. In an individual prediction, the contribution of Polyuria, Polydipsia, and Gender is constant whether it is a wrong or correct classification across the models. If a person is marked as Yes (1) for Polyuria, it indicates a strong likelihood of being affected by diabetes; the same applies to Polydipsia. On the other hand, the probability of getting affected by diabetes increases for females (0) Gender. These three features are crucial; if they are altered, there is a high probability of misclassification. Despite not being the top contributors, Irritability, Muscle stiffness, and Sudden weight loss could be valuable features for reducing the misclassification rate.

Previous studies have also aligned with our findings. Previous studies have also mentioned that Polyuria and Polydipsia are good symptoms of diabetes and influential features for diabetes classification. An editorial paper conducted by Dr. A. Ramachandran, India Diabetes Research Foundation & Dr. A. Ramachandran's Diabetes Hospitals 28, Marshalls Road, Egmore, Chennai 600 008, Tamil Nadu, India has mentioned that, Polyuria, polydipsia, and polyphagia are classic symptoms commonly seen in type 1 diabetes and in type 2 diabetes with severe hyperglycemia⁴³. Polyuria-polydipsia syndrome involves

excessive urination and thirst, widely seen in diabetes⁴⁴. Another study has directly mentioned that polydipsia and polyuria are symptoms of diabetes⁴⁵. Even in some research work that has classified diabetes, it has been confirmed that Polyuria, Polydipsia, and Gender have the highest impact on the classifier models.^{5,6,24} have conducted classification tasks separately, and all of them have found that Polyuria, Polydipsia, and Gender are the top three important features for diabetes classification. Therefore, our findings have aligned with the previous research works, maintaining higher performance with minimal features.

Future work

The findings of this research will contribute to diabetes classification using minimum features while maintaining the performance. Our proposed framework can be used to develop an advanced ML-enhanced Diabetes diagnosis toolkit. With minimal training and testing time, our proposed frameworks have been able to deliver results within a very short period, thereby reducing patient complications and accelerating the diagnosis process.

Limitations

The ESDRP dataset contains only 520 samples, which represents a relatively small sample size and serves as the primary limitation of this study. Additionally, we have implemented only a few swarm-based optimization algorithms (FOX, HBA, and TSO), whereas other potentially more effective optimizers were not explored within the scope of this work. This can be considered another limitation. Furthermore, our model has not been externally validated using an independent dataset due to the unavailability of a similar dataset, which also constitutes a limitation of this study. We aim to address these limitations in future research by incorporating a larger and comparable dataset, if available.

Conclusion

Diabetes is a chronic condition of high blood sugar that can damage organs like the heart, eyes, and kidneys. Type 2 diabetes has occurred more frequently than Type 1 and gestational diabetes. Early detection with minimal features has reduced the burden of suffering and the financial strain. Machine learning has shown its ability to classify diabetes based on the available dataset. This research has introduced three novel frameworks, FOX_RF, HBA_RF, and TSO_RF, to classify diabetes based on the Early Stage Diabetes Risk Prediction dataset. Our models have not only achieved the highest performance but have also reduced the number of features used in forecasting. TSO_RF has achieved 100% Accuracy without cross-validation, including a perfect F-score, Precision, and Recall of 100%. However, we have also performed cross-validation to evaluate the performance. The highest cross-validation mean accuracy of 98.14% has been attained by the TSO_RF. Besides, TSO_RF has also demonstrated its supremacy by achieving the highest F-score of 98.47% and recall of 98.54%. However, FOX_RF has outperformed other models with the highest Precision of 98.43%. In terms of feature reduction, the HBA_RF has been found as more effective by selecting only 10 features out of 16. However, FOX_RF and TSO_RF have respectively deleted 13 and 14 features. TSO_RF has achieved the highest mean AUC of 0.9988, indicating its superiority across the proposed CV models. HBA_RF has proved its potential implication on real-world applications, such as real-time diabetes monitoring devices, with its lowest fold-wise testing time of 9.04 milliseconds only. Each of our proposed models has shown consistency in finding the most influential feature by selecting Polyuria, Polydipsia, and Gender. Polyuria is the most highly impacted feature for diabetes prediction. The SHAP evaluation has introduced the overall model's nature, showing the mean SHAP value of each feature. Individual prediction of a True Positive, True Negative, False Positive, and False Negative has been introduced using feature-wise SHAP value. So, the findings of this study can be implemented in developing IoT-based applications for smart diabetes monitoring systems.

As we have performed classification both with and without cross-validation, the framework shows potential for maintaining scalability on other datasets. In our models with cross-validation, we have a sufficient gap between the training and testing Accuracy (Minimum: 1.8%; Maximum: 2.76%), indicating no presence of overfitting issues. However, our models have certain limitations; for instance, the dataset contains only 520 samples. Developing a model with such a small dataset may lead to generalizability issues.

Data availability

The dataset used in this study is publicly available at <https://archive.ics.uci.edu/dataset/529/early+stage+diabetes+risk+prediction+dataset>, accessed on: 13th of April, 2025.

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Author contributions

PS perceived the idea, conducted formal analysis, worked on the visualization, developed the methodology, and authored draft. AAN investigated and supervised the study. KC administrated and funded the project, and participated in reviewing the initial draft, modified it, and considerably improved it. MAS curated the data, administrated the project, and participated in reviewing the initial draft, modified it, and considerably improved it.

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Declarations Competing interests

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The authors declare no competing interests.

Additional information

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OPEN A copula based supervised filter for feature selection in machine learning driven diabetes risk prediction

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Effective feature selection is critical for building robust and interpretable predictive models, particularly in medical applications where identifying risk factors in the most extreme patient strata is essential. Traditional methods often focus on average associations, potentially overlooking predictors whose importance is concentrated in the tails of the data distribution. In this study, we introduce a novel, computationally efficient supervised filter that leverages a Gumbel copula implied uppertail concordance score (λ_U , a monotone transformation of Kendall's τ) to rank features by their tendency to be simultaneously extreme with the positive class. We evaluated this method against four standard baselines (Mutual Information, mRMR, ReliefF, and L1/Elastic-Net) across four classifiers on two diabetes datasets: a large-scale public health survey (CDC, $N=253,680$) and a classic clinical benchmark (PIMA, $N=768$). Our analysis included comprehensive statistical tests, permutation importance, and robustness checks. On the CDC dataset, our method was the fastest selector and reduced the feature space by $\approx 52\%$. While this resulted in a minimal but statistically significant performance trade-off compared to using all 21 features, our filter significantly outperformed standard filters (Mutual Information, mRMR) and was statistically indistinguishable from the strong ReliefF baseline. On the PIMA dataset (8 predictors), our method's ranking produced the numerically highest ROC-AUC, despite paired DeLong tests showing no statistically significant differences versus strong baselines. PIMA thus serves as a ranking-only sanity check that our upper-tail criterion behaves sensibly in a low-dimensional clinical setting. Across both datasets, the Gumbel- λ_U selector consistently identified clinically coherent and impactful predictors. We conclude that feature selection via upper-tail dependence is an efficient and interpretable screening approach that can complement standard feature-selection baselines in public health and clinical risk prediction.

Keywords Copula-based feature selection, Tail dependence, Gumbel copula, Risk prediction, Machine learning, Supervised feature selection, Public health, Diabetes

Diabetes mellitus is one of the biggest public health challenges of our time. The International Diabetes Federation estimates that 1 in 9 adults worldwide have diabetes, and this could rise to nearly 1 in 8 by 2050, which is about a 46% increase¹. As more people are affected, there is a growing need for reliable and easy-to-use risk prediction tools. These tools can help clinicians and public health systems find people at high risk and offer targeted prevention.

Machine learning has emerged as a powerful paradigm for developing such tools, leveraging large-scale health datasets like the CDC Diabetes Health Indicators² survey to uncover complex patterns associated with disease risk. A critical step in building effective and trustworthy predictive models is feature selection, a process fundamental to both performance and scientific utility³. For a non-specialist audience, feature selection (FS) is the principled process of identifying the most informative subset of predictors from a larger

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pool. The goals are threefold: (i) improve predictive performance by eliminating irrelevant or noisy variables that promote overfitting; (ii) reduce cost and latency by limiting variables to measure, store, and process; and (iii) enhance interpretability to yield actionable domain insight.

Traditional filter-style selectors measure how features and labels are related overall. Pearson correlation looks at average linear relationships, while information theoretic measures like mutual information (MI)^{4,6} capture more general, non-linear connections. However, these methods do not focus on the extremes, which are often most important for identifying high-risk cases. For example, a strong correlation between BMI and diabetes does not show what happens when BMI is at its highest levels. Our approach instead ranks features using the Gumbel copula's upper-tail dependence coefficient (λ_U)⁷, which highlights when extreme values of a feature and the outcome occur together.

In risk modeling, the most important information often comes from the extremes rather than the average cases. To focus on these situations, we use ideas from extreme-value theory and tail dependence⁸ within the copula framework. Sklar's theorem allows copulas to separate how variables are related from their individual distributions, so we can directly measure how they move together, especially in the extremes (tails)^{9–11}.

We introduce and validate a supervised filter for feature selection based on the Gumbel copula's⁷ upper-tail dependence coefficient λ_U , used here as a tail-sensitive concordance score derived from rank-based dependence. Each predictor is ranked by its $\lambda_U(X, Y)$ with the diabetes label, prioritizing features whose highest values cooccur with positive outcomes. To our knowledge, this is the first study to operationalize a copula tail-dependence coefficient as a direct, standalone criterion for supervised feature selection in clinical risk prediction.

We validate on two public datasets. On the large CDC survey, our method demonstrates a favorable trade-off between parsimony and performance. By reducing the feature set by 52% (from 21 to 10), our Gumbel-selected model achieves a competitive ROC–AUC of 0.823 and is the fastest selector. Most importantly, it demonstrates its utility as a competitive filter by achieving statistically higher AUC scores than standard filter methods like Mutual Information and mRMR, while performing on par with the strong ReliefF baseline. To demonstrate external validity beyond a large, survey-style cohort (CDC), we also evaluate on the PIMA Indians Diabetes dataset, a small clinical benchmark with *eight* predictors. Because PIMA contains all and only eight covariates, this experiment isolates *feature ranking* rather than dimensionality reduction: all methods use the same variables. The Gumbel ranking combined with a Random Forest attains the highest ROC–AUC of 0.867, with paired DeLong tests showing no significant differences across selectors (DeLong $p > 0.05$). As such, PIMA functions as a ranking-only sanity check that our upper-tail criterion yields clinically coherent orderings without harming discrimination. Overall, our results show that focusing on the extremes can create models that are simple, easy to understand, and effective for clinical and public health use.

One might wonder why we chose the Gumbel copula and did not use a copula goodness-of-fit test. Our goal is to focus on the property of interest, not just on how well a model fits. We need a dependence measure that captures when both a predictor and the label are high at the same time. The Gumbel copula provides this, since it has nonzero upper-tail dependence and zero lower-tail dependence, giving us the coefficient we need. By contrast, the Clayton copula only shows lower-tail dependence, and the Gaussian and Frank copulas are tail-independent. Even if a goodness-of-fit test favored these other models, they would not help us understand synchrony in the upper tail. We are not fitting a full generative copula here. Instead, we use the Gumbel mapping as a scoring function that is monotonic in Kendall's tau, so we can rank features by their upper-tail dependence. In the future, other upper-tail families, such as Joe, Student's t, or BB1, could be compared using the same property-focused approach.

The next section reviews related work. We then present the preliminaries, introducing the methods used in this study. After that, we describe the datasets. The methodology section outlines the experimental pipeline. The results section reports our findings and includes an in-depth discussion of the medical implications. Finally, the conclusion summarizes the study and highlights directions for future research.

Related work

Feature selection methods are usually grouped as filters, wrappers, or embedded approaches³. Wrappers¹² use a learning algorithm to evaluate feature subsets, such as genetic search or RFE, and often achieve strong results but require significant computation^{13,14}. Embedded methods combine selection with model training. Well-known examples include LASSO¹⁵ and elastic-net, which encourage sparsity through regularization^{16,17}.

Our work introduces a filter method where features are scored directly from the data, without relying on any downstream model¹⁸. Classic filters include mutual information (MI), which measures non-linear dependence but evaluates features one at a time⁵. The mRMR method addresses this by selecting features that are both highly relevant and minimally redundant^{19,20}. ReliefF^{21,22} focuses on local class separation by using nearest neighbors. However, these methods still mainly assess overall association. Unlike correlation-based filters (which are tail-independent), our score targets joint extremes via the Gumbel upper-tail coefficient λ_U rather than average association.

A recent area of research uses copulas to create more fair and robust ways to measure dependence. Copulabased filters like Copula Correlation (Ccor) and Robust Copula Dependence (RCD) help reduce functional bias compared to traditional metrics^{23,24}, which aligns with the goals of the MIC literature^{25,26}. Other advances include using copulas for dimensionality reduction²⁷, entropy-based selection^{28,29}, hybrid filters that combine copula and information-theoretic scores to balance relevance and redundancy³⁰. However, most copula-based studies still focus on measuring overall dependence or redundancy between features.

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Importantly, tail-dependence coefficients, which directly measure the probability of joint extremes, have not yet been used as the main supervised ranking criterion in feature selection. As seen from the literature, a Gaussian copula cannot capture tail dependence and have called for methods that use tail-dependent families. Our work addresses this need. By using the Gumbel copula's upper-tail coefficient λ_U ^{7,11}, we shift the selection focus from overall association to joint extremal behavior. This approach provides a simple, interpretable, and computationally efficient supervised filter that (i) identifies clinically meaningful predictors linked to high-risk cases, (ii) achieves performance comparable to strong baselines on both the large survey and clinical benchmark datasets, and (iii) remains fast compared to standard baselines like MI, mRMR, and ReliefF, making it practical for real-world use.

Finally, to specifically assess the effect of tail-aware ranking when dimensionality reduction is moot, we include the 8-variable PIMA benchmark where all methods use the same variables. Next, we present the preliminaries section.

Preliminaries

In this section, we introduce the main concepts and notation used in the paper, including copulas, taildependence coefficients, types of feature-selection methods, baseline selectors, classification models and metrics, permutation importance, and paired statistical tests.

Copulas

A copula $C : [0, 1]^2 \rightarrow [0, 1]$ is a joint distribution with uniform marginals that captures dependence^{10,11}. By Sklar's theorem⁹, for any bivariate cdf H with continuous independently of the marginal distributions marginals F_X, F_Y there exists a unique copula C such that

$$H(x, y) = C(F_X(x), F_Y(y)). \quad (1)$$

Consequently, if we convert data to pseudo-observations

$$U = \frac{\text{rank}_n(X) - 1}{n - 1}, \quad V = \frac{\text{rank}_n(Y) - 1}{n - 1}, \quad (2)$$

any measure computed from C depends only on the dependence structure, not on the marginal scales. Copulas are organized in families (e.g., elliptical, Archimedean); we will later specialize to the Gumbel⁷ copula, which is part of the Archimedean family, because it exhibits positive upper-tail clustering, which is the signal we target for feature screening. **Tail dependence in copulas**

"Tail dependence" expresses the probability that one variable is extreme given the other is extreme. Tail dependence is symmetric in the variables (i.e., invariant to swapping U and V in the bivariate case). For the upper tail, the coefficient $\lambda_U \in [0, 1]$ is defined as

$$\lambda_U = \lim_{q \rightarrow 1^-} \frac{\Pr(V > q \mid U > q)}{\Pr(U > q)}. \quad (3)$$

Intuitively, λ_U is the asymptotic co-occurrence probability of simultaneous highs. Values near 1 indicate strong synchronization at extreme levels; values near 0 indicate little to no joint extremal behavior. The lower-tail analogue $\lambda_L = \lim_{q \rightarrow 0^+} \Pr(V \leq q \mid U \leq q)$ exists but is not used in this work.

Scope. Tail-dependence concepts are most naturally defined for continuous margins; in this paper, we use the Gumbel mapping as a rank-based, tail-sensitive concordance score (a monotone transformation of Kendall's τ) for feature ranking, rather than as a generative copula tail parameter. Throughout the paper, references to λ_U (upper-tail dependence) are to this Gumbel-implied, rank-based concordance score derived from Kendall's τ , used for ranking rather than for generative copula modeling.

Feature Ranking via λ_U . For each feature X_j and the binary label Y , we form pseudo-observations U_j, V from ranks, estimate Kendall's $\tau(U_j, V)$, and (under the Gumbel family) map τ to the copula parameter and λ_U then to λ_U . The resulting λ_U serves as a feature score: features with larger λ_U exhibit stronger joint extremes with the outcome and are ranked higher.

Throughout, "extremes" and "movement in the tails" refer to the same concept: the synchronized cooccurrence of upper-tail values of a feature and the label, quantified by the Gumbel upper-tail dependence coefficient λ_U .

The Gumbel (Gumbel-Hougaard)^{7,31} copula is an Archimedean copula with generator

$$C_{\theta}(u, v) = \exp \left\{ - \left[(-\log u)^{\theta} + (-\log v)^{\theta} \right]^{1/\theta} \right\}, \quad (u, v) \in (0, 1)^2. \quad (4)$$

Gumbel exhibits upper-tail but not lower-tail dependence:

$$\lambda_U(\theta) = 2 - 2^{1/\theta}, \quad \lambda_L(\theta) = 0. \quad (5)$$

Kendall's tau for Gumbel is

$$\tau(\theta) = 1 - \frac{1}{\theta} \iff \theta = 1 - \frac{1}{\tau} \quad (\tau > 0). \quad (6)$$

Gumbel copula

Thus, $\tau \in \tau[0(X, 1)_j, Y; \tau)$ on pseudo-observations; if $\tau = 0$ corresponds to independence ($\tau > 0$ we map $\theta =$

1). In our scoring, for each feature X_j we estimate Kendall's $\hat{\theta} =$

$$\hat{\theta}_j = 1 - \hat{\tau}_j, \quad \hat{\lambda}_{U(j)} = 2 - 2^{1/\hat{\theta}_j},$$

and if $\hat{\tau} \leq 0$ we set $\hat{\lambda}_{U(j)} := 0$ (no positive upper-tail co-occurrence under Gumbel).

We use a property-first approach and require a dependence measure with nonzero λ_U . The Gumbel copula meets this requirement, since $\lambda_U > 0$ and $\lambda_L = 0$. In contrast, the Clayton copula only has lower-tail dependence ($\lambda_U = 0$), and the Gaussian and Frank copulas are tail-independent ($\lambda_U = \lambda_L = 0$). Even if a goodness-of-fit test favored those other families, they would not address our question about upper-tail synchrony. For this reason, we use the Gumbel mapping as a monotone transform of τ to rank features by upper-tail dependence. We are not claiming a full generative copula fit. While the classical upper-tail dependence coefficient is defined asymptotically for continuous margins, here it is used as a monotone transformation of Kendall's τ to construct a tail-sensitive concordance score for ranking; we do not claim estimation of asymptotic tail probabilities or a fully specified generative copula model. (for further details, see section 2.2.2 of⁴⁵).

Computational cost. Let n be the number of samples and d the number of features. Our selector does *not* fit a model per feature. For each X_j we compute pseudo-observation ranks ($O(n \log n)$; V is ranked once), compute

Kendall's τ and $\lambda_U = 2^{\tau} (U_{-j}, V_{2^{1/\theta}})$ via an $O(\tau)$ for the Gumbel copula (τ is used), with much smaller constants than

wrappers/embedded methods that ($n \log n$) exact algorithm^{10,11}. Thus the total selection cost is³², and apply

the closed-form maps $O(dn \log n)$ (and $\theta = 1 - \tau$ if $\tau > 0$) an approximate/linear-time

fit models. Empirically (Results, Tables 7, 12), Gumbel- λ_U is the fastest selector on CDC and comparable to MI/mRMR on PIMA. Pearson correlation is cheaper but tail-agnostic and cannot target joint extremes (e.g., Gaussian/Frank are tail independent). **Families of feature selection methods**

Feature selection (FS) methods are commonly grouped as:

- **Filter methods**³: model-agnostic scoring of features using data statistics (e.g., MI, correlation, ReliefF, our λ_U score). They do not fit the predictor during selection and are computationally efficient.
- **Wrapper methods**¹⁸: search over subsets guided by a predictive model (e.g., forward selection, RFE); usually costly.
- **Embedded methods**³: selection occurs within model training (e.g., L1/elastic-net logistic regression, treebased splits).
- **Hybrid methods**³³: combination of filter and wrapper methods. The filter approach selects the candidate feature set from the original feature set and refines the candidate feature set by the wrapper methods. It brings together the strengths of both methods.

Feature selection approaches can also be categorized according to selection criteria into four groups³⁴: statistical measure-based, information theory-based, similarity measure-based, and sparse learning-based.

Our proposed Gumbel- λ_U procedure is a supervised filter method. In comparisons, MI, mRMR, and ReliefF are also filters; L1/Elastic-Net is an embedded selector. Benchmarking the proposed method against wrapper and hybrid methods will be left for future work.

Mutual Information (MI): MI^{4,5} is a filter method that measures how far the joint behavior of two variables deviates from what would be expected if they were independent. If X and Y were independent, the joint distribution would factor as $p(x,y) = p(x)p(y)$. MI quantifies how the actual joint relationship between two variables differs from this independence baseline; a greater deviation indicates a stronger dependence, whether linear or non-linear.

$$E. \quad (7) \quad \left[\log \frac{p(X,Y)}{p(X)p(Y)} \right] = \sum_x \sum_y p(x,y) \log \frac{p(x,y)}{p(x)p(y)} \quad I(X;Y) =$$

$I(X;Y) \geq 0$, and $I(X;Y) = 0$ if and only if X and Y are independent. In feature selection, MI is used to rank features by their dependence with the target variable, typically by computing $I(X_j;Y)$ for each feature X_j .

mRMR (Minimum-Redundancy, Maximum-Relevance): mRMR^{19,20} is a filter method that builds a feature set step-by-step. At each step, it prefers a feature that is (i) relevant to the label (informative on its own) and (ii) not redundant with features already chosen (brings new information). A common score is the difference between “MI with Y ” and the “average similarity to selected features”. In practice, similarity is often measured by absolute correlation for computational efficiency. In feature selection, mRMR builds the feature set greedily by adding the feature with the highest “relevance–redundancy” score, i.e., maximizing $I(X_j;Y)$ minus its average correlation with already-selected features, and stops after selecting k features.

ReliefF: ReliefF^{21,22} is a filter method that takes a nearest-neighbor view of feature usefulness. For each data point, it finds the closest point in the same class (a “hit”) and the closest point in the other class (a “miss”). A feature receives a higher score if, along that feature, the point is close to its hit but far from its miss, meaning the feature helps separate classes in local neighborhoods. This approach captures non-linear structure and some feature interactions. In feature selection, ReliefF evaluates many sampled points by comparing each feature’s average difference to nearest neighbors of the other class with that of the same class. Features are then ranked or selected based on the resulting average score. Larger scores indicate more discriminative features.

L1 / Elastic-Net Logistic Regression (L1EN):

L1EN^{15–17} is an embedded method that uses both $L1$ and $L2$ regularization penalties, with the emphasis

on $L1$ for feature selection. The $L1$ (lasso) penalty, $\|\beta\|_1 = \sum_j |\beta_j|$, tends to set many coefficients exactly to

zero, producing a sparse model where selected features correspond to the non-zero coefficients. The $L2$ (ridge) penalty,

$\|\beta\|_2^2 = \sum_j \beta_j^2$, shrinks coefficients toward zero but rarely makes them exactly zero, stabilizing estimates when $\alpha \|\beta\|_1 + \frac{1-2\alpha}{2} \|\beta\|_2^2$, providing sparsity from $L1$ and stability from $L2$. When a penalized logistic regression is fitted with these penalties, the $L1$ component determines which coefficients are exactly zero, with a non-zero coefficient β_j indicating that feature j is retained and a zero coefficient effectively removing it. Feature selection can then be performed by ranking

features according to the absolute values of their coefficients, $|\beta_j|$, with larger values indicating greater importance, and selecting the top k features based on this ranking.

Classifiers used for evaluation

Random forest (RF): RF³⁵ creates many decision trees using different random samples of the data and by choosing random subsets of features for each split. The final prediction is made by combining the results from all the trees, either by voting or averaging. RF is effective for both classification and regression tasks. It reduces variance through averaging, handles non-linearities and feature interactions, and performs well on tabular data with mixed signals.

Gradient Boosting (GB): GB³⁶ constructs a sequence of trees, with each new tree trained to correct the errors of the preceding ones. The final model aggregates the predictions of all trees. GB is applicable to both classification and regression tasks. It offers strong predictive performance on tabular data, with fine control over overfitting through the learning rate and tree depth.

Extreme Gradient Boosting (XGB): XGB³⁷ is a high-performance, regularized implementation of gradient boosting that incorporates system-level optimizations such as parallelism and sparsity-aware learning. It is typically fast and accurate on medium to large tabular datasets, and its regularization mechanisms help mitigate issues when features are correlated.

Logistic Regression (LR): LR³⁸, despite its name, is primarily used for classification tasks with two or more classes. It computes a linear combination of the features and applies a logistic or softmax function to convert this score into class probabilities. LR is simple and interpretable, often provides well-calibrated probabilities, and with regularization, it resists overfitting and can serve as an embedded feature selector.

Performance metrics

We use quantitative performance metrics to evaluate how well machine learning models work. These metrics help us compare different models and choose the best one for a specific task. In classification problems, the most common metrics are accuracy, precision, recall, F1 score, and area under the ROC curve (AUC).

Accuracy: Accuracy measures the proportion of correctly classified observations among all observations:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \quad (8)$$

where TP = true positives, TN = true negatives, FP = false positives, and FN = false negatives.

Accuracy is intuitive and easy to compute. In imbalanced datasets, high accuracy may be misleading because the model can achieve high scores simply by predicting the majority class.

Precision:

Precision measures the proportion of predicted positives that are truly positive:

$$\text{Precision} = \frac{TP}{TP + FP} \quad (9)$$

Precision is useful when false positives are costly (e.g., diagnosing a disease when the patient is healthy).

Recall:

Recall (also called sensitivity or true positive rate) measures the proportion of actual positives that were correctly identified:

$$\text{Recall} = \frac{TP}{TP + FN} \quad (10)$$

Recall is useful when false negatives are costly (e.g., failing to detect a disease in a patient who has it). **F1 Score:**

The F1 score combines precision and recall into a single metric, defined as the harmonic mean:

$$F1 = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}} \quad (11)$$

F1 balances precision and recall. It is especially valuable when dealing with imbalanced data, where accuracy alone can be misleading.

ROC and AUC (ROC–AUC): The ROC (Receiver Operating Characteristic) is a curve: it plots the true positive rate (TPR, i.e., recall) against the false positive rate (FPR) as the decision threshold sweeps from 1 to 0. Here

$$\text{TPR} = \frac{TP}{TP + FN}, \quad \text{FPR} = \frac{FP}{FP + TN} \quad (12)$$

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The AUC is a number: the Area Under that ROC curve, with discrimination: it equals the probability that a randomly chosen positive instance receives a higher model score $0 \leq \text{AUC} \leq 1$. AUC summarizes threshold-free than a randomly chosen negative instance,

$$\text{AUC} = \Pr(s^+ > s^-) + \frac{1}{2} \Pr(s^+ = s^-). \quad (13)$$

Thus, ROC–AUC jointly refers to the ROC curve and its area. Values near 1 indicate strong ranking ability; 0.5 corresponds to random guessing. Because it does not depend on a single threshold (and is less sensitive to class imbalance than accuracy), ROC–AUC is our primary metric. **Permutation importance (model-agnostic interpretation)**

For a fitted model f and test data $(X_{\text{test}}, y_{\text{test}})$, the permutation importance of feature j measures the drop in performance when the j -th column is randomly permuted:

$$I_j = E[\text{AUC}(f; X_{\text{test}}, y_{\text{test}})] - E[\text{AUC}(f; X_{\text{test}(j)}, y_{\text{test}})]. \quad (14)$$

We report the mean and standard deviation over n_{repeats} permutations (default 500), using ROC–AUC as the scorer. We compute permutation importance for the best Gumbel configuration (GB on CDC; RF on PIMA).

Paired statistical tests

To compare models fairly on the same test split, we use two paired tests:

DeLong’s test for AUC. DeLong’s test³⁹ is a non-parametric method for comparing the Area Under the Curve (AUC) of two correlated ROC curves, which arise when two different models are evaluated on the same test set. The test accounts for the covariance between the models’ predictions to provide a more accurate variance estimate. The null and alternative hypotheses are:

- **H₀ (Null Hypothesis):** The AUC of Model A is equal to the AUC of Model B.
- **H₁ (Alternative Hypothesis):** The AUC of Model A is not equal to the AUC of Model B.

A high p-value (e.g., $p > 0.05$) suggests that any observed difference in AUC is not statistically significant. It does not give the direction. Direction comes from the numerical AUCs themselves.

McNemar’s test for error discordance. McNemar’s test⁴⁰ is a non-parametric test used to compare the classification errors of two models. It operates on a 2x2 contingency table that cross-classifies the instances based on whether each model’s prediction was correct or incorrect. The test determines if there is a significant difference in the models’ disagreement rates. We use the continuity-corrected version. The null and alternative hypotheses are:

- **H₀ (Null Hypothesis):** The two models have the same error rate (i.e., their disagreements are symmetric).
- **H₁ (Alternative Hypothesis):** The two models have different error rates.

A low p-value (e.g., $p < 0.05$) indicates that one model is making significantly more errors on a different set of instances than the other, suggesting a meaningful difference in their predictive behavior.

In the next section, we describe the two datasets used in our study.

Dataset description

In this section, we describe the two datasets we have used in this study.

Dataset description (CDC diabetes health indicators)

The first dataset used in this study is the CDC Diabetes Health Indicators dataset, comprising 253,680 individuals and all 21 original features. These span demographic, lifestyle, and health-related characteristics (Table 1), providing a comprehensive basis for diabetes risk analysis.

The dataset was accessed via the UCI Machine Learning Repository CDC Diabetes Health Indicators² using the `ucimlrepo` Python library. All features were validated to contain no missing or constant values.

Target Variable: The dataset includes a binary label `Diabetes_binary`, where 0 means no diabetes and 1 means prediabetes or diabetes. Our study treats diabetes risk prediction as a binary classification problem, in line with the repository label. This definition enables focused evaluation of risk factors and classification performance.

Features: The dataset includes 21 features spanning five categories: demographic variables, lifestyle factors, health indicators, general and mental health indicators, and healthcare access. These are summarized in Table 1, highlighting the comprehensive scope of the dataset.

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Data Characteristics: The dataset is clean, with no missing or constant values. The features are well-defined, encompassing diverse factors relevant to the study’s objective.

Purpose: Created by the Centers for Disease Control and Prevention (CDC), this dataset is specifically designed to examine the interplay between lifestyle, healthcare accessibility, and diabetes prevalence in the

Category	Features
Demographic variables	Age, sex, income, education level
Lifestyle factors	Smoking status, physical activity, fruit and vegetable consumption, alcohol consumption
Health indicators	high blood pressure, high cholesterol, coronary heart disease or heart attack, stroke history
General and mental health indicators	Self-reported general health, days of poor mental health, days of poor physical health
Healthcare access	Health insurance coverage, instances of being unable to see a doctor due to cost

Table 1. Summary of features in the CDC Diabetes Health Indicators dataset. Features are grouped into thematic categories relevant to risk prediction and healthcare modeling

Feature	Description
Pregnancies	Number of pregnancies.
Glucose	2-hour plasma glucose concentration (oral glucose tolerance test).
BloodPressure	Diastolic blood pressure (mm Hg).
SkinThickness	Triceps skinfold thickness (mm).
Insulin	2-hour serum insulin (mu U/ml).
BMI	Body mass index ($\frac{kg}{m^2}$).
DiabetesPedigreeFunction	Proxy for hereditary risk (family history signal).
Age	Age in years.

Table 2. Summary of features in the PIMA dataset. Brief one-line descriptions of each diagnostic predictor.

US population. The insights derived from this dataset have broad implications for public health strategies and intervention design. **Dataset description (PIMA)**

We additionally evaluate our method on a second dataset namely the *Pima Indians Diabetes Database*, a widely used clinical benchmark originating from the National Institute of Diabetes and Digestive and Kidney Diseases (NIDDK) and described by Smith *et al.*⁴¹. The dataset comprises 768 adult female patients (age ≥ 21) of Pima Indian heritage, with **eight** diagnostic predictors and a binary diabetes label `Outcome`. A curated copy of this dataset was obtained from Kaggle.

Target Variable: The dataset includes a binary label `Outcome` (0 = no diabetes, 1 = diabetes). We use this label as provided by the repository—no relabeling or class collapsing was performed. Our study treats diabetes risk prediction as a binary classification problem, in line with the repository label. This definition enables focused evaluation of risk factors and classification performance.

Features: The eight predictors are: *Pregnancies*, *Glucose*, *BloodPressure*, *SkinThickness*, *Insulin*, *BMI*, *DiabetesPedigreeFunction*, and *Age*. For completeness, Table 2 lists the variables and brief descriptions.

Data Characteristics: The working CSV (768 $\times$ 9) contains no NA/missing values. As is standard for this dataset, several physiological fields may contain literal zeros (e.g., *Glucose*, *BloodPressure*, *SkinThickness*, *Insulin*, *BMI*); these zeros are present in the released files and are not NA placeholders. (Handling of such values, if any, is documented in the Methodology.)

Provenance and reference: The dataset traces to NIDDK and the study of Smith *et al.*⁴¹; we cite the original source and also provide the Kaggle page link (in the additional information section below) that was used to obtain the CSV.

Note on scope. Since PIMA has exactly eight predictors, all methods ultimately use the same variables. We report rankings only for interpretability; statistical tests (paired DeLong and McNemar) verify that any observed performance differences are not significant.

In the next section, we describe the methodology. This includes Gumbel- λ -based feature selection (Alg. 1), benchmark comparators, model fitting, threshold selection, and evaluation.

Methodology

For tree ensembles (RF/GB/XGB), predicted probabilities were calibrated via 5-fold isotonic regression^{42–44}. Logistic regression was left uncalibrated. Calibration was fit on the training folds only and applied to the validation/test sets. We calibrated the tree-ensemble probabilities because uncalibrated trees are typically miscalibrated (often overconfident), whereas regularized logistic regression is generally close to calibrated; therefore, we left LR uncalibrated. Logistic regression used standardized features, while tree-based models

used raw inputs. Decision thresholds were selected on the validation set to maximize F1 and then held fixed for test evaluation. All experiments used a fixed random seed and identical train/validation/test splits.

We standardize inputs only where the downstream method requires distance or coefficient comparability: (i) LR is trained on standardized predictors (scaler fit on train only and applied to val/test) to stabilize the optimizer and penalty. (ii) ReliefF relies on Euclidean nearest neighbors, so we z-score each feature on the training split only before its hit/miss computations and apply the learned mean/SD to val/test. (iii) L1EN also operates on features that have been scaled to have a zero mean and unit variance. By contrast, (iv) tree ensembles (RF/GB/ XGB) are scale-invariant and use raw inputs; and (v) the MI and mRMR filters are used without scaling: MI here is univariate $I(X_j; Y)$, so linear rescaling of X_j does not change the information content, and the mRMR redundancy term uses Pearson correlation, which is itself scale-invariant. All scalars are train-only to avoid leakage.

Research question: Which predictors exhibit upper-tail co-occurrence with diabetes status, i.e., when a feature is high, how likely is the outcome to be positive and does selecting features by this property improve predictive performance over standard baselines?

For each feature X_j , we compute the Gumbel upper-tail dependence score $\lambda_U(X_j, Y)$ from rank-based pseudo-observations and *rank features by λ_U* . We say a feature shows a positive upper-tail association when the 95% bootstrap percentile confidence interval for $\lambda_U(X_j, Y)$, computed from $B = 1000$ bootstrap resamples of the *training split* (with replacement), lies strictly above 0. We keep the top- k features and train models on that set. To judge usefulness, we compare these models with baselines (MI, mRMR, ReliefF, and L1/Elastic-Net) on the same test split, using ROC-AUC as the primary metric (with PR-AUC, accuracy, recall, and F1 as secondary metrics). For ROC-AUC, we also report 95% bootstrap percentile confidence intervals on the test set using $B = 1000$ resamples of test indices. For formal paired comparisons we use DeLong's test for ROC-AUC and McNemar's test for error profiles.

Guided by the research question above, we rank features by λ_U , fix k per dataset (CDC: $k=10$; PIMA: $k=8$), and evaluate the resulting sets alongside baselines (MI, mRMR, ReliefF, L1EN) using four classifiers (RF, XGB, LR, GB) with class-balanced training and probability calibration. We report Accuracy, Precision, Recall, F1, and ROC-AUC; we compute permutation importance on the held-out test set; and we use DeLong and McNemar tests for paired comparison. Computationally, computing Kendall's τ per feature with a closed-form transform yields an $O(d \log n)$ selector with no model fitting in the selection loop.

We study two public datasets: CDC Diabetes Health Indicators dataset and the PIMA Indians Diabetes dataset. For CDC, we use all 21 features as provided. The dataset has no missing values. For PIMA, we use the common practice of treating zeros as missing for five physiological variables: Glucose, BloodPressure, SkinThickness, Insulin, and BMI. In the training split only, we replace zeros in these columns with NaN and fit a median imputer. We then apply the learned medians to both the training and test splits.

We split each dataset once into training and test sets using an 80/20 stratified split. From the training portion, we hold out a validation subset, which is 20% of the training data. This subset is used only to select decision thresholds; we never tune models on the test set. We address class imbalance in two ways. First, when training logistic regression, random forests, and gradient boosting, we set class weights to “balanced”.

For XGBoost, we calculate `scale_pos_weight` as the ratio of negatives to positives in the training data. Second, instead of fixing the probability cutoff at 0.5, we choose the operating threshold on the validation data by scanning a fine grid of probability quantiles and selecting the cutoff that maximizes the F1 score. This approach keeps operating points comparable across different class prevalences and reduces the risk that an imbalanced base rate affects the differences in accuracy.

Our feature selection stage compares four standard baselines with the proposed Gumbel- λ_U criterion, all calculated on the training data.

1. The mutual information (MI) baseline ranks each feature using `mutual_info_classif`, which employs a k -nearest-neighbors entropy estimator with the library's default settings (effectively $k = 3$ and continuous features) under our fixed random seed. Features are sorted by the resulting $I(X_j; Y)$.
2. The elastic-net baseline fits a logistic regression using the SAGA solver on standardized predictors. It runs for up to 4,000 iterations with `class_weight="balanced"` and ranks features by the absolute coefficient magnitudes at the end.
3. The mRMR implementation is a simple greedy method that selects the first feature with the highest mutual information to the label (using the same estimator as before). It then adds the feature that maximizes “relevance minus redundancy”, where redundancy is the mean absolute Pearson correlation with the already chosen features.
4. ReliefF uses the standard binary method: we z-score each feature and then find ten nearest “hits” (same class) and ten nearest “misses” (opposite class) for every training instance using Euclidean distance. We accumulate a score equal to the average miss-distance minus the average hit-distance across all rows. Larger values indicate stronger class separation.

Our method is a supervised filter feature selector. It is supervised because the score for each feature X_j uses the label Y (via $\lambda_U(X_j, Y)$). It is a filter because features are scored and ranked without fitting any predictive model during selection; the score depends only on the data's dependence structure (via pseudo-observations and the Gumbel mapping), not on a particular classifier. Because the score is computed from pseudo-observations (ranks), it is invariant to monotone rescalings of X_j and does not assume common measurement units. By contrast, wrappers (e.g., forward selection, RFE) repeatedly train a model to evaluate subsets, and embedded methods (e.g., L1/Elastic-Net) perform selection inside model training.

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Our method focuses on upper-tail dependence between each feature and the outcome (response/label) variable. We convert both variables into pseudo-observations using their ranks. We map each variable x to pseudo-observations using equation (15) and we use average ranks for ties. The same process applies to the binary label y .

$$u_i = \frac{\text{rank}(x_i) + 1}{n+1}, \quad i = 1, \dots, n, \quad (15)$$

We calculate Kendall's τ for each feature against the label. Then, we map τ to the Gumbel copula parameter

with the formula $\lambda_U = 2 - 2\theta^{\frac{1}{1-\tau}}$ whenever $\tau > 0$. We bootstrap the training split 1000 times with replacement, recompute τ .

After that, we convert λ_U . To quantify uncertainty in the tail-dependence scores, we θ^* into the upper-tail dependence coefficient λ_U for each resample, and report the

. We rank the features based on

bootstrap the training split bootstrap mean and the 2.5th-97.5th percentile interval.

Because Gumbel models positive upper-tail clustering, we set $\lambda_U^{(j)} = 0$ when $\tau(U_j, V) \leq 0$. Intuitively, a nonpositive τ indicates no positive concordance (or an inverse association), which does not constitute upper-tail co-occurrence; such features are therefore de-prioritized by design. In CDC, several variables had $\tau < 0$ (e.g., PhysActivity, Fruits, Veggies, HvyAlcoholConsump, Education, Income) and so received $\lambda_U = 0$. In PIMA, all eight predictors had $\tau > 0$, so all received strictly positive λ_U . If protective (lower-tail) signals are of interest, a lower-tail coefficient (e.g., λ_L via Clayton or a symmetric family with both tails) could be used, which we leave for future work.

for all CDC analyses. For PIMA, no For the CDC dataset, we chose the number of features via a small sweep, k -sweep was performed: the dataset has exactly eight predictors, so all $k \in \{3, 5, 7, 10\}$, and fixed $k = 10$ methods use the same eight variables and our criterion serves as a ranking-only check. Models were trained on the training split, the operating threshold was selected on a validation split using F1, and test performance was summarized by ROC-AUC (primary) and related metrics. See Aich⁴⁵ for additional CDC sweep details.

We evaluate a diverse set of classifiers on each feature set. Logistic regression (LR) is used with balanced class weights, while random forests (RF) are also applied with balanced class weights. Gradient boosting (GB) is implemented using the standard `scikit-learn` version with a fixed random seed, and XGBoost (XGB) is employed with its standard setup, handling class weights as described below.

To address class imbalance during model training, we incorporate specific weighting strategies. For LR and RF, the `class_weight="balanced"` parameter is used, which automatically adjusts weights to be inversely proportional to the class frequencies present in the training data. For GB, we emulate class balancing by passing per-sample weights proportional to the inverse class frequencies. For XGB, we set the `scale_pos_weight` parameter equal to the ratio of the number of negative samples to positive samples.

For the tree-based models (RF, GB, XGB), we use raw feature values, while for logistic regression (LR), we standardize predictors by fitting a scaler on the training data. LR is not calibrated, whereas RF, GB, and XGB models use isotonic calibration. Our evaluation pipeline then follows a careful two-stage fit to prevent data leakage:

1. **Stage 1 (Threshold Selection):** We first train an initial model on a smaller training subset (80% of the full training data), applying the class balancing strategies defined above. For RF, GB, and XGB models, we also perform five-fold isotonic calibration on this subset. This model is then used to generate probability scores on a held-out validation set (the remaining 20% of the training data), from which we select the optimal probability threshold that maximizes the F1-score.
2. **Stage 2 (Final Fit and Testing):** With the threshold now fixed, we train a final model on the entire training set, which combines the smaller training and validation subsets. The model architecture, balancing, and calibration strategies remain identical to Stage 1, but the feature scalers and probability calibrators are re-fit on this larger dataset. This final model is then evaluated on the held-out test set, using the pre-determined threshold to assign class labels.

This two-stage process ensures the test set remains completely unseen and that the operating threshold is selected without using the final model's training data. We report ROC-AUC, accuracy, precision, recall, and F1-score on the test set.

To check if the observed differences are statistically significant, we run paired tests on the test set based on the Gumbel selection. For ROC-AUC, we use the DeLong test to compare the best model trained on Gumbel-selected features with the same model trained on each baseline feature set. This gives us a p-value for each paired AUC difference. At the same time, we compare binary correctness vectors between the

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anchored model and alternatives using McNemar's test with a continuity correction. This test determines whether the two classifiers disagree symmetrically on test instances.

We evaluate the model's behavior using permutation importance on the held-out test inputs. First, we rank all (feature set, model) combinations by ROC–AUC. We select the single best pair, fit its model to the entire training data as described earlier, and calculate permutation importance by randomly permuting each feature 500 times. We measure the average drop in ROC–AUC; we apply the same preprocessing and calibration used during training before scoring. Next, we repeat this process specifically for the Gumbel top- k feature set, using the best-performing model within Gumbel, again with 500 permutations per feature. These two approaches allow us to verify that the selected variables are significant for generalization performance.

Ordering for “All” vs. ranked sets. For CDC, “All” uses every feature. For PIMA, all methods use the same eight predictors; column order has no effect for LR/RF/GB/XGB. We include rankings only for interpretability; as expected, AUCs are statistically indistinguishable across selectors (DeLong $p > 0.05$).

Finally, we examine how robust our model is under light, controlled changes to the training and test inputs. We use the gradient boosting classifier with the Gumbel top- k features. Starting with a baseline run using the original data, we look at three stressors.

1. The first is label noise, where we randomly flip 5% of the training labels.
2. The second is feature noise, where we add zero-mean Gaussian noise equal to 10% of each feature's training-set standard deviation to both the training and test sets.
3. The third is MCAR missingness (missing completely at random), where we randomly set 10%, 20%, and 30% of the entries to missing per feature and imputed them using the training median of that feature. In this paper, we report results only for the 10% missingness scenario, while the 20% and 30% results are provided in the dissertation⁴⁵.

We evaluate each change using the same training process, and we record ROC–AUC scores on the unchanged test labels.

Algorithm

In this section, we present two related algorithms. Algorithm 1 outlines our method for selecting features based on the Gumbel copula's upper-tail dependence. Each feature is turned into pseudo-observations and linked to the label using Kendall's τ . These are then mapped to the Gumbel parameter θ and ranked by λ_U . We select the top- k features (CDC: $k = 10$; PIMA: $k = 8$) for the predictive pipeline. Algorithm 2 summarizes the complete experimental protocol. This includes stratified splits, validation-based thresholding using the F1 score, calibrated models that handle class imbalance, test-set metrics and curves, permutation importance measured by the ROC–AUC scorer with $n_repeats=500$, bootstrap confidence intervals for λ_U ($B = 1000$) and robustness checks, and significance testing. Together, these algorithms match our implementation and the results presented in the paper.

Require: Dataset $\mathcal{D} = \{(\mathbf{x}_i, y_i)\}_{i=1}^n$ with $\mathbf{x}_i \in \mathbb{R}^d$, $y_i \in \{0, 1\}$; desired k

Ensure: Top- k features ranked by upper-tail dependence λ_U

- 1: **Pseudo-observations.** For each feature X_j and the label Y , define

$$U_j \leftarrow \frac{\text{rank}(X_j)}{n+1}, \quad V \leftarrow \frac{\text{rank}(Y)}{n+1}.$$

- 2: **Association (Kendall's τ).** For each $j = 1, \dots, d$, compute Kendall's $\tau_j = \tau(U_j, V)$.
- 3: **Gumbel parameter.** Map τ_j to the Gumbel copula parameter

$$\theta_j \leftarrow \begin{cases} \frac{1}{1-\tau_j}, & \text{if } \tau_j > 0, \\ \text{undefined}, & \text{if } \tau_j \leq 0, \end{cases}$$

- 4: **Upper-tail dependence.** For each j , set

$$\lambda_U^{(j)} \leftarrow \begin{cases} 2 - 2^{1/\theta_j}, & \text{if } \theta_j \geq 1, \\ 0, & \text{otherwise.} \end{cases}$$

- 5: **Rank & select.** Sort features by $\lambda_U^{(j)}$ (descending) and return the indices of the top- k .
-

Algorithm 1. Feature Selection via Gumbel Copula Upper-Tail Dependence

Require: Dataset $\mathcal{D}$, random seed, k (CDC: $k=10$; PIMA: $k=8$), models $\mathcal{M} = \{\text{RF}, \text{XGB}, \text{LR}, \text{GB}\}$

Ensure: Test performance across feature sets; permutation importance (PI); λ_U CIs; robustness; significance tests

- 1: **Split.** Stratified train/test (80/20). From train, hold out a validation split (20% of train) for threshold tuning.
- 2: **(PIMA hygiene, only for PIMA).** Treat zeros as missing in Glucose, BloodPressure, SkinThickness, Insulin, BMI; fit a *median* imputer on train and transform both train and test (no leakage).
- 3: **Feature scoring on train only.** Compute λ_U table by Alg. 1; let Gumbel top- k be those features. Also form baselines: MI, L1/ElasticNet, mRMR, ReliefF, and the All set.
- 4: **Bootstrap CIs for λ_U (top- k).** For $B=1000$ resamples of train, recompute λ_U for the selected features; report 2.5–97.5 percentile CIs.
- 5: **for** each feature set $S \in \{\text{All}, \text{Gumbel}, \text{MI}, \text{L1EN}, \text{mRMR}, \text{ReliefF}\}$ **do**
- 6: **for** each model $m \in \mathcal{M}$ **do**
- 7: **Build model.** Use class balancing (`class_weight='balanced'` for RF/LR/GB; `scale_pos_weight` for XGB). Apply isotonic calibration (5-fold) for RF/XGB/GB. Standardize inputs for LR only.
- 8: **Fit on train, tune threshold on val.** Fit on train (with sample weights for GB). On val, choose the probability threshold that maximizes F1 (default; Youden optional).
- 9: **Final fit & test.** Refit on full train with S , apply the chosen threshold to test scores; record ROC-AUC, accuracy, precision, recall, F1; save ROC/PR curves and confusion matrix.
- 10: **end for**
- 11: **end for**
- 12: **Pick best configuration.** By primary metric (ROC-AUC), identify the best (feature set, model).
- 13: **Permutation importance (PI).**
 - Compute PI on the test set for the *overall best* (set, model) using `permutation_importance` with `scoring='roc_auc'`, `n_repeats=500`.
 - Compute PI again for the Gumbel top- k features using the *best model within Gumbel* (same scorer, `n_repeats=500`).
- 14: **Robustness checks.** Re-evaluate the best Gumbel model under small perturbations: label flip 5%, feature noise $\sigma=0.10$, MCAR missingness 10% (median impute).
- 15: **Significance tests.** For the best model, compare the Gumbel set against other sets using paired DeLong tests on ROC-AUC and McNemar's test on correctness.

Algorithm 2. End-to-End Pipeline (Train/Val/Test, Thresholding, PI, CIs; Robustness)

Reproducibility

All experiments were conducted using Google Colab Pro. The runtime environment was configured with Python 3 as the language runtime, CPU as the hardware accelerator with Colab Pro high-memory option enabled, approximately 51 GB of available RAM, and approximately 226 GB of available disk space. Because experiments were executed on Google Colab Pro (shared CPU resources), absolute wall-clock times can vary slightly across sessions. However, we consistently observe the same relative ordering of methods (the “pattern” of faster/slower selectors remains unchanged).

In the next section, we present empirical results obtained with the above pipeline and we highlight performance across the CDC and the PIMA datasets.

Results

In this section, we present the results of our Gumbel- λ_U feature selection method, which was tested on two datasets: the large-scale CDC Diabetes Health Indicators survey and the clinical PIMA Indians Diabetes Database. For each dataset, we list the top features chosen by our method and the four baseline methods. Next, we compare predictive performance using key classification metrics, present statistical significance tests, and analyze feature importance for the best models. Finally, we discuss the computational complexity of each feature selection method.

Results for the CDC dataset

After running our feature selection methods on the CDC dataset, we identified the top 10 predictors from each approach:

- **Gumbel- λ_U :** GenHlth, HighBP, DiffWalk, HighChol, BMI, HeartDiseaseorAttack, Age, PhysHlth, Stroke, CholCheck
- **MI:** GenHlth, HighBP, AnyHealthcare, PhysActivity, CholCheck, HighChol, Veggies, Fruits, BMI, Age

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- **mRMR:** GenHlth, Sex, AnyHealthcare, HvyAlcoholConsump, CholCheck, Veggies, HighChol, PhysActivity, Stroke, BMI
 - **ReliefF:** GenHlth, Age, BMI, Education, Income, HighBP, HighChol, Sex, Smoker, MentHlth
 - **L1EN:** GenHlth, BMI, Age, HighBP, HighChol, CholCheck, HvyAlcoholConsump, Sex, Income, HeartDiseaseorAttack
- Summary:** A clear pattern emerges from this comparison: GenHlth is unanimously selected as the top predictor by all five methods. There is also strong consensus for HighBP, HighChol, BMI, and Age, which recur across multiple lists. Notably, our Gumbel- λ_U selector elevates DiffWalk and HeartDiseaseorAttack. Additionally, our method also identifies PhysHlth, Stroke, and CholCheck, which are features plausibly tied to severe health status and consistent with its focus on upper-tail co-occurrence with the label. The mRMR method, by contrast, prioritizes several lifestyle and healthcare-access variables (e.g., AnyHealthcare, HvyAlcoholConsump, Veggies), reflecting its criterion that penalizes redundancy among selected features.

Next, we present the classification results.

Classification performance

The predictive performance for each feature selection method across the four classifiers is detailed in Table 3.

We report ROC-AUC with 95% bootstrap confidence intervals ($B = 1000$ resamples) to quantify uncertainty under resampling and to complement paired significance tests.

Using all 21 features gives the best overall performance, which is common in datasets with a moderate number of predictors. Moreover, the A_{11} , Gumbel- λ_U , and L1EN configurations yield closely matched ROC-AUC values, and their 95% bootstrap confidence intervals frequently overlap, indicating that the observed differences are small relative to resampling variability. Our main focus, however, is to assess how well the selection methods perform. The results in Table 3 show a clear ranking: our Gumbel- λ_U method and the L1/Elastic-Net (L1EN) baseline consistently perform among the top methods across classifiers, with strong and nearly identical results. For example, with Gradient Boosting (GB), Gumbel- λ_U attains a ROC-AUC of 0.823, which is close to L1EN’s 0.826. Within the Gumbel- λ_U feature set, GB achieves the highest ROC-AUC (0.823), making it the bestperforming classifier for our proposed selected features on the CDC dataset.

The Gumbel- λ_U feature set shows a clear advantage over several other filter methods. Its ROC-AUC of 0.823 is numerically higher than the scores for Mutual Information (MI) at 0.821 and mRMR at 0.794. As the formal statistical tests in the next section will confirm, our method is not only competitive but statistically higher ROC-AUC than MI and mRMR on this split, suggesting that the upper-tail criterion can be a useful screening signal in this setting.

To provide a visual comparison of these results, Fig. 1 displays the superimposed ROC curves for the top-performing Gradient Boosting model across each feature set. The plot clearly shows the curves for our

Set	Model	Accuracy	Precision	Recall	F1-Score	ROC-AUC (95% CI)
All	GB	0.804	0.854	0.804	0.822	0.827 [0.823, 0.832]
	LR	0.791	0.852	0.791	0.813	0.820 [0.815, 0.825]
	RF	0.779	0.847	0.779	0.804	0.801 [0.796, 0.806]
	XGB	0.809	0.853	0.809	0.826	0.827 [0.823, 0.832]
Gumbel	GB	0.813	0.850	0.813	0.827	0.823 [0.819, 0.828]
	LR	0.797	0.849	0.797	0.817	0.816 [0.812, 0.821]
	RF	0.732	0.826	0.732	0.766	0.741 [0.735, 0.746]
	XGB	0.810	0.851	0.810	0.826	0.822 [0.817, 0.827]
MI	GB	0.821	0.848	0.821	0.832	0.821 [0.817, 0.826]
	LR	0.793	0.849	0.793	0.814	0.815 [0.810, 0.820]
	RF	0.733	0.833	0.733	0.768	0.755 [0.748, 0.760]
	XGB	0.814	0.849	0.814	0.828	0.820 [0.815, 0.825]
mRMR	GB	0.781	0.843	0.781	0.804	0.794 [0.788, 0.799]
	LR	0.763	0.843	0.763	0.791	0.789 [0.784, 0.795]
	RF	0.745	0.839	0.745	0.777	0.763 [0.757, 0.769]
	XGB	0.781	0.842	0.781	0.804	0.792 [0.787, 0.797]
ReliefF	GB	0.805	0.851	0.805	0.822	0.822 [0.818, 0.827]
	LR	0.791	0.850	0.791	0.813	0.815 [0.810, 0.820]
	RF	0.748	0.835	0.748	0.779	0.766 [0.760, 0.771]
	XGB	0.814	0.849	0.814	0.828	0.821 [0.816, 0.826]
L1EN	GB	0.801	0.855	0.801	0.820	0.826 [0.822, 0.831]
	LR	0.804	0.850	0.804	0.822	0.819 [0.815, 0.824]
	RF	0.733	0.833	0.733	0.768	0.753 [0.747, 0.759]

	XGB	0.809	0.852	0.809	0.825	0.825 [0.820, 0.830]
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Table 3. Predictive performance on the CDC dataset test set for each feature selection method and classifier. The highest ROC–AUC within each feature set is bolded. ROC–AUC is reported with 95% bootstrap confidence intervals

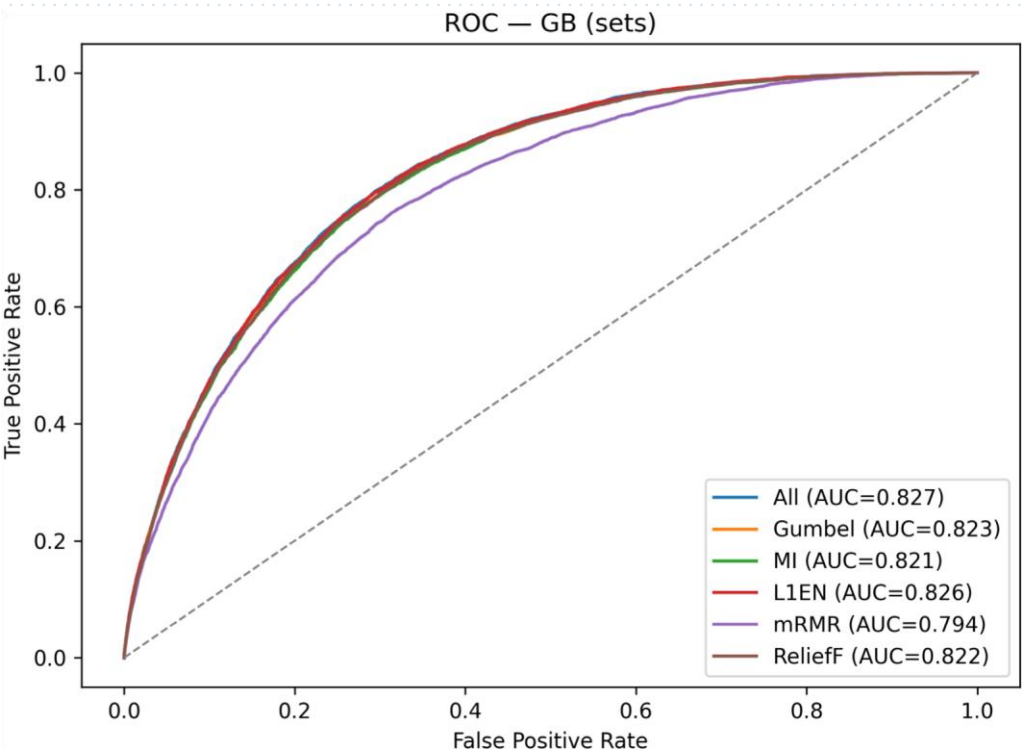


Figure 1. ROC curves for GB across feature sets on CDC (AUCs in legend); Gumbel- λ_U closely tracks the top performers.

Gumbel- λ_U method, MI, L1EN, ReliefF and the All Features baseline clustered tightly together, visually confirming their very similar performance.. In contrast, the curve for mRMR falls visibly below this top group, consistent with their lower AUC score reported in Table 3. This visualization reinforces the conclusion that our proposed method effectively identifies a feature set that is highly competitive with the best-performing baselines.

To determine if the small differences between our method and the top baselines are statistically significant, we now present the results of paired statistical tests.

Statistical tests

To validate the Gumbel- λ_U scores and assess the statistical significance of our model’s performance, we conducted bootstrap analysis and paired statistical tests. First, we generated 95% bootstrap confidence intervals for the λ_U scores of the top-10 selected features (Fig. 2). The intervals for all top-ranked predictors lie strictly above zero, indicating that our method is capturing a genuine upper-tail dependence signal rather than noise.

Now we use a covariance-aware DeLong’s test to compare ROC–AUCs in a paired manner. The model trained on all 21 features achieved the highest performance (ROC–AUC = 0.827). As expected given the modest full set (21 variables) and our emphasis on parsimony, we focus comparisons among selectors that return exactly 10 features ($k = 10$).

From Table 4, we see that our Gumbel- λ_U filter reduced the feature space by $\approx 52\%$ ($21 \rightarrow 10$) and yielded a competitive ROC–AUC of 0.823. The difference vs. All is statistically significant, but the key comparison is among 10-feature selectors: Gumbel- λ_U is better than MI ($p = 1.47 \times 10^{-5}$) and mRMR ($p < 0.001$), and statistically indistinguishable from ReliefF ($p = 0.154$). L1EN shows a statistically detectable but numerically

small AUC difference relative to Gumbel ($p = 2.69 \times 10^{-9}$; $\Delta\text{AUC} \approx 0.003$), consistent with its embedded, model-coupled nature.

Finally, McNemar’s test on paired error profiles is highly significant for Gumbel vs. each baseline (Table 4), indicating the models make different types of mistakes even when AUCs are similar.
Note. Extremely small p-values are reported as $p < 0.001$ instead of 0.
Next, we look at the permutation importance within the Gumbel set.

Permutation importance within the gumbel set (CDC)

To understand which variables our method uses, we computed permutation importance for our best CDC setup: Gradient Boosting trained on the Gumbel top-10 features (Fig. 3). On the held-out CDC test set, we permuted one feature at a time (500 repetitions) and measured the mean drop in ROC–AUC. Larger drops mean the model relies more on that feature.
The results, visualized in Fig. 3 and detailed in Table 5, reveal a clear hierarchy of feature importance. GenHlth is the most influential predictor by a wide margin. BMI and Age follow with strong contributions, and HighBP and HighChol provide additional signal. Smaller but non-zero effects are observed for CholCheck

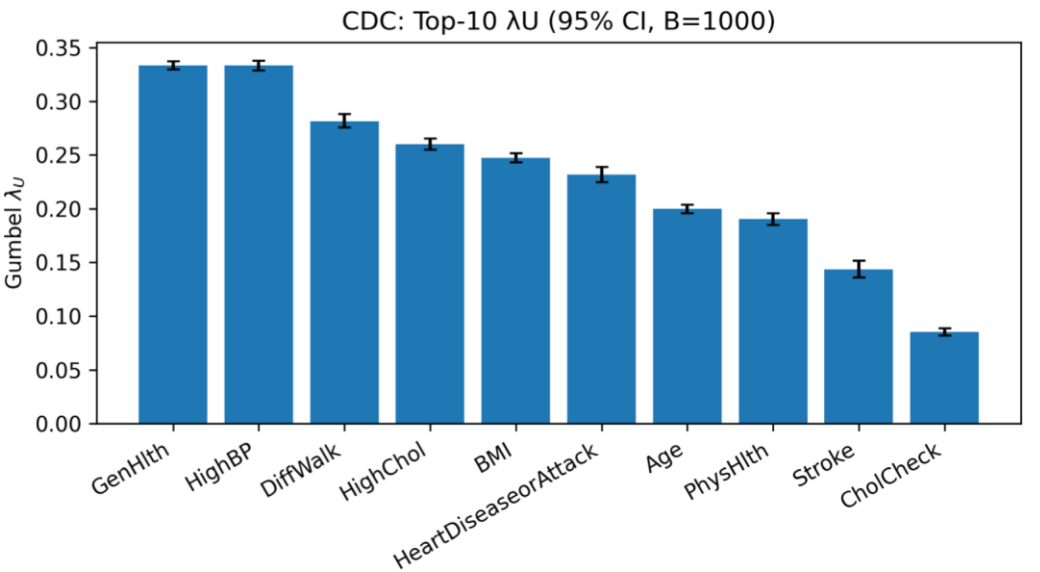


Figure 2 CDC: top-10 Gumbel λ_U with 95% bootstrap confidence intervals ($B = 1000$).

Comparison	DeLong p	McNemar χ^2	McNemar p
Gumbel vs. All	< 0.001	94.86	2.05×10^{-22}
Gumbel vs. MI	1.47×10^{-5}	135.96	2.03×10^{-31}
Gumbel vs. mRMR	< 0.001	496.05	6.89×10^{-110}
Gumbel vs. ReliefF	0.154	70.30	5.10×10^{-17}
Gumbel vs. LIEN	2.69×10^{-9}	158.02	3.07×10^{-36}

Table 4. Paired tests on the CDC test set comparing the Gumbel-selected feature set against alternatives (Gradient Boosting). DeLong compares ROC–AUC; McNemar compares error profiles.

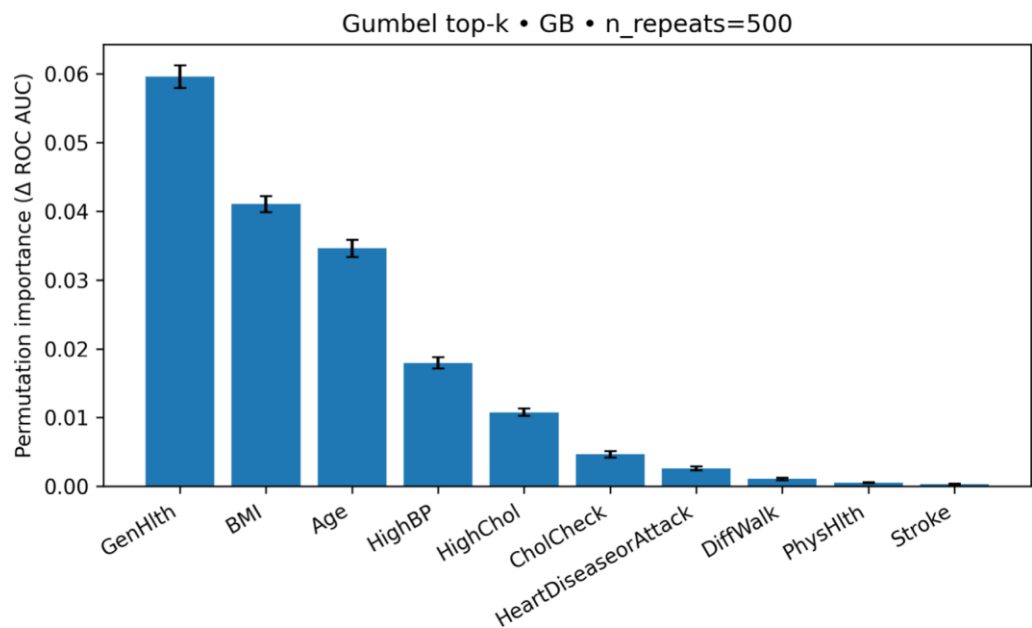


Figure 3. Permutation importance (mean Δ ROC–AUC over 500 permutations) for gradient boosting on the Gumbel top-10 features (CDC test set). Larger values indicate greater importance.

Feature	Mean importance (Δ ROC–AUC)	SD
GenHlth	0.059574	0.001662
BMI	0.041051	0.001183
Age	0.034596	0.001264
HighBP	0.017929	0.000814
HighChol	0.010755	0.000545
CholCheck	0.004652	0.000475
HeartDiseaseorAttack	0.002608	0.000268
DiffWalk	0.001073	0.000197
PhysHlth	0.000501	0.000082
Stroke	0.000303	0.000108

Table 5. Permutation importance of the top-10 features selected by the Gumbel- λ_U method for the CDC dataset. Importance is measured as the mean decrease in ROC–AUC over 500 permutations on the test set, using the trained Gradient Boosting model.

Type	Level	ROC–AUC
Baseline	0.00	0.82321
Label flip	0.05	0.82230
Feature noise (σ)	0.10	0.82270
MCAR	0.10	0.81266

Table 6. CDC robustness: GB on Gumbel top-10.

Selector	Time (s)
Gumbel- λ_U	0.332
L1EN	3.102
MI	20.205
mRMR	20.515

ReliefF	925.591
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Table 7. Feature-selection wall-clock time (seconds) on the CDC dataset.

and HeartDiseaseorAttack, while DiffWalk, PhysHlth, and Stroke contribute modestly. All ten Gumbel-selected features have positive mean importance, confirming that each contributes meaningfully to the model’s performance.

After this, we evaluated the stability of our Gumbel-selected feature set under several common data perturbations.

Robustness checks

We stress-tested the Gradient Boosting (GB) classifier trained on the Gumbel top-10 features under three perturbations (Table 6). Relative to the baseline ROC–AUC of 0.823, performance remained within ≈ 0.001 under label-flip 5% (0.8223) and feature noise $\sigma=0.10$ (0.8227), and declined by ≈ 0.011 under MCAR 10% (0.8127), confirming robustness under light noise.

Having confirmed that our method’s performance is stable under noisy conditions, we now check the computational complexity of the feature selectors.

Runtime of feature selectors

Asymptotically the selector is $O(dn\log n)$ as detailed in Preliminaries; wall-clock times below reflect that efficiency on CDC.

We recorded wall-clock time for each selector on the CDC training split (Table 7). Gumbel- λ_U is fastest (0.332 s) slower ($\sim$, about 2.8×10^3 faster than LIEN and). $\approx 61 \times$ faster than MI and mRMR; ReliefF is orders of magnitude

Comment on CDC feature type and model behavior. $\times$ The CDC dataset contains a large number of binary/

discrete predictors. In such settings, a linear model can perform competitively because much of the apparent nonlinearity is effectively represented through indicator coding, which helps explain why logistic regression can match or exceed tree-based models in our benchmarks. As a result, while the CDC experiment provides strong evidence that the proposed upper-tail screening is effective and efficient, further evaluation on datasets with richer continuous predictors and stronger interaction-driven nonlinearities is a natural direction for future work.

Next, we move to the PIMA dataset.

Results for the PIMA dataset

PIMA serves as a complementary clinical benchmark where no dimensionality reduction is possible (eight total predictors). Consequently, all methods use the same variables; differences reflect sampling and calibration randomness (column order has no effect for LR/RF/GB/XGB). Under this setup, the Gumbel ranking paired with Random Forest achieved the *numerically highest* ROC–AUC (0.867). Formal paired tests are reported in the Statistical Tests subsection.

After running our feature?selection methods on the PIMA dataset, we identified the top 8 predictors (which are same as all the features) from each approach:

- **Gumbel- λ_U :** Glucose, BMI, Age, Insulin, SkinThickness, Pregnancies, BloodPressure, DiabetesPedigreeFunction
 - **MI:**Glucose, BMI, Age, BloodPressure, Insulin, Pregnancies, SkinThickness, DiabetesPedigreeFunction
 - **mRMR:**Glucose, Pregnancies, BMI, DiabetesPedigreeFunction, Insulin, BloodPressure, Age, SkinThickness
 - **ReliefF:**Glucose, BMI, SkinThickness, BloodPressure, Age, Pregnancies, DiabetesPedigreeFunction, Insulin
 - **LIEN:**Glucose, BMI, Pregnancies, DiabetesPedigreeFunction, Insulin, BloodPressure, Age, SkinThickness
- Summary:**A clear pattern emerges for PIMA: Glucose stands out as the top predictor in all five methods. There is also strong agreement on the importance of BMI and Age, which consistently rank near the top, while BloodPressure and Pregnancies appear frequently as well. Our Gumbel- λ_U selector ranks Insulin and SkinThickness higher than some other methods, which matches its focus on upper-tail co-occurrence with the label, meaning it highlights extreme metabolic readings that align with positive outcomes. In contrast, mRMR produces a slightly different order by bringing DiabetesPedigreeFunction forward, reflecting its approach to reducing redundancy. ReliefF gives more weight to SkinThickness because of its ability to separate classes locally. Since the PIMA dataset has eight predictors, all methods keep the full set, so the

main difference is in the ranking. The Gumbel method highlights clinically relevant factors like glycemic control and adiposity as key drivers of diabetes risk.

After this, we present the classification results.

Classification performance

PIMA has only 8 predictors, so feature selection here is more about ranking than reducing dimensions. We include it as a clinical benchmark to check that our tail-dependence criterion works as expected on a small, wellstudied dataset, to compare with standard selectors using the same training and thresholds, and to show that our method performs as well as or better than others without needing many features. This approach complements the CDC results, where dimensionality reduction plays a larger role.

The predictive performance for each feature ranking on the PIMA dataset is presented in Table 8. A unique characteristic of this benchmark is that it contains only eight predictors, so every method ultimately ranks the full feature set. Because every model trains on exactly the same eight variables, AUCs are expected to be very close. The small differences observed reflect sampling/calibration randomness rather than meaningful effects, which, as later confirmed by our DeLong tests (all $p > 0.05$), are not significant.

We report ROC–AUC with 95% bootstrap confidence intervals ($B = 1000$ resamples) to quantify uncertainty under resampling and to complement paired significance tests.

On PIMA (8 predictors), dimensionality reduction is not the goal, so this experiment primarily serves as a ranking sanity check in a low-dimensional clinical setting. In Table 8, the best-performing configurations across All, Gumbel- λ_U , and strong baselines such as Relieff and LIEN are very close, and the 95% ROC–AUC bootstrap confidence intervals overlap substantially, consistent with the expectation that differences largely reflect sampling variability (and modest model-fitting/calibration effects) rather than screening gains. Nevertheless, Gumbel- λ_U paired with Random Forest achieves the numerically highest ROC–AUC (0.867; 95% CI [0.804, 0.920]), indicating that the tail-aware ranking remains competitive and clinically coherent even when dimensionality reduction is not possible.

The wider ROC–AUC confidence intervals on the Pima dataset are expected due to its smaller sample size, which increases sampling variability; conversely, the larger CDC cohort yields tighter intervals.

To visualize the PIMA results, Fig. 4 shows superimposed ROC curves for the Random Forest model across all feature sets. The curves are tightly clustered, indicating very similar discrimination. The Gumbel- λ_U set attains the numerically highest AUC (0.867), with mRMR, All, LIEN, and Relieff close behind and MI slightly lower.

A key finding is that even with a small number of features where feature selection is not needed to manage model complexity, our upper-tail dependence ranking still performs at a high level. It highlights clinically relevant factors such as Glucose, BMI, and Age, supporting CDC findings that focusing on joint extremes can be informative.

To determine if the small differences between our method and the top baselines are statistically significant, we now present the results of paired statistical tests.

Note: PIMA has 8 predictors; all methods use the same variables. Reported differences therefore arise from ranked ordering and model refitting/calibration rather than feature removal.

Set	Model	Accuracy	Precision	Recall	F1	ROC–AUC (95% CI)
All	GB	0.773	0.806	0.773	0.778	0.860 [0.791, 0.916]
	LR	0.773	0.768	0.773	0.766	0.862 [0.796, 0.913]
	RF	0.766	0.792	0.766	0.771	0.861 [0.798, 0.918]
	XGB	0.779	0.786	0.779	0.782	0.840 [0.767, 0.899]
Gumbel	GB	0.682	0.790	0.682	0.685	0.858 [0.789, 0.914]
	LR	0.773	0.768	0.773	0.766	0.862 [0.796, 0.913]
	RF	0.753	0.801	0.753	0.759	0.867 [0.804, 0.920]
	XGB	0.760	0.804	0.760	0.765	0.824 [0.748, 0.885]
MI	GB	0.773	0.806	0.773	0.778	0.859 [0.789, 0.914]
	LR	0.773	0.768	0.773	0.766	0.862 [0.796, 0.913]
	RF	0.812	0.817	0.812	0.813	0.859 [0.794, 0.912]
	XGB	0.747	0.776	0.747	0.752	0.842 [0.769, 0.902]
mRMR	GB	0.760	0.799	0.760	0.765	0.863 [0.795, 0.919]
	LR	0.773	0.768	0.773	0.766	0.862 [0.796, 0.913]
	RF	0.760	0.799	0.760	0.765	0.863 [0.796, 0.917]

	XGB	0.753	0.784	0.753	0.759	0.839 [0.768, 0.898]
ReliefF	GB	0.779	0.815	0.779	0.784	0.866 [0.799, 0.920]
	LR	0.773	0.768	0.773	0.766	0.862 [0.796, 0.913]
	RF	0.812	0.813	0.812	0.812	0.860 [0.795, 0.915]
	XGB	0.799	0.804	0.799	0.801	0.835 [0.765, 0.896]
L1EN	GB	0.760	0.817	0.760	0.765	0.864 [0.799, 0.918]
	LR	0.773	0.768	0.773	0.766	0.862 [0.796, 0.913]
	RF	0.792	0.809	0.792	0.796	0.861 [0.800, 0.913]
	XGB	0.753	0.775	0.753	0.758	0.833 [0.763, 0.894]

Table 8. Predictive performance on the PIMA dataset test set for each feature-ranking and classifier. We report ROC–AUC with 95% bootstrap confidence intervals. The highest ROC–AUC within each feature set is bolded.

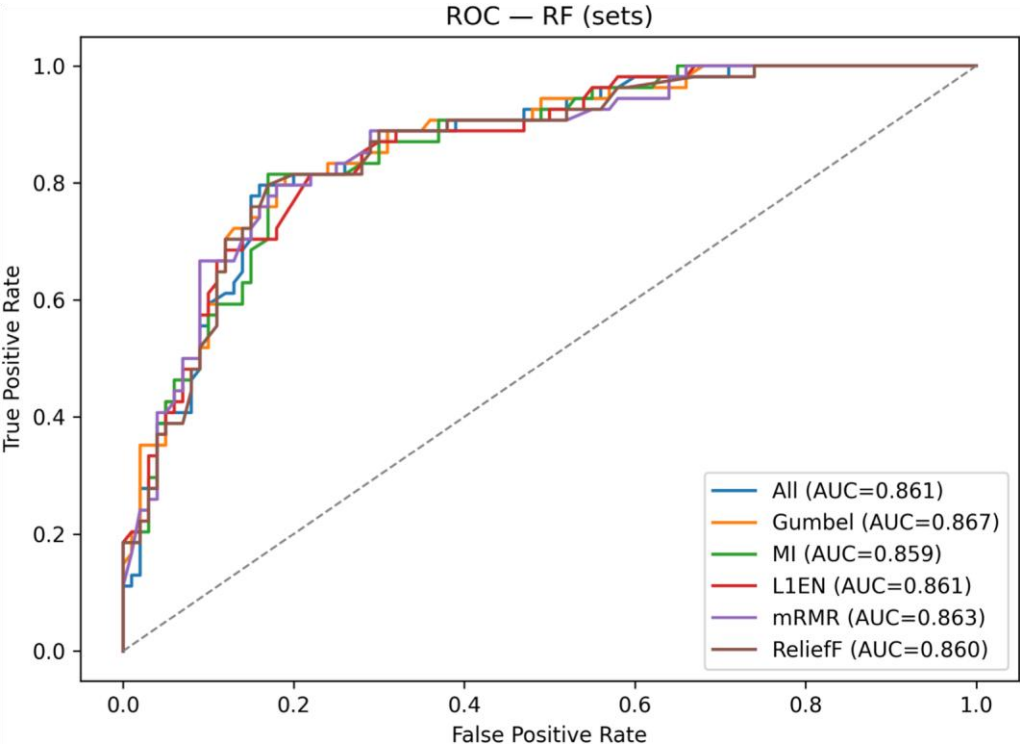


Figure 4. Superimposed ROC curves for random forest on PIMA across feature sets. Gumbel- λ_U yields the top AUC (0.867); all others are close, underscoring minimal sensitivity to feature set size on this 8-variable benchmark.

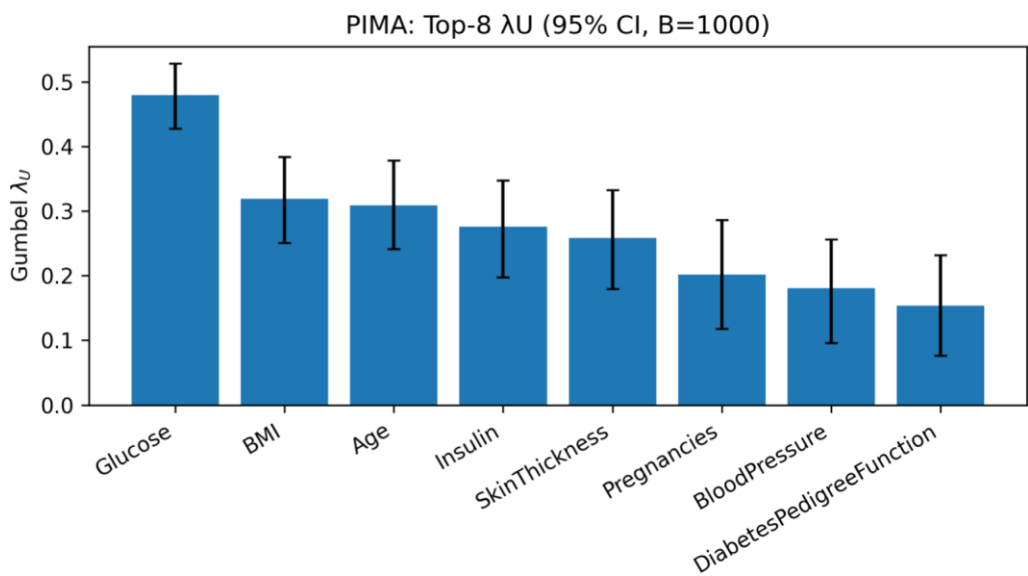


Figure 5 PIMA: Top-8 λ_U with 95% bootstrap CIs ($B = 1000$).

Comparison	DeLong <i>p</i> -value	McNemar χ^2	McNemar <i>p</i> -value
Gumbel versus All Features	0.253	0.13	0.724
Gumbel versus MI	0.095	3.37	0.066
Gumbel versus mRMR	0.504	0.00	1.000
Gumbel versus ReliefF	0.101	2.78	0.095
Gumbel versus L1EN	0.255	2.08	0.149

Table 9. Paired statistical test results comparing the Gumbel-ranked feature set against all others on the PIMA test set, using the Random Forest classifier.

Statistical tests

To validate the Gumbel- λ_U feature rankings and assess the statistical significance of our model’s performance on the PIMA dataset, we again conducted bootstrap analysis and paired statistical tests. First, we generated 95% bootstrap confidence intervals for the λ_U scores of the top-ranked features (see Fig. 5). As with the CDC data, the confidence intervals for the clinically important predictors like Glucose, BMI, and Age are strictly greater than zero, confirming that our method is capturing a genuine and statistically significant upper-tail dependence signal.

Next, we used a covariance-aware DeLong’s test to compare the ROC–AUC of the best-performing classifier (Random Forest) trained on the Gumbel- λ_U features against each baseline. As shown in Table 9, all pairwise comparisons yielded $p > 0.05$. Therefore, we fail to reject the null in every case and conclude there is no statistically significant difference in AUC. This supports that, in this low-dimensional clinical setting, the Gumbel- λ_U ranking achieves performance statistically indistinguishable from strong baselines.

Interestingly, as seen from Table 9, in contrast to the CDC results, McNemar’s test also returned nonsignificant p -values ($p > 0.05$) for all comparisons. This indicates that on this smaller, more homogeneous clinical dataset, the different feature rankings not only produce similar overall accuracy but also share a similar pattern of classification errors.

After this, we look at the permutation importance within the Gumbel set.

Permutation importance within the gumbel set (PIMA)

We computed permutation importance on the PIMA test set using the best configuration, Random Forest trained on the eight Gumbel- λ_U features, by permuting each feature 500 times and measuring the mean drop in ROC–AUC (error bars show ± 1 SD across permutations). As shown in Fig. 6 and detailed in Table 10, Glucose dominates: its importance is a little over three times that of the next features (BMI and Age). Insulin, DiabetesPedigreeFunction, and Pregnancies provide smaller but positive contributions, while SkinThickness is marginal and BloodPressure is effectively zero. These results align with clinical expectations and confirm that the Gumbel- λ_U ranking concentrates signal on the most informative predictors.

Next, we evaluated the stability of our Gumbel-selected feature set under several common data perturbations.

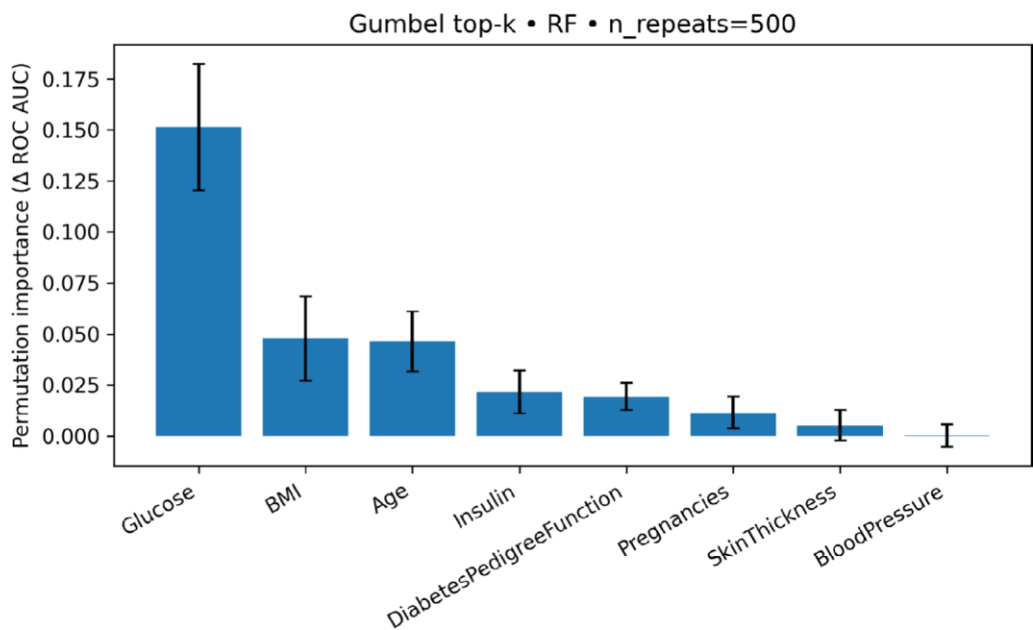


Figure 6. Permutation importance (mean Δ ROC-AUC; ± 1 SD over 500 permutations) for Random Forest on the PIMA Gumbel- λ_U feature set.

Feature	Mean importance (Δ ROC-AUC)	SD
Glucose	0.1514	0.0310
BMI	0.0477	0.0208
Age	0.0463	0.0147
Insulin	0.0216	0.0105
DiabetesPedigreeFunction	0.0193	0.0066
Pregnancies	0.0114	0.0078
SkinThickness	0.0054	0.0075
BloodPressure	0.0002	0.0055

Table 10. Permutation importance of the Gumbel-ranked features for the PIMA dataset. Importance is measured as the mean decrease in ROC-AUC over 500 permutations on the test set, using the trained Random Forest model.

Type	Level	ROC-AUC
Baseline	0.00	0.8666
Label flip	0.05	0.8519
Feature noise (σ)	0.10	0.8676
MCAR	0.10	0.8597

Table 11. PIMA robustness: RF on Gumbel-ranked features.

Robustness checks

We stress-tested the Random Forest trained on the Gumbel top-8 features under three perturbations (Table 11). Relative to the baseline ROC-AUC of 0.8666, performance declined by about 0.015 with label-flip 5% (0.8519), was essentially unchanged (slightly higher) with feature noise $\sigma = 0.10$ (0.8676), and declined by about 0.007 under MCAR 10% (0.8597). These small (≤ 0.015 AUC) shifts indicate the model remains robust to light label noise, moderate feature perturbations, and modest missingness.

Having confirmed that our method’s performance is stable under noisy conditions, we now check the computational complexity of the feature selectors.

Selector	Time (s)
L1EN	0.0049
MI	0.0221
mRMR	0.0283
Gumbel- λ_U	0.0291
ReliefF	0.4711

Table 12. Feature-selection wall-clock time (seconds) on the PIMA dataset.

Runtime of feature selectors

As above, the selector is $O(dn \log n)$; on the small PIMA split, all filter methods are within the same order of magnitude. We recorded the wall-clock time for each selector on the PIMA training split (Table 12). On this smaller dataset, L1/Elastic-Net is the fastest at 0.0049 s, while ReliefF remains the slowest at 0.4711 s. Crucially, our Gumbel- λ_U method is comparably fast to other standard filters like MI and mRMR (all in the $\sim 0.02\text{--}0.03$ s range), demonstrating competitive predictive performance on this dataset without any significant computational penalty.

This successful validation on the PIMA clinical benchmark concludes our quantitative analysis. Next, we discuss the medical and public health implications of our findings for both datasets.

Limitation. PIMA’s low dimensionality limits conclusions about feature reduction; future work will assess richer multi-center EHR cohorts to evaluate tail-dependent selection under higher dimensionality and domain shift.

Two regimes, complementary evidence. On CDC (21 features), upper-tail filtering reduces the feature set by $\sim 52\%$. This introduces a minor but statistically significant performance trade-off against using the full feature set, but our method’s key advantage is highlighted by its statistically higher performance compared to other filter baselines, demonstrating its efficiency and ranking power. On PIMA (8 features), where selection is not applicable, our criterion yields clinically coherent rankings and maintains discrimination (no significant AUC differences), i.e., it does no harm in low-dimensional clinical settings. Together, these results support the method’s robustness, interpretability, and practicality across dataset types.

Medical and public health implications

Our feature selector ranks variables using the Gumbel copula upper-tail dependence $\lambda_U(X, Y)$. This measures how likely it is that a predictor X and the diabetes label Y are both extreme at the same time. A high λ_U highlights variables that appear with diabetes in the highest-risk groups, which is exactly where screening and intervention are most important.

Clinical practice vs study contribution. The points below (for both CDC and PIMA) are not presented as new clinical guidelines. We emphasize that our study does not propose new interventions; rather, it provides a dependence-based prioritization lens from the upper-tail ranking that complements, and does not replace, established clinical practice and clinician judgment. Instead, we (i) reference established screening/management practices commonly used in diabetes and cardiometabolic care to aid interpretation, and (ii) highlight study-informed implications that arise specifically from our upper-tail dependence ranking, which is intended to prioritize attention toward the highest-risk strata rather than to propose new interventions.

CDC (Top 10 by Gumbel- λ_U : GenHlth, HighBP, DiffWalk, HighChol, BMI, HeartDiseaseorAttack, Age, PhysHlth, Stroke, CholCheck).

1. **GenHlth** (self-reported general health). Poor global health concentrates in the diabetic upper tail. Implication: Use brief self-rated health screens to triage patients into accelerated diabetes risk assessment and counseling.
2. **HighBP** (hypertension). Upper-tail co-occurrence reflects cardiometabolic clustering. Implication: Integrate BP control with diabetes prevention programs and treat thresholds aggressively in high-risk clinics.
3. **DiffWalk** (difficulty walking/climbing). Functional limitation aligns with severe metabolic disease. Implication: Pair screening with referrals to physical therapy/fall-prevention and structured activity programs.
4. **HighChol** (hypercholesterolemia). Atherogenic profiles spike with diabetes risk. Implication: Integrate lipid management (statins, diet) into diabetes prevention workflows.
5. **BMI**. Extreme adiposity is strongly tail-linked to diabetes. Implication: Prioritize intensive lifestyle/weight-management services for patients above BMI cut-points.
6. **HeartDiseaseorAttack** (CHD/MI history). Macrovascular disease clusters with diabetic risk. Implication: Coordinated cardiology-endocrinology pathways and tight control of BP, lipids, and glucose.
7. **Age**. Risk concentrates in older strata. Implication: Age-adjusted screening cadence (e.g., lower testing thresholds after 45–50 years).
8. **PhysHlth**. Days of poor physical health align with high risk. Implication: Flag for care coordination and rehab referrals.

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9. **Stroke.** Vascular disease clusters with diabetes. Implication: Prioritize cardio-metabolic management.
10. **CholCheck.** Screening/healthcare engagement signal. Implication: Leverage for targeted outreach.

These signals help guide a two-step process in population health. First, brief intake items like self-rated health, blood pressure, BMI, and mobility are used to identify patients at higher risk. Next, those flagged are directed to bundled interventions that address nutrition, blood pressure and lipid control, and mobility support. Since λ_U highlights the most extreme cases, these steps focus on patients who are most likely to face serious health issues soon.

PIMA (Top 8 by Gumbel- λ_U : Glucose, BMI, Age, Insulin, SkinThickness, Pregnancies, BloodPressure, DiabetesPedigreeFunction).

1. **Glucose.** The dominant tail feature; high OGTT glucose co-occurs with diabetes by design and in practice. Implication: Prioritize confirmatory testing and rapid care escalation when screening glucose is extreme.
2. **BMI.** Strong adiposity signal in the upper tail. Implication: Implement intensive weight-management and lifestyle programs for patients with high BMI.
3. **Age.** Risk concentrates with advancing age. Implication: Tighten screening intervals in older adults.
4. **Insulin.** Elevated 2-hour insulin may reflect insulin resistance. Implication: Pair education on diet/activity with evaluation for metabolic syndrome.
5. **SkinThickness.** A proxy for subcutaneous fat. Implication: Reinforce anthropometric assessment when lab testing is unavailable; use as an adjunct risk flag.
6. **Pregnancies.** Parity is associated with later diabetes risk (e.g., via gestational diabetes history). Implication: Capture obstetric history and screen earlier/more often in women with multiple pregnancies or prior GDM.
7. **BloodPressure.** Hypertension co-clusters with diabetes. Implication: Integrate BP control with diabetes prevention/management visits.
8. **DiabetesPedigreeFunction.** Familial risk concentrates in the tail. Implication: Adopt earlier screening for individuals with strong family history, even at moderate BMI.

The PIMA signals match what clinicians usually expect: glucose, BMI, and age are the most important factors. Our ranking method highlights that patients with the highest or lowest values should be prioritized for further testing and targeted risk-reduction support.

Summary. In both datasets, Gumbel- λ_U identifies a clear group of upper-tail risk factors: global health and cardiometabolic burden in the CDC data, and glucose, adiposity, and age in the PIMA data. These findings suggest that public health efforts should focus on targeted screening and bundled interventions for people in the highest percentiles, where diabetes is most likely to co-occur and prevention can have the greatest impact.

Conclusion and future work

In this study, we developed a new supervised filter method for feature selection using the Gumbel copula's uppertail dependence coefficient, λ_U interpreted as a rank-based tail concordance score. By focusing on how extreme values in predictors and the target label occur together, our method provides a fresh way to find features that matter most in high-risk situations. Our thorough evaluation, supports that this approach is statistically wellmotivated, effective in our experiments, and computationally efficient.

We tested our method on two different diabetes datasets and found strong support for its usefulness. On the large CDC public health dataset, our Gumbel- λ_U selector was competitive and was the fastest method, running much quicker than other approaches. It also reduced the number of features by approximately 52% (from 21 to 10), and on this parsimonious set it achieved statistically higher performance than standard filters such as Mutual Information and mRMR, while being statistically indistinguishable from the strong ReliefF baseline. On the PIMA dataset (8 predictors), our ranking achieved the numerically highest ROC-AUC while remaining statistically indistinguishable from strong baselines, functioning as a ranking-only clinical sanity check. Although it was not the fastest on this smaller dataset, its speed was still close to other standard filters, showing that its strong performance did not come at a high computational cost. In both cases, our method consistently highlighted predictors that are clinically plausible and consistent with prior knowledge, showing it can identify meaningful and understandable signals.

This work suggests several promising directions for future research, including new methods and applications.

Limitation. The proposed screening procedure scores each predictor individually (purely marginally) via its upper-tail dependence with the outcome and therefore does not explicitly capture interaction effects among predictors. As a result, a feature that is weak marginally (or marginally uncorrelated) but informative through joint effects may be under-ranked or missed. This is a known limitation of marginal screening and is explicitly discussed in the SIS literature⁴⁶. Group-based screening extensions have been proposed to better capture joint effects; for example, DC-SIS can be directly employed to screen grouped predictors (and multivariate responses)⁴⁷. In future work, we plan to extend our framework to interaction-aware dependence screens (e.g., groupwise tail-dependence screening for small predictor subsets and conditional tail-dependence criteria) while retaining the interpretability and computational efficiency of the current filter.

Methodological Extensions.

This work suggests many ways to extend the methodology. Since we are focused on upper-tail co-occurrence, we chose the Gumbel family with nonzero λ_U based on its properties, instead of selecting a copula based on model fit. One next step is to look beyond the Gumbel family and compare selectors from other copulas with upper-tail dependence, such as the Joe, Student's t, or BB1 families. It may also be useful to

explore new generators like the new A1 and A2 copulas^{48,49}, which are designed for very strong dependence. The dependence criterion could be broadened to use the lower-tail coefficient (λ_L) to find protective factors, or a combined metric for a more balanced view of risk. To make the method more rigorous, it could be combined with stability selection to control false discoveries in high-dimensional data, analyzed for finite-sample complexity, and improved with automatic feature selection using nested cross-validation or stability-based rules. Future work could also focus on making the approach more robust to covariate shift and complex missing data patterns, such as missing at random (MAR) or missing not at random (MNAR), by using copula-aware imputation. The framework could be adapted for multi-class, ordinal, and time-to-event survival data, not just binary outcomes. Finally, performance might be improved by creating hybrid and ensemble selectors that combine λ_U scores with embedded methods like L1EN, or by using decision-analytic thresholding based on clinical utility and net benefit instead of just maximizing F1.

New Application Domains in Bioinformatics and Clinical Science.

The idea of finding features that are extreme alongside high-risk outcomes can be used in many areas of biomedical research, especially where there is a lot of data. For example, in genomics and transcriptomics, this approach can help identify genes that are highly active in aggressive cancers. In proteomics and metabolomics, it can find molecules that are most abundant in severe disease or in response to treatment. This method can also improve clinical trial analysis by spotting unusual lab values that predict which patients might benefit most or be at higher risk, making it easier to focus on the right groups. In neuroimaging, it provides a new way to look at fMRI or PET scans by highlighting brain regions with unusually high activity linked to certain neurological disorders, adding to what we learn from average-based analyses. All of these uses aim to build more accurate or refined personalized risk models, helping to identify patients who are most at risk so they can get early and targeted care.

Practical Deployment and Ethics.

To make this method truly effective in real-world settings, future research should tackle the challenges of practical deployment. Because important decisions often depend on spotting rare but serious risks, our model needs to be reliable when predicting extreme cases, not just accurate on average. This means we should create calibration methods that pay special attention to these rare events, so the model's confidence is justified for the highest-risk patients. We also need to carry out thorough fairness checks to make sure focusing on these extremes does not unfairly affect vulnerable groups. Building trust in the model depends on transparency, which we can support by making the selector a straightforward, repeatable preprocessing step and by recording all key details, such as random seeds, feature rankings, and confidence intervals. This approach helps ensure the process is both auditable and reproducible.

In summary, using copula-based upper-tail dependence offers an efficient and interpretable way to select features. By prioritizing predictors associated with outcomes in the upper tail and performing as well as or better than strong baselines on both large and small datasets in our experiments, this method can be a useful tool for machine learning applications in health data. The suggested extensions above outline ways to strengthen the theory, handle real-world data challenges, and evaluate the approach on additional biomedical settings.

Data availability

The CDC Diabetes Health Indicators dataset has been accessed from the UCI Machine Learning Repository using the link <https://doi.org/10.24432/C53919>. The PIMA Indians Diabetes dataset originates from NIDDK and Smith et al.; a curated copy used for this work was accessed via Kaggle at <https://www.kaggle.com/datasets/uciml/pima-indians-diabetes-database>.

Code availability

All code to reproduce the analyses is available at <https://github.com/agnivibes/copula-feature-selection>.

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Author contributions

A.A.: Conceived and designed the study, developed the methodology, ran the experiments, analyzed the results, and wrote the manuscript. M.M.M. and S.H.: Supported the literature review and assisted in manuscript preparation. A.M.: Interpreted the results in the public health context.

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Declarations Competing interests

The authors declare no competing interests.

Additional information

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